TEL AVIV UNIVERSITY

The Iby and Aladar Fleischman Faculty of Engineering
School of Electrical and Computer Engineering

Deep Learning Architectures for Two-Hop Relay Communication: A Comparative Study of Classical and Neural Network Relay Strategies

A thesis submitted toward the degree of
Master of Science in Electrical and Electronic Engineering
at Tel Aviv University

by

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This research work was carried out at Tel-Aviv University
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Abstract

This thesis compares classical and learned processing in a two-hop, single-relay communication system without a direct source–destination path. It asks where a small learned relay improves on the implemented classical baselines, and how that comparison depends on channel information and computation.

The canonical experiment uses uncoded Gray-QPSK, SISO i.i.d. Rayleigh fading, and receiver-side channel knowledge. Eight relay strategies are compared by Monte Carlo simulation. Learned feedforward relays improve on the implemented block-normalized AF at low SNR, but against conventional finite-power AF the MLP's 0 dB margin is only 0.003–0.007, depending on the gain rule; these relays track DF, which matches or exceeds them from 6 dB upward. Replicated size sweeps identify small configurations meeting the specified degradation criteria; they do not establish an overfitting mechanism or universal minimum architecture.

Separate experiments introduce ISI, amplifier nonlinearity, and uncertain channel information. On the selected fixed three-tap BPSK channel, a 170-parameter windowed MLP substantially lowers BER relative to AF and zero-threshold DF. Matched Viterbi MLSE remains ahead over the resolved range, with target-dependent required-SNR penalties of 0.26–2.53 dB. Fixed-budget MLP validation extends through 16 dB; the higher-SNR tail is not validated. In a separate composite-channel pilot sweep at 10 dB, the MLP trails pilot-aided detection at twenty pilots, is not resolved from it at ten and leads at five. These results concern the trained impairment families, not unseen model classes.

A separately configured coded-QPSK memory study uses a 220-parameter relay with 208 MACs per symbol. An exactly factorized MLSE needs fewer multiplications than this relay up to six taps and 1.23 times as many at seven, and retains the lower BER; the count of the implemented joint trellis, which crosses the relay's between three and five taps, is a diagnostic of that implementation. In the coded study, the learned MLP relays' high-SNR advantage over block decode-and-forward disappears once the decoders weight each symbol by its channel reliability: with matched metrics, block decode-and-forward has the lower error rate at every tested SNR. The QPSK BCJR control measures relay-output errors only.

The evidence supports bounded benefits against specified classical pipelines. It does not establish universal learned-relay superiority, generalization outside the training families, or hardware latency and energy savings.

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List of Symbols

Symbols are grouped as Latin capitals, Latin lowercase, Greek letters and other notation. Where a symbol carries more than one meaning, every meaning is listed and the context that disambiguates it is named.

Latin capitals

C	Ergodic Rayleigh (Shannon) capacity, bits per channel symbol
E_b	Energy per information bit
E_s	Average energy per transmitted symbol
G	Goodput, information bits per channel symbol; $G = \eta(1 - \mathrm{FER})$; *also* AF amplification gain where explicitly stated
K	Constraint length of the convolutional code
L	FIR channel length (number of taps; memory order $L - 1$); *also* block length in symbols where explicitly stated
$L_{\max}$	Idealized buffer-readiness budget in the coded study, in symbols
M	Constellation order (points per constellation)
N_0	One-sided noise power spectral density
P_e	Bit error probability
P_b	Destination bit error probability of the relay-design problem (Section 4.1.5)
$Q(\cdot)$	Gaussian Q-function
R	Code rate, k/n
S	Number of trellis states, 2^{K-1} for a rate-$1/n$ code

Latin lowercase

$f_\theta(\cdot)$	Relay processing function; the only block varied in the canonical comparison
g	Unknown channel gain in the flat-channel control
$h_1[i]$, $h_2[i]$	Hop-1 and hop-2 fading coefficients at symbol index i
i	Symbol (time) index
k	Information bits per code input group; *also* bits per symbol, $\log_2 M$
n	Coded output bits per input group
$n_1[i]$, $n_2[i]$	Additive noise on hops 1 and 2
w	Window half-width; the relay observes $2w + 1$ samples
$x[i]$	Symbol transmitted by the source

$x_R[i]$	Symbol transmitted by the relay
$y_R[i]$, $y_D[i]$	Signal received at the relay and at the destination

Greek letters

$\alpha_k(s)$	BCJR forward recursion at trellis state s, step k
$\beta_k(s)$	BCJR backward recursion at trellis state s, step k
γ	Linear signal-to-noise ratio, $\gamma = 10^{\mathrm{SNR_{dB}}/10} = E_s/N_0$
$\bar{\gamma}$	Average signal-to-noise ratio
$\gamma_k(s', s)$	BCJR branch transition-and-observation probability; distinct from the SNR γ above, disambiguated by its arguments
η	Spectral efficiency, $\eta = R \log_2 M$ bits per symbol
θ	Learned relay parameters; *also* unknown carrier phase in the flat-channel control (Section 6.2.1)
σ^2	Real-noise variance in real AWGN sections; total complex-noise power where $n \sim \mathcal{CN}(0, \sigma^2)$, giving variance $\sigma^2/2$ per real axis

Other notation

$\mathcal{CN}(0, 1)$	Circularly-symmetric complex Gaussian distribution, unit variance
$\mathbb{E}[\cdot]$	Expectation
$\mathbb{1}[\cdot]$	Indicator function
$\hat{\cdot}$	Estimate of the quantity beneath the hat

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Chapter 1

Introduction

Relay communication uses an intermediate node to deliver information from a source to a destination. Classical amplify-and-forward (AF) and decode-and-forward (DF) provide well-studied baselines. Capacity results for coded block-DF apply under particular relay-channel assumptions and asymptotically long codes; they do not automatically characterize the uncoded symbol-wise DF studied here [1, 2, 3, 4, 5].

Neural processing motivates a narrower experimental question: *when does a small learned relay improve on the implemented classical alternatives, and what information and computation does that comparison require?* This thesis investigates that question by simulation, rather than proposing a universally superior relay or proving a capacity result.

Nine strategies are implemented, with eight evaluated in the canonical comparison: AF, symbol-wise DF, MLP, Hybrid, VAE, Transformer, and simplified implementations of Mamba-S6 and Mamba-2 SSD. A conditional GAN is implemented but excluded from that comparison on cost grounds. The source, channel models and destination are fixed within the canonical study; the learned models' architectures, training and checkpoints are part of the strategies being compared.

A sequence of bounded comparisons. The canonical memoryless study establishes a reference under known receiver CSI. Separate fixed-ISI, composite-impairment, finite-pilot and blind studies then examine model mismatch and information limitations. These groups form a conceptual organization, not a one-factor-at-a-time ablation: modulation, fading, architecture and budgets differ where stated. A supplementary coded study investigates forwarding, soft read-out, rate selection and idealized readiness constraints.

1.1 Scope and Contributions

The system has one half-duplex SISO relay and no direct source–destination path. Table 1.1 distinguishes the principal experiments. Chapter 4 specifies their implementation and Chapter 3 states the hypotheses.

Table 1.1: Primary experimental groups; conclusions apply to their stated configurations

Study	Configuration and measured question
Canonical	Uncoded Gray-QPSK, i.i.d. per-symbol complex Rayleigh fading on both hops, receiver CSI; destination BER versus per-hop E_s/N_0 for eight relays.
Unknown/mismatched channel	Fixed BPSK ISI, faded-ISI QPSK, and specified composite families; model information, pilot budgets and blind comparators. Different groups are not causal controls for one another.
Coded supplement	Finite-length convolutional codes and a QPSK/16-QAM grid; implemented coded forwarding, rate selection, and idealized buffering. A coded-QPSK memory sweep also compares arithmetic and BER.

In the canonical study, coherent equalization uses receiver CSI before the learned function is applied. No explicit channel coefficient is passed to that function. This is not a CSI-free chain. The unknown-channel MLPs are likewise trained on the evaluated impairment families, even when analytic parameters and test-time pilots are withheld.

The main contributions are:

1. A controlled canonical relay comparison, supplemented by equal-budget and replicated size studies. It distinguishes measured BER ordering from untested explanations involving capacity or overfitting.
2. Bounded experiments under ISI, nonlinearity and imperfect channel information, including matched Viterbi references, a pilot-count crossover at one SNR, and separately preserved fixed-budget rare-event validation.
3. A coded forwarding and memory/cost study that distinguishes symbol decisions, channel-sequence detection and error-correction decoding. With reliability-weighted (matched) decoder metrics, which give the relay decoder per-symbol channel reliability that the learned relays do not receive, decoding at the relay outperforms the two learned MLP read-outs evaluated at every tested SNR, reversing the high-SNR advantage they showed under the originally implemented unweighted metric; the coded Mamba relay was not rerun with matched metrics.

The canonical DF measurements agree with the stated two-hop closed form within 3.6% relative error at every tabulated SNR, after the QPSK E_s/N_0 conversion (Table 5.2). This is a simulator calibration check, not proof of all implementations or results. The additional controls and their endpoints are documented in their experiment sections.

Table 1.2: Assumptions that limit interpretation of the results

Dimension	Evidence boundary
Processing	Symmetric learned windows use future samples; power normalization uses blocks. Neither this nor operation counts establishes causal streaming or measured end-to-end latency.
Channel knowledge	Canonical receiver CSI and training-family knowledge are available. Realizations vary within specified families; arbitrary family shift is untested.
Comparators	Viterbi MLSE optimizes sequence error, not necessarily bit error. QPSK BCJR is a relay-output control. Coded results are reported with both the unweighted and the matched decoder metric.
Uncertainty	Fixed-checkpoint trials, replicated training, paired tests and nominal bit-level bounds have different units. Replication budgets are stated per study; null results do not prove equivalence and multiple comparisons are not jointly corrected.
Resources	Arithmetic and reference-program timing are implementation-specific. Parameter storage excludes runtime buffers and software. Energy, quantized hardware execution and deployment robustness are unmeasured.

References to Viterbi in the coded supplement mean convolutional-code decoding; references to Viterbi MLSE in the ISI study mean channel-sequence detection. The combined coded-memory study uses both. The general relay channel with a direct link and cooperative-diversity results in the literature review are background, not the operating topology of this thesis.

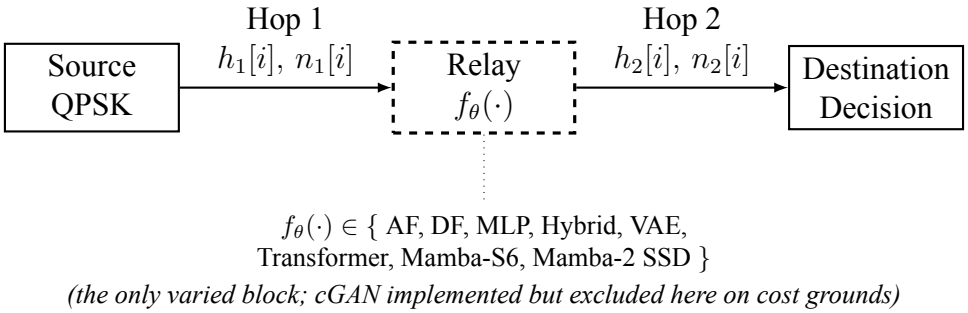


Figure 1.1: Canonical two-hop relay benchmark. The source, the channel model on both hops, and the destination are the same in every experiment of the canonical comparison; only the relay processing function $f_\theta(\cdot)$ (dashed) is varied among the eight candidate strategies compared there. The fading itself is not static: $h_1[i]$ and $h_2[i]$ are redrawn every symbol. The comparison evaluates these implemented relay strategies, including their trained checkpoints

Chapter 2

Background and Related Work

2.1 Classical Relay Strategies

The fundamental question that motivates this thesis is: can a learned relay function $f_\theta(\cdot)$ complement classical relay functions (AF scaling, DF regeneration) when their channel assumptions are mismatched or the relevant mapping is difficult to specify analytically? On the matched memoryless channel, the relevant symbol statistics are already captured by the classical MAP rule; the potential advantage of learning must therefore be established empirically in low-SNR or mismatched regimes rather than presumed.

2.1.1 Amplify-and-Forward (AF)

The AF relay amplifies the received signal by a gain factor G that normalizes the output power: (Equation 2.1)

$$G = \sqrt{\frac{P_{\text{target}}}{\mathbb{E}[|y_R|^2]}} \tag{2.1}$$

For CSI-assisted variable-gain AF, the standard dual-hop end-to-end SNR is [3, 6]: (Equation 2.2)

$$\text{SNR}_{\text{eff}}^{\text{AF}} = \frac{\text{SNR}_1 \cdot \text{SNR}_2}{\text{SNR}_1 + \text{SNR}_2 + 1} \tag{2.2}$$

This form assumes *CSI-assisted variable-gain* AF (Laneman et al.), in which the relay computes a fresh gain from its instantaneous first-hop channel state. That is different from the implementation in this thesis: its gain is a single block-adaptive scalar estimated from the empirical received power over the simulated block. It is therefore data- and noise-dependent and requires the block to be available before the gain is fixed; it is neither conventional statistical fixed-gain AF nor

per-symbol CSI-assisted variable-gain AF. Equations 2.2–2.18 are included only as standard background and are not closed forms for the simulated AF results, which come entirely from Monte Carlo evaluation.

This reveals AF's fundamental limitation: the effective SNR lies below that of the weaker hop even when the other has high SNR, because the relay amplifies the first hop's noise along with its signal, accumulating both at the destination.

2.1.2 Decode-and-Forward (DF)

The DF relay demodulates the received signal, recovers the transmitted bits, and re-modulates clean symbols:

$$\hat{b}_R = \text{demod}(y_R), \quad x_R = \text{mod}(\hat{b}_R) \tag{2.3}$$

The end-to-end BER for DF is: (Equation 2.4)

$$P_e^{\text{DF}} = P_{e,1}(1 - P_{e,2}) + (1 - P_{e,1}) \cdot P_{e,2} \tag{2.4}$$

where $P_{e,1}$ and $P_{e,2}$ are the BER of the first and second hops, respectively: an end-to-end bit error occurs exactly when *one* of the two hops flips the bit (two flips cancel), so this expands to $P_{e,1} + P_{e,2} - 2P_{e,1}P_{e,2}$, the same expression used in the AWGN analysis (Equation 2.20). For BPSK modulation over AWGN (real-valued noise with variance $1/(2\gamma)$), each hop's BER is $P_e = Q\left(\sqrt{2\gamma}\right)$, where the simulator's $\gamma = E_s/N_0$ equals E_b/N_0 for BPSK. DF provides clean regeneration at high SNR but suffers from error propagation when the first-hop BER is non-negligible. As discussed in Remark 4.2, the "DF" baseline used here is the uncoded symbol-wise slicer, not practical finite-length coded block-DF; its optimality claims are therefore confined to the uncoded setting.

2.1.3 Other Classical Relay Techniques

AF and DF are two canonical relay strategies: AF linearly forwards an analog observation, whereas DF regenerates a detected symbol or decoded block. Compress-forward, estimate-forward, and compute-forward address different network models and objectives and are not intermediate points on a single AF–DF scale. The focus on AF and symbol-wise DF is deliberate but limited: they are representative relay-function baselines for the canonical benchmark, not strict performance bounds or an exhaustive account of classical relaying.

2.2 Machine Learning in Wireless Communication

Applying machine learning to physical-layer wireless communication has gained significant momentum, driven by neural networks' ability to learn complex non-linear mappings directly from data where analytical solutions are intractable or suboptimal: channel estimation [7], signal detection [8], autoencoder-based end-to-end communication [9], and resource allocation [10].

2.2.1 Theoretical Basis: Universal Approximation and Denoising

The theoretical justification for applying neural networks to relay signal processing rests on two pillars. First, **universal approximation** results show that feedforward networks can approximate continuous functions on compact domains to arbitrary accuracy, given sufficient width: for sigmoidal activations [11], for bounded nonconstant activations [12], and, in the sharpest form, for exactly the non-polynomial activations [13]; see also [14]. This existence result does not guarantee finite-sample learning, optimization, or generalization. For relay denoising, the target function maps noisy observations to clean transmitted symbols:

$$f^* : \mathbb{R}^{2w+1} \to [-1, 1], \quad f^*(y_{i-w}, \ldots, y_{i+w}) = \mathbb{E}[x_i \mid y_{i-w}, \ldots, y_{i+w}] \tag{2.5}$$

This conditional expectation f^* is the Bayes-optimal denoiser: it minimizes the mean squared error (MSE) over all possible estimators. It is optimal for the local estimation problem at the relay, but it is not in general the relay mapping that minimizes the destination bit error rate, which is what the experiments evaluate; Section 4.1.5 states that problem and the sense in which MSE and cross-entropy training stand in for it. For BPSK symbols corrupted by AWGN, f^* reduces to the posterior mean:

$$f^*(y) = \tanh\left(\frac{y}{\sigma^2}\right) \tag{2.6}$$

which is a smooth sigmoid approaching the hard-decision signum function as $\sigma^2 \to 0$. A single tanh unit with the right input scaling represents this posterior *exactly* for a fixed σ^2. The deployed relay, however, is trained at SNR $\in \{5, 10, 15\}$ dB and evaluated over 0–20 dB, so one set of weights must approximate a whole *family* of posteriors indexed by σ^2, and on the fading channel a mixture over random effective SNRs. That a 169-parameter network approximates that family closely enough to track the classical baselines is an empirical finding of this thesis, not a consequence of the exact single-σ^2 representation.

Second, the **bias-variance decomposition** provides a framework for understanding model complexity:

$$\mathbb{E}[(\hat{x} - x)^2] = \text{Bias}^2(\hat{x}) + \text{Var}(\hat{x}) + \sigma^2_{\text{irreducible}} \tag{2.7}$$

For the denoising problem considered here the irreducible term is the Bayes risk $\mathbb{E}[\text{Var}(X \mid Y)]$, the conditional uncertainty that remains after observing Y, rather than the channel-noise variance σ^2 itself. The decomposition is used as a conceptual framework; the irreducible term should be read that way. Increasing capacity can change both bias and variance, but it does not necessarily reduce one or increase the other in modern regularized networks. Since the target f^* is a smooth low-dimensional mapping in the memoryless special case, the working hypothesis is that additional capacity may buy little. Whether that occurs is an empirical question; Section 7.2 reports only what was measured and does not infer overfitting without train–test-gap evidence.

2.2.2 Prior Work in Deep Learning for Physical-Layer Processing

Ye et al. [7] demonstrated joint DNN-based channel estimation and signal detection in OFDM systems using received pilot and data information. Their approach replaces separately implemented estimation and equalization stages and was reported to be robust under several modeled impairments; it does not eliminate pilots.

Dorner et al. [9] proposed treating the modulator, channel, and demodulator as an autoencoder trained end-to-end with a differentiable message-reconstruction objective and evaluated by error rate. This approach jointly optimizes the communication chain rather than directly minimizing the nondifferentiable BER.

The sampled relay-assisted literature includes learning for channel estimation and network-level decisions rather than only the forwarding map. Akdemir et al. [15] use learned channel-coefficient forecasts within conventional DF systems and compare SNR-based, threshold-based and all-relay participation protocols. This is not evidence that a neural relay-selection classifier directly replaces the relay's signal processing. No systematic search establishes the prevalence of either approach across the field.

Model-based learned detectors place neural components inside classical recursions: Tsai et al. [16] learn channel-dependent likelihoods for neural BCJR detection, while Karanov et al. [17] study ISI with bursty noise. Chen et al. [18] report benefits under imperfect channel knowledge, with robustness depending on channel variation. These are examples of the importance of model match, not proof of a universal boundary. Section 6.2.3 supplies a QPSK relay-output BCJR control; it is not an end-to-end learned-relay comparison.

A broader perspective on machine learning's role across wireless protocol layers, including cooperative communications, is given in [19]. A separate line of work exploits the structural similarity between AF relay networks and neural networks, treating relay nodes as neurons and tuning relay gains with deep-learning optimization to exploit hardware nonlinearities [20, 21];

the gains reported there are on analog gain optimization, not symbol-level detection.

Classical AF/DF frameworks remain well understood and widely adopted [5] but rely on fixed linear forwarding or hard-decision rules. Learning the relay's symbol-level input-output mapping itself has drawn comparatively little attention, motivating the data-driven alternative studied here.

2.2.3 Neural Network Relay Processing

For relay processing specifically, a supervised learning approach trains a neural network f_θ to minimize the empirical mean-squared error: (Equation 2.8)

$$\mathcal{L}(\theta) = \frac{1}{N}\sum_{i=1}^{N}(\hat{x}_i - x_i)^2, \quad \hat{x}_i = f_\theta(y_{i-w:i+w}) \tag{2.8}$$

where $\hat{x}_i = f_\theta(y_{i-w:i+w})$ is the network output based on a sliding window of $2w+1$ received symbols, and x_i is the clean transmitted symbol. This window-based approach provides temporal context that enables the network to exploit statistical dependencies in the noise-corrupted signal.

A well-known example of $f_\theta(\cdot)$ is the *multilayer perceptron (MLP)*, a feedforward artificial neural network composed of an input layer, one or more fully connected hidden layers with nonlinear activation functions, and an output layer, trained using backpropagation to model nonlinear input–output relationships. The MLP equation is shown in equation 2.9:

$$\mathbf{h} = \mathrm{ReLU}(\mathbf{W}_1\mathbf{y}_i + \mathbf{b}_1), \quad \hat{x}_i = \tanh(\mathbf{W}_2\mathbf{h} + \mathbf{b}_2), \quad \mathbf{y}_i = y_{i-w:i+w} \tag{2.9}$$

The sliding-window input $\mathbf{y}_i$ is retained for architectural consistency with the unknown-ISI experiments of Chapter 6. Under the canonical i.i.d. per-symbol fading model, adjacent observations carry no additional information about the current symbol, and the window is redundant there by construction. For hard BPSK/QPSK slicing, phase-corrected equalization preserves the decision boundary. A soft estimator also depends on the instantaneous reliability: the equalized sample alone is not generally sufficient for the CSI-conditioned posterior when the channel magnitude is discarded. The unknown-channel study separately examines temporal context under ISI.

Multi-SNR training is a key design choice: training on data generated at multiple SNR levels (5, 10, 15 dB here) yields a single estimator whose denoising function generalizes across operating conditions rather than specializing to one noise level. In decision-theoretic terms the training objective averages the loss over the sampled SNRs, so what it minimizes is *Bayes* risk under the empirical SNR distribution, not the worst-case (minimax) risk over that range: nothing in the

objective protects the SNR at which the network happens to do worst.

For resource-limited relays, it is useful to test whether a modest model suffices for the chosen processing task. Universal approximation and bias–variance reasoning do not predict a mandatory BER upturn as neural-network width increases. The capacity sweep therefore treats plateauing or degradation as empirical hypotheses, not theoretical consequences.

2.2.4 Positioning Against the Closest Work

Three lines of prior work are closest to this thesis, and each already contains an idea the thesis uses.

Estimate-and-forward. Gomadam and Jafar [22] show that, among memoryless relay functions, forwarding a minimum-mean-square uncorrelated-error estimate of the transmitted symbol maximizes the destination SNR. They call the scheme estimate-and-forward and show that it outperforms both AF and demodulate-and-forward, while for Gaussian inputs all three coincide. A soft relay trained with mean squared error (Section 4.1.5) is therefore a learned approximation of a conditional-mean forwarding rule of this kind, not a new forwarding principle. It differs in that it is fitted from samples rather than derived from known statistics, it is not given the fading reliability, and it is evaluated by destination BER rather than SNR. Estimate-and-forward itself is not implemented as a baseline here.

Learned detection without explicit CSI. Farsad and Goldsmith [23] train a sliding bidirectional recurrent detector that needs no channel state information after training and, on Poisson channels, report lower BER than Viterbi detection with imperfect CSI. ViterbiNet [24] keeps the Viterbi recursion and learns its likelihoods from data, which makes it robust to channel uncertainty while retaining the trellis cost. Both establish that a data-driven detector can replace or approximate model-based sequence detection and should be compared with detectors given perfect and estimated CSI — the questions Chapter 6 asks for a relay, together with the cost of each detector.

What this thesis adds. The contribution is a measured comparison, not a new relay principle, and it differs from the work above in four respects. First, the learned map sits at a relay whose output is power-normalized and sent over a second noisy hop, so the objective is destination BER rather than the detector's own error rate, and the two coincide only for hard-output relays (Eq. (4.10)). Second, several architecture families are compared in one framework under declared parameter budgets. Third, resources are accounted explicitly as decision delay and multiplications per symbol, against both the implemented joint trellis and its exact I/Q factorization (Section 2.6). Fourth, on the ISI channel a non-monotonic scalar MAP relay separates the gain available from a better symbol-wise decision rule from the gain that needs temporal context (Section 6.2): the large improvements over zero-threshold DF are not by themselves evidence for learning. Table 3.2 compares named studies on these properties.

2.3 Generative Models for Signal Processing

Generative models offer an alternative to directly learning a denoising function (the discriminative approach above): they instead learn the distribution of clean signals $p(\mathbf{x})$. Bayesian denoising provides the general probabilistic framing:

$$p(\mathbf{x}|\mathbf{y}) \propto p(\mathbf{y}|\mathbf{x})p(\mathbf{x}) \tag{2.10}$$

The generative approach is theoretically appealing because it separates the modeling of the signal prior from the noise model, potentially enabling better generalization to unseen noise conditions. The VAE implemented in this thesis does not compute that posterior explicitly. It is a latent-variable generative architecture trained supervised, with noisy observations at the encoder and the clean symbol as the reconstruction target, under a KL regularizer; Eq. (2.10) frames the problem it addresses rather than describing the operation it performs.

2.3.1 Variational Autoencoders

Variational Autoencoders (VAEs) [25] learn a latent representation by maximizing the evidence lower bound (ELBO). The generative model posits that data $\mathbf{x}$ is generated from a latent variable $\mathbf{z} \sim p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$ through a decoder $p_\theta(\mathbf{x}|\mathbf{z})$. Since the true posterior $p(\mathbf{z}|\mathbf{x})$ is intractable, an encoder network $q_\phi(\mathbf{z}|\mathbf{x})$ approximates it. The ELBO objective is derived from the log-evidence decomposition: (Equation 2.11)

$$\log p_\theta(\mathbf{x}) = \underbrace{\mathbb{E}_{q_\phi}\left[\log \frac{p_\theta(\mathbf{x}, \mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})}\right]}_{\text{ELBO}(\theta,\phi;\mathbf{x})} + \underbrace{D_{KL}(q_\phi(\mathbf{z}|\mathbf{x})\|p_\theta(\mathbf{z}|\mathbf{x}))}_{\geq 0} \tag{2.11}$$

Since the KL divergence is non-negative, the ELBO is a lower bound on the log-evidence. Maximizing the ELBO simultaneously trains the encoder to approximate the true posterior and the decoder to reconstruct the data. The ELBO decomposes into a reconstruction term and a regularization term:

$$\mathcal{L}_{\text{VAE}} = \underbrace{\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z})]}_{\text{Reconstruction quality}} - \underbrace{\beta \cdot D_{KL}(q_\phi(\mathbf{z}|\mathbf{x})\|p(\mathbf{z}))}_{\text{Latent space regularity}} \tag{2.12}$$

The reconstruction term encourages accurate recovery while the KL term regularizes the latent distribution toward the prior. The β weight in Eq. (2.12) changes that tradeoff [26]. Choosing $\beta < 1$ puts more relative weight on reconstruction; it does not by itself establish a BER benefit for this relay task. Larger β motivates more constrained representations, but disentanglement is not guaranteed or measured here.

The *reparameterization-trick* enables gradient-based optimization through the stochastic sampling layer:

$$\mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}, \text{diag}(\boldsymbol{\sigma}^2)) \tag{2.13}$$

Instead of sampling from equation 2.13, the sample is expressed as:

$$\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon} \tag{2.14}$$

where $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. This moves the stochasticity outside the computational graph, allowing backpropagation through the encoder.

The implemented relay is a supervised denoising VAE rather than the unconditional model above: its encoder receives the noisy observation window, while reconstruction is evaluated against the clean target. The latent regularization supplies a generative inductive bias, but the denoising property follows from this noisy-input/clean-target training objective rather than from the standard VAE derivation alone. Sampling $\mathbf{z}$ at inference also injects variance into an otherwise deterministic estimation task; Section 2.3.2 reports the measured outcome without attributing a causal effect to that sampling.

2.3.2 Generative vs. Discriminative Paradigms for Relay Processing

The application of generative models to relay processing has, to the best of our knowledge, not been extensively studied. The key question is whether the generative inductive bias (learning the data distribution rather than just the input-output mapping) provides any advantage for the relay denoising task. For BPSK, the clean signal distribution is trivially simple (a discrete distribution on $\{-1, +1\}$), suggesting that the generative overhead may not be justified. This thesis compares VAE-based relay processing against classical and supervised learning methods on a common benchmark, and the results show that the generative paradigm provides no significant advantage for this particular task. A conditional GAN relay is implemented and described in Chapter 4 but is excluded from the reported comparison on cost grounds, so no claim is made about it here. On the canonical setup the VAE tracks the supervised MLP and symbol-wise DF rather than trailing them, reaching 0.00972 at 20 dB against DF's 0.00972 and the MLP's 0.00992 (Table 5.2), and it stays inside the feedforward group at every SNR in the equal-parameter comparison as well (Table 5.3). The generative bias is therefore neither an advantage nor a handicap here: it costs parameters and training time to arrive where a 169-parameter regressor already is. One qualification remains, unchanged by this: the relay samples $\mathbf{z} \sim q_\phi(\mathbf{z} \mid \mathbf{x})$ stochastically at inference, and the deterministic variant that substitutes the posterior mean $\boldsymbol{\mu}$ at test time was not evaluated, so nothing here bears on whether that variant would do better.

2.4 Sequence Models: Transformers, State Space Models

Two sequence-modeling paradigms are evaluated as relay functions in this thesis, and what matters here is their inductive bias and their cost rather than their internal machinery, which is standard and is not reproduced. **Transformers** [27] use multi-head self-attention. In the bidirectional encoder configuration used here, every position attends to every other through learned query, key and value projections, giving a global receptive field over the buffered window; a causal (decoder) Transformer instead attends only to the past, and would not see the right-hand half of the symmetric window this thesis feeds its relays; the two simplified Mamba relays evaluated here (Section 4.2) are causal in this sense (Remark 4.1). The cost is $O(n^2)$ time and memory in the window length n. Because attention is permutation-equivariant, sinusoidal positional encodings supply the ordering the relay needs to distinguish symbols within its window. For the 11-symbol window used throughout, the quadratic term is negligible; what the architecture costs is parameters: query, key and value projections for every head and a shared output projection, $4d_{\text{model}}^2$ weights per layer.

Structured state space models [28] take the opposite view, treating the sequence model as a discretized linear dynamical system — a learned recursive filter, linear in the sequence length, with the HiPPO initialization supplying a principled memory of the past. **Mamba** [29] makes that system time-varying by conditioning its parameters on the input, so the state-transition term acts as a learned gate that can reset or retain the state per symbol; **Mamba-2** [30] recasts a structured selective SSM through its duality with a structured attention matrix. SSD uses a more restrictive scalar-identity transition structure than a general diagonal SSM; that restriction enables the faster implementation, so Mamba-2 is not simply the same function class with a faster kernel. The classical analogue is adaptive filtering, and that connection becomes relevant only in the unknown-channel study of Chapter 6; on the canonical memoryless channel there is no temporal structure for a recurrence to exploit.

2.4.1 State Space Models for Signal Processing

The application of state space models as the *relay-side* processing function is, to the best of our knowledge, largely unexamined. Transformers and state-space models have already been applied to a range of wireless receiver and signal-processing tasks; what the literature reviewed here did not turn up is a controlled comparison of these architectures against one another, at a matched parameter budget, in the two-hop relay benchmark studied in this thesis.

2.5 Theoretical Foundations: Channel Models

This section presents the theoretical BER analysis for each channel model used in this study, derives the closed-form expressions, and validates them against Monte Carlo simulation. The

theoretical-simulative comparison serves two purposes: (i) it verifies the correctness of the simulation framework, and (ii) it establishes the baseline performance that AI relays must beat. The coverage is partial: the closed forms derived here validate the memoryless components to which they apply — the AWGN and Rayleigh symbol-error expressions, the two-hop DF composition built from them, and the channel and noise conventions every experiment shares. They do *not* validate every classical baseline used later. The AF relay implemented in this thesis normalizes by a block-adaptive empirical-power scalar rather than an instantaneous CSI-assisted gain, so Eq. (2.2) is background for it and not a prediction of its curve (Section 2.1.1). Viterbi MLSE, the BCJR/APP detector and CMA have no closed-form BER on the channels they are run on, and are audited instead by implementation-specific controls stated where they appear: that exact ML decoding returns an admissible path with likelihood at least as large as that of the transmitted sequence, that it reduces to the nearest-symbol slicer when the ISI is removed, and that the BER-optimal rule agrees with symbol-MAP where the two provably coincide. The comparisons that follow therefore set a learned relay against classical detectors verified in these two different ways, by closed form where one exists and by control otherwise, rather than against unaudited simulations of them.

2.5.1 AWGN Channel: Theoretical Analysis

The AWGN channel adds zero-mean Gaussian noise to the transmitted signal: (Equation 2.15)

$$y = x + n, \quad n \sim \mathcal{N}(0, \sigma^2), \quad \sigma^2 = \frac{P_s}{2\,\gamma} \tag{2.15}$$

where $P_s = \mathbb{E}[|x|^2] = 1$ for unit-power BPSK symbols and $\gamma = 10^{\mathrm{SNR_{dB}}/10} = E_s/N_0$. The factor of two is the standard $N_0/2$ variance per real dimension.

2.5.1.1 Single-hop BER

For BPSK ± 1 symbols and real-valued AWGN with variance $\sigma^2 = 1/(2\gamma)$, an error occurs when the noise pushes the received sample past the decision boundary at the origin. The theoretical BER is: (Equation 2.16)

$$P_e^{\mathrm{AWGN}} = Q\left(\frac{1}{\sigma}\right) = Q\left(\sqrt{2\gamma}\right) = \frac{1}{2}\,\mathrm{erfc}(\sqrt{\gamma}) \tag{2.16}$$

where $Q(x) = \frac{1}{2}\mathrm{erfc}(x/\sqrt{2})$ is the Gaussian Q-function. The simulator's SNR label is the normalized E_s/N_0 axis, not the raw ratio P_s/σ^2 in a single real dimension.

Remark 2.1 (Noise convention)**.** The standard textbook result $P_e = Q(\sqrt{2E_b/N_0})$ assumes a one-sided noise PSD N_0, hence variance $N_0/2$ per real dimension. The implementation uses

$\sigma^2 = P_s/(2\gamma)$, so for BPSK ($E_s = E_b = P_s$) its BER is $Q(\sqrt{2\gamma})$, exactly the textbook expression with $\gamma = E_b/N_0$. For QPSK, $E_b = E_s/2$, so the same simulator label corresponds to an E_b/N_0 value 3.01 dB lower.

For complex baseband signals the same convention gives total noise power $N_0 = P_s/\gamma$, split equally between I and Q; after coherent equalization each real decision dimension therefore has variance $P_s/(2\gamma|h|^2)$. The Rayleigh expressions use that per-dimension variance and consequently retain the standard factor of two in the Q-function argument.

2.5.1.2 Two-hop AF BER

For the analytical CSI-assisted variable-gain AF model, let the per-hop SNRs be symbol energy divided by complex noise power. With equal per-hop SNR γ, the effective symbol SNR is:

$$\mathrm{SNR}_{\mathrm{eff}}^{\mathrm{AF}} = \frac{\mathrm{SNR}_1 \cdot \mathrm{SNR}_2}{\mathrm{SNR}_1 + \mathrm{SNR}_2 + 1} = \frac{\gamma^2}{2\gamma + 1} \tag{2.17}$$

where $\gamma = \mathrm{SNR}$ is the per-hop SNR. The resulting BER is

$$P_e^{\mathrm{AF,BPSK}} = Q\left(\sqrt{2\,\mathrm{SNR}_{\mathrm{eff}}}\right) \tag{2.18}$$

for coherent BPSK, using the $N_0/2$ noise variance per decision dimension (Remark 2.1). For Gray-QPSK the per-bit expression is $Q(\sqrt{\mathrm{SNR}_{\mathrm{eff}}})$. At high SNR,

$$\mathrm{SNR}_{\mathrm{eff}} \approx \frac{\gamma}{2} \tag{2.19}$$

confirming the well-known 3 dB penalty of AF relaying. This penalty is a property of the analytical CSI-assisted variable-gain model above. The AF relay implemented in this thesis uses a single block-adaptive scalar estimated from the empirical received power, so Eq. (2.19) is background rather than a closed-form prediction of the simulated AF curve, and the 3 dB figure should not be read as explaining it.

2.5.1.3 Two-hop DF BER

For a decode-and-forward relay with equal-SNR hops, an error occurs at the destination if exactly one hop introduces an error:

$$P_e^{\mathrm{DF}} = P_{e,1} + P_{e,2} - 2P_{e,1}P_{e,2} \tag{2.20}$$

With equal hops (i.e. $P_{e,1} = P_{e,2} = P_e^{\text{AWGN}}$), this becomes

$$P_e^{\text{DF}} = 2P_e(1 - P_e) \tag{2.21}$$

At high SNR, $P_e \ll 1$ and

$$P_e^{\text{DF}} \approx 2P_e \tag{2.22}$$

i.e., the two-hop penalty is approximately a factor of 2 in BER.

2.5.2 Rayleigh Fading Channel: Theoretical Analysis

The Rayleigh fading channel models non-line-of-sight (NLOS) propagation where the signal undergoes multiplicative fading:

$$y = hx + n, \quad h \sim \mathcal{CN}(0, 1), \quad n \sim \mathcal{CN}(0, \sigma^2) \tag{2.23}$$

The fading coefficient h is a circularly-symmetric complex Gaussian random variable, so its magnitude $|h|$ follows a Rayleigh distribution:

$$f_{|h|}(r) = 2r \cdot e^{-r^2}, \quad r \geq 0 \tag{2.24}$$

with $\mathbb{E}[|h|^2] = 1$ (unit average power). The instantaneous SNR after equalization ($\hat{x} = y/h$) becomes $\gamma_h = |h|^2 \cdot \text{SNR}$, which is exponentially distributed.

2.5.2.1 Single-hop BER

Averaging the conditional BER $P_e(\gamma_h) = Q(\sqrt{2\gamma_h})$ over the exponential distribution of γ_h yields the closed-form [31, Ch. 13] (Equation 2.25):

$$P_e^{\text{Rayleigh}} = \frac{1}{2}\left(1 - \sqrt{\frac{\bar{\gamma}}{1 + \bar{\gamma}}}\right) \tag{2.25}$$

where $\bar{\gamma} = \gamma_b = \bar{E}_b/N_0$ is the average SNR *per bit*. This distinction is load-bearing, because the canonical experiments plot $\gamma_s = E_s/N_0$ per *symbol*: for QPSK, $\gamma_b = \gamma_s/2$, a 3.01 dB offset. Substituting gives the per-bit Rayleigh expression in terms of the plotted axis, $P_b = \frac{1}{2}\left(1 - \sqrt{\gamma_s/(\gamma_s + 2)}\right)$, which is the form the DF-theory column of Chapter 5 evaluates. At high SNR ($\bar{\gamma} \gg 1$):

$$P_e^{\text{Rayleigh}} \approx \frac{1}{4\bar{\gamma}} \tag{2.26}$$

This $1/\text{SNR}$ decay is fundamentally slower than the exponential decay of AWGN ($Q(\sqrt{2\gamma}) \sim e^{-\gamma}/\sqrt{4\pi\gamma}$), explaining why Rayleigh fading is significantly more challenging. The channel is **diversity-limited**: deep fades (where $|h| \approx 0$) cause errors regardless of the average SNR, and a single link of this kind has diversity order one. Cooperative relaying is often introduced as the remedy, by giving the destination a second, independently faded look at the same message. That is not the mechanism at work here, and it should not be read as the motivation for the architectures this thesis studies. The canonical setup has *no* direct source–destination path (Table 1.2), so the source's message reaches the destination along one route only and no combining is possible; the two-hop DF expression derived immediately below, $P_e^{\text{DF}} = 2P_e(1 - P_e)$, is itself first order in P_e and therefore describes a diversity order of one, not two. What relaying buys on such a link is a shorter hop and hence a higher per-hop SNR at fixed transmit power, at the cost of a second error opportunity. The question this thesis asks is a different one again: given that geometry, what the relay should *compute* on the signal it receives.

2.5.2.2 Two-hop DF BER

The two-hop DF relay over Rayleigh fading follows the same composition rule as AWGN:

$$P_e^{\text{DF,Rayleigh}} = 2P_e^{\text{Rayleigh}}(1 - P_e^{\text{Rayleigh}}) \tag{2.27}$$

2.6 Channel Coding: Convolutional Codes and Trellis Decoding

The channel models above assume an uncoded link. Chapter 5's supplementary coded study (Section 5.5) departs from that, so this section fixes the three quantities that study reports on. The underlying coding theory is standard and is not reproduced here; see Proakis and Salehi [31] for convolutional encoding and the Viterbi algorithm [32], and Bahl *et al.* [33] for the forward–backward (BCJR) recursion. Two properties of those two decoders matter later and are used without re-derivation: Viterbi returns the maximum-likelihood *sequence* while BCJR returns per-symbol posteriors, so the two optimize different criteria and neither dominates the other on every metric (Section 6.2.3 measures this distinction); for a binary rate-$1/n$ code, a constraint-length-K shift register has a 2^{K-1}-state trellis. The principal coded study uses $k = 1$, $n = 2$, $K = 3$ and obtains rates $2/3$ and $3/4$ from that mother code by puncturing. A separate code-strength control also evaluates $K = 5$ and $K = 7$.

Exact I/Q separation for real-tap QPSK ISI. For real-valued channel taps and circular white Gaussian noise, the squared-distance MLSE metric [32, 31] separates:

$$\sum_n |y_n - (h * x)_n|^2 = \sum_n \big(y_{I,n} - (h * x_I)_n\big)^2 + \sum_n \big(y_{Q,n} - (h * x_Q)_n\big)^2. \tag{2.28}$$

For an unrestricted QPSK sequence search, the two binary-axis sequences can therefore be minimized separately. Two binary trellises replace the joint QPSK trellis exactly, not by reduced-state approximation. Each has 2^{L-1} states and 2^L branches for L real taps. With precomputed expected samples and one multiplication per real squared residual, the combined metric cost is 2^{L+1} multiplications per QPSK symbol, versus $2 \cdot 4^L$ for the joint implementation. Both counts exclude additions, comparisons and traceback. The same argument applies to any square M-QAM constellation, whose points are the Cartesian product of two $\sqrt{M}$-ary amplitude alphabets: two $\sqrt{M}$-ary trellises with $2M^{L/2}$ branch metrics in total replace the joint trellis's M^L, a count that is still exponential in L but with half the exponent; only the QPSK case was checked numerically. This separation applies to the channel equalizer preceding code decoding; it does not imply separation of a joint code-constrained search, and does not generally hold for complex taps or coupled noise. The separation was also checked in the implementation used by the coded-memory study (`factorized_mlse_equivalence.py`): over 264,000 symbols, spanning three, five and seven taps, that study's six SNRs from 0 to 20 dB, its seven traceback depths, and full-sequence decoding at three and five taps, the two binary trellises returned the joint trellis's decisions on every symbol. The measured error rates of the joint trellis therefore apply unchanged to the factorized detector; only the arithmetic count differs.

A code's rate $R = k/n$ and the constellation size M together set the spectral efficiency

$$\eta = R \cdot \log_2 M \quad \text{[bits/symbol]} \tag{2.29}$$

so a rate-$1/2$ code halves the information rate of whatever modulation carries it, and comparing a coded link against an uncoded one at the same SNR without accounting for this is not a like-for-like comparison (Section 5.5.4 revisits Chapter 5's coded-versus-uncoded comparison on that basis). Spectral efficiency counts bits transmitted rather than bits that arrived, so the quantity a link actually adapts on is the goodput

$$G = \eta \cdot (1 - \mathrm{FER}) \quad \text{[information bits/symbol]} \tag{2.30}$$

Maximizing G over a grid of modulation-and-coding schemes at each SNR — the standard adaptive modulation and coding (AMC) problem — traces the link-adaptation envelope that Section 5.5.5 measures.

Chapter 3

Research Objectives

3.1 Main Objective

To determine *under which conditions* a learned relay surpasses classical relay processing in two-hop communication, and where it does not. A single canonical setup (SISO on both hops, i.i.d. Rayleigh fast fading, complex baseband, QPSK, uncoded BER) is the controlled baseline on which classical and AI-based strategies are compared as a function of SNR regime and computational constraints. Two deliberate departures from that baseline then test what the canonical comparison cannot: a coded chain, which tests whether the conclusion survives when the classical relay is allowed to decode a channel code (Section 5.5), and channels whose structure lies outside the model class the classical relays assume, which is where the learned relay's measured benefit lies (Chapter 6). The canonical setup is therefore the control, not the scope.

3.1.1 Two Research Questions

The main objective resolves into two questions that differ only in whether a budget is imposed, and which for that reason have different answers. Throughout, the classical reference is named explicitly rather than left as "classical processing": symbol-wise DF and AF where the channel is memoryless, Viterbi MLSE where it has memory, and block DF, a channel code decoded and re-encoded at the relay, where the relay is permitted to spend a frame of delay.

O1 (unconstrained). On a channel with memory, and with no budget on latency or computation, in which regime and under what conditions can a learned relay complement the classical method? This is a question about accuracy and about what each detector needs to know. Chapter 6 examines it in separate fixed-memory, finite-pilot and random-channel studies, not cumulative controlled relaxations of one system.

O2 (constrained). The same question when the relay must also meet a latency budget and a computational budget. Here the classical method may be the more accurate detector and still be unavailable, because a trellis search that grows exponentially in the channel mem-

ory eventually cannot be afforded and a relay that decodes a codeword cannot decide before the codeword ends.

The studies examine two resources separately: *channel-estimate reliability* in the pilot-budget sweep of Section 6.3, and *arithmetic and decision delay* in the coded-memory study of Section 6.5.1. Their channel models, modulation, endpoints and resource axes differ. Each supplies a configuration-specific comparison; together they do not measure a common boundary or a quantitative distance between boundaries.

Latency is singled out for measurement because it is more often assumed than measured. A sequence detector is commonly treated as latency-expensive on the grounds that it searches a trellis, and a windowed relay as cheap on the grounds that its window is short; neither follows from the structure of the algorithms, since a trellis search need not run to the end of the block and a window must be filled on both sides of the symbol it decides. Section 6.5.1 therefore measures the decision delay each detector actually imposes, rather than inferring it.

How the questions, hypotheses and experiments fit together. The two research questions are answered by Chapter 6, which is why that chapter carries the principal contribution. O1 is tested as hypothesis H5, by the studies around it and, for channel information, by the pilot-budget sweep; O2 has no numbered hypothesis and is answered descriptively, by the delay and arithmetic studies. Hypotheses H1–H4 answer neither question directly. They belong to the canonical comparison of Section 5.2, a memoryless control on which the classical answer is known in closed form: it calibrates the framework, and it compares the learned architectures on a task where memory and channel information play no role. The coded study tests whether the canonical conclusion survives a channel code. The studies are separate experiments, not successive relaxations of one system: modulation, fading, training and evaluation budgets change between them. Table 3.1 maps each question and hypothesis to the sections that answer it; the outcomes are collected in Chapter 8.

Table 3.1: Research questions and hypotheses, and the experiments that answer them. Outcomes are summarized in Chapter 8 and Table 8.1

Question or hypothesis	Experiment	Sections
O1 / H5: memory, no budget	Memoryless control; fixed three-tap ISI with genie MLSE, scalar MAP and BCJR references; QPSK re-check; composite channel; blind regime with MMSE and CMA baselines; sequence models on memory	6.1, 6.2, 6.2.1–6.2.4, 6.4–6.4.2
O1: channel information	Pilot-budget sweep against LS-plus-MLSE	6.3

Question or hypothesis	Experiment	Sections
O2: delay and arithmetic	Decision delay and multiplications per symbol in one coded chain, across traceback depth and channel length	6.5, 6.5.1
H1, H2 (control)	Canonical eight-relay comparison; conventional AF reference	5.2, 7.1.1
H3 (control)	Complexity study; replicated size sweep	5.3, 5.4
H4 (control)	Equal-budget comparison at about 3,000 parameters	4.4, 5.3
Canonical result under coding	Block DF with a convolutional code against coded-aware learned relays	5.5

3.2 Research Gap and Motivation

Despite growing interest in AI for wireless communication, the literature reviewed for this thesis did not identify a controlled comparison of the particular relay-processing paradigms evaluated here. Because this was not a preregistered systematic review, the gap statements below are scoped to the cited and reviewed literature rather than asserted as exhaustive:

1. **Gap 1: Limited cross-paradigm relay comparison.** The cited work studies related receiver or end-to-end tasks, generally one architecture family at a time (Section 2.2.2 reviews it): the reviewed literature did not identify a comparison of supervised feedforward networks, variational autoencoders, attention-based Transformers, and state space models (Mamba [29], Mamba-2 SSD [30]) on the same relay denoising task under controlled conditions. The cGAN is implemented here but excluded from the reported comparison, so the adversarial paradigm is not quantitatively covered. State space models in particular have begun to be applied to receiver-side physical-layer tasks, such as channel state prediction [34], but not, to the best of our knowledge, to the relay processing function itself, where the node must re-transmit rather than merely decide.

2. **Gap 2: Limited characterization of the relay complexity–performance relationship.** Model-size trade-offs have been examined for other physical-layer tasks, including automatic modulation classification [35], but the reviewed literature did not identify a corresponding controlled size sweep for this relay task. In the memoryless BPSK-AWGN special case, recovering one symbol is a smooth, one-dimensional estimation problem, so it is not evident that additional capacity helps. An equal-parameter comparison can remove raw parameter count as one explanation, although architecture, depth, width, state dimension, window, and optimization remain confounded.

This thesis addresses both gaps through a unified framework that implements nine relay strategies and compares eight of them on the canonical setup, with both original and normalized parameter counts, paired statistical tests, and a dedicated complexity study; Gap 1's adversarial paradigm (cGAN) is implemented but, being excluded from the canonical comparison on cost grounds, is addressed qualitatively rather than quantitatively. The following table summarizes the positioning of this work relative to the literature:

Table 3.2: Research positioning: named studies closest to this thesis, the node and task each addresses, what is learned, and what it is compared against (Section 2.2.4)

Study	Node and task	Learned component	Comparators and measures
Gomadam and Jafar [22]	Relay; memoryless forwarding	None: an analytic estimate from known statistics	AF and demodulate-and-forward; destination SNR
Farsad and Goldsmith [23]	Receiver; sequence detection	Sliding bidirectional RNN detector, no CSI after training	Viterbi with perfect and imperfect CSI; BER on Poisson channels
Shlezinger et al. [24]	Receiver; sequence detection	Likelihoods inside a Viterbi recursion	Viterbi with perfect and uncertain CSI; BER
Akdemir et al. [15]	DF relay system; relay selection	Channel-coefficient forecasts	SNR-based, threshold-based and all-relay participation protocols
Bergel et al. [20]	AF relay network; gain setting	Relay gains, with relays treated as neurons	Analog gain optimization exploiting hardware nonlinearity; no symbol-level detection
This thesis	Two-hop relay; symbol-level forwarding	Eight compared relay maps, six learned, under declared parameter budgets	AF, symbol-wise DF, genie and pilot-aided MLSE, BCJR, scalar MAP, block DF; destination BER, decision delay and multiplications per symbol

Relationship between the canonical study and the extension study. Hypotheses H1–H4 concern the canonical relay scenario introduced in Chapter 1: a memoryless two-hop channel with white noise and uncoded transmission. Hypothesis H5 studies a deliberately different regime in which the canonical model no longer represents the true channel. It is therefore a robustness extension rather than a direct continuation of the canonical experiments.

3.3 Research Hypotheses

Based on the theoretical background of Chapter 2, this thesis tests the following hypotheses. H1–H4 are posed and answered on the canonical setup. H5 is the central hypothesis of the thesis's principal contribution (Chapter 6), posed on channels *outside* the canonical family:

1. **H1 (AI advantage over AF at low SNR).** On the canonical setup, some AI-based relay strategies achieve lower BER than AF at low SNR (0–4 dB), where the non-linear denoising learned by the neural network provides an advantage over AF's noise amplification.

   Rationale: The scalar BPSK-AWGN posterior mean, $f^*(y) = \tanh(y/\sigma^2)$, provides qualitative intuition for a smooth nonlinear relay map, but it is not the exact estimator for the canonical Gray-QPSK Rayleigh channel. There, the per-axis posterior depends on the channel-compensated observation and its fading-dependent reliability. For BPSK in AWGN, moreover, hard detection of the posterior mean is identical to the MAP decision, since $\text{sign}(\mathbb{E}[X \mid Y]) = \text{sign}(Y)$; any relay-level benefit therefore arises from preserving soft reliability information for the second hop rather than from improving the single-hop hard decision. In either case, the local MMSE estimator is not necessarily the end-to-end BER-optimal relay function, so H1 is an empirical hypothesis rather than a theorem.

2. **H2 (DF leads at medium-to-high SNR).** The symbol-wise DF relay (Remark 4.2) achieves the lowest BER at medium-to-high SNR ($\geq$ 6 dB), outperforming all AI methods.

   Rationale: In the scalar AWGN limit the posterior mean approaches a sign decision; on the canonical QPSK channel the corresponding high-SNR rule is nearest-neighbor slicing after channel compensation. This motivates DF as the strongest evaluated hard-regeneration baseline, not as a proof that it is end-to-end BER-optimal among all power-constrained per-symbol relay mappings.

3. **H3 (An error minimum at intermediate capacity).** There exists a model size for relay denoising beyond which BER stops improving and begins to rise, so that the error-versus-capacity curve has an interior minimum. The hypothesis as tested is descriptive and says only that the curve turns; *overfitting* is the candidate explanation for such a turn, and it is a separate claim requiring evidence this protocol does not collect — an upturn could equally reflect optimization difficulty at larger widths, or a training budget held fixed while capacity grows. Specifically, models with $\sim$100–200 parameters achieve performance comparable to models with 10–100$\times$ more parameters. Only the second, weaker part of this hypothesis was testable with the protocol originally adopted: establishing *overfitting* as the mechanism behind any upturn would require seed replication and train/test-gap measurement. The seed replication is performed for the MLP size axis by the sweep of Section 5.4, and the hypothesis is adjudicated there and in Section 7.2.1; the train/test-gap measurement remains unperformed; separate cross-architecture seed studies are described

in Section 7.4.

Rationale: The low-dimensional target in the memoryless special case motivates testing whether modest capacity is sufficient. The effect of excess capacity on predictive variance is not fixed a priori and must be established by replicated training and held-out evaluation.

4. **H4 (Architecture convergence at an equal parameter budget).** When all AI models are normalized to the same parameter count ($\sim$3,000), the performance differences between architectures become negligible over the evaluated SNR range. This is an equal-*budget* comparison rather than an isolation of architecture from capacity: as Table 4.3 records, meeting the budget required adjusting width, depth, state dimension and input window size together, so those degrees of freedom co-vary with architecture and are not separately controlled.

   Rationale: The evaluated models have enough representational capacity for the tested mapping. Equalizing parameter count tests whether raw capacity explains observed differences, while leaving architecture and several hyperparameters confounded.

Unlike H1–H4, which assume the canonical memoryless channel throughout, H5 deliberately introduces impairments not present in the system model of Chapter 1, including channel memory, ISI, and nonlinear distortion.

5. **H5 (Principal contribution, unknown-channel mitigation).** On the evaluated channels with memory (intersymbol interference) or asymmetry (biased nonlinearity), the *memoryless* classical relays (AF, symbol-wise DF) exhibit error floors or severe degradation, whereas a small windowed MLP (169–193 parameters) mitigates the impairment after training on the relevant channel family, without per-block CSI or online adaptation.

   Rationale: AF forwards the impairment and symbol-wise DF applies a memoryless zero-threshold decision; neither explicitly compensates channel memory. For a known matched FIR channel, Viterbi is the maximum-likelihood *sequence* detector; with estimated taps, LS-plus-Viterbi is a plug-in detector rather than an optimal one. The hypothesis is not that learning beats a BER-optimal classical receiver, but that a fixed learned architecture can approach matched-MLSE performance on an unseen realization drawn from the impairment family represented during training, without per-block CSI, pilots, explicit identification, or online adaptation. Its arithmetic is fixed only for a fixed deployed window; spanning longer memory generally widens that window and raises cost roughly linearly, whereas full-state MLSE evaluates M^L branches per symbol for an L-tap channel, a count that stays exponential in L even when a square constellation over real taps is factorized into two $\sqrt{M}$-ary trellises (Section 2.6). Chapter 6 determines the empirical scope of this claim.

3.4 Specific Objectives

1. **Implement nine relay strategies and compare eight of them** spanning four paradigms among those compared — no learning (AF, DF), supervised learning (MLP, Hybrid), generative modeling (VAE), and sequence modeling (Transformer and simplified Mamba-S6 and Mamba-2 SSD implementations, Section 4.2); the ninth, a conditional GAN (cGAN), is implemented but excluded from the comparison on cost grounds.
2. **Evaluate on the canonical setup:** SISO Rayleigh fast fading with QPSK, with symbol-wise DF checked against the closed-form two-hop composition on that same channel as the analytical calibration.
3. **Investigate the complexity–performance trade-off** by testing model architectures ranging from 0 parameters (classical) to 26,179 parameters (Mamba-2 SSD), and by conducting a normalized comparison at approximately 3,000 parameters.
4. **Test whether the canonical conclusion survives a coded chain** by replacing the uncoded symbol-wise DF baseline with a genuine block-DF relay, a convolutional code decoded and re-encoded at the relay, and comparing it against coded-aware learned relays at equal E_s/N_0 and again at equal throughput, since a code that halves the information rate changes what "better" means (Section 5.5).
5. **Delineate the conditions under which a learned relay surpasses classical processing** on channels whose structure the classical relays do not model — unknown memory, nonlinear asymmetry, and composite cascades of both — through separate bounded comparisons rather than a demonstration of capability: a memoryless control where learning is expected to buy nothing, a matched genie-CSI MLSE comparator, a pilot-budget sweep to find the crossover, and a blind regime without an explicit per-block channel estimate, where Bayesian inference would require marginalizing latent channel parameters (Chapter 6).
6. **Measure accuracy and resources within a coded-memory system**, with every relay placed inside the same coded chain and cost reported as decision delay in symbols and declared arithmetic per symbol (Section 6.5.1). Distinguish the implemented joint trellis from an exactly factorized channel detector. These comparisons do not put the separate pilot study on a common resource axis.
7. **Identify practical deployment recommendations** for selecting the appropriate relay strategy given specific operational constraints (SNR range, computational budget, channel environment).

3.5 Scope and limitations

The following limitations define the scope of the canonical comparison, confined throughout to the setup below (SISO, Rayleigh fast fading, complex baseband, QPSK, uncoded); the supplementary coded and unknown-channel studies depart from it deliberately, as noted where relevant:

- **Modulation:** QPSK for the canonical relay comparison; the unknown-channel study (Chapter 6) mainly uses BPSK, with a QPSK re-check of the unknown-ISI result (Section 6.2.2), and 16-QAM enters only as a modulation-and-coding rung in the link-adaptation study. Higher-order constellations as a relay design question, and the joint 2D N-class formulation they require, are deferred to future work.
- **Channel knowledge:** Perfect CSI in the canonical core; estimation error is not modeled there. Chapter 6 lifts this with finite-pilot estimation, a swept pilot budget, and a zero-CSI blind regime.
- **Topology:** A single relay, two hops, half-duplex, SISO on both hops. Multi-relay, full-duplex and multi-antenna (MIMO) configurations are deferred to future work.

These delimitations isolate the relay processing function from protocol and system effects.

The scope that bounds every result in this thesis, and the generalizations it does not license, is set out once in Section 1.1 and not restated per chapter. The delimitations listed above are the ones specific to the objectives of this chapter.

Chapter 4

Methods

4.1 System Model

The system under study is a two-hop relay network with a single relay node. All methods in this chapter are described for the canonical setup of Chapter 1: SISO on both hops, i.i.d. Rayleigh fast fading, complex baseband, QPSK, uncoded BER. *Fast fading* here means the strict memoryless case, $h[i] \sim \mathcal{CN}(0,1)$ drawn independently for every symbol — i.i.d. per-symbol Rayleigh fading, not a fading process with a finite Doppler spectrum. Every later use of the term in this thesis carries that meaning. QPSK on the canonical Rayleigh channel is the pairing used here and throughout the canonical comparison; the coded and unknown-channel studies additionally use the pairings set out in Table 4.1.

Per-symbol information. In the canonical memoryless channel, neighboring observations add no information about the current symbol conditional on its receiver-side CSI. The equalized observation preserves the local hard BPSK/QPSK decision boundary. For soft inference, however, the channel magnitude also specifies reliability; discarding it does not preserve the full CSI-conditioned posterior. Learned canonical relays consume equalized windows without explicit channel magnitudes, while the coherent front end still uses CSI.

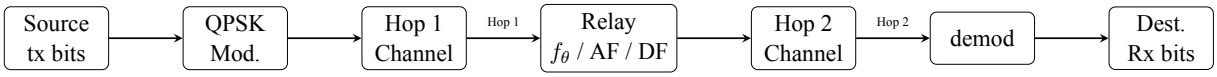


Figure 4.1: Two-hop relay system model: source transmits through two channels with a neural network relay in between. Each hop applies the channel operator followed by one-tap zero-forcing equalization in the canonical setup

Source bits are QPSK-modulated on the canonical setup, passed through Hop 1 (SISO Rayleigh), processed by the relay (classical or neural), transmitted over Hop 2 (SISO Rayleigh), and demodulated at the destination after one-tap zero-forcing equalization at the corresponding receiver. Chapter 6 reuses this chain with BPSK (DBPSK where phase must be removed by differencing), replacing Hop 1 with unknown ISI, a composite cascade, or a per-block redraw; Hop 2

is AWGN for the unknown-ISI comparison, Rayleigh otherwise. In the canonical setup, both hop channels are known perfectly at their respective receivers (Chapter 1); the unknown-ISI departure keeps perfect CSI, while the finite-pilot and blind departures relax it.

$$\text{Source} \xrightarrow{\text{Hop 1}} \text{Relay} \xrightarrow{\text{Hop 2}} \text{Destination} \tag{4.1}$$

4.1.1 Modulation

QPSK carries data in every canonical relay comparison; BPSK carries data mainly in the unknown-channel study (Chapter 6), which also re-checks the unknown-ISI result with QPSK (Section 6.2.2), 16-QAM additionally carries data as a rung of the coded link-adaptation study (Section 5.5), and denser constellations appear in Figure 4.2 for orientation only. The real-valued base case is BPSK, which maps a single bit to a real-valued symbol,

$$x = 1 - 2b, \quad b \in \{0, 1\} \implies x \in \{-1, +1\} \tag{4.2}$$

with hard-decision demodulation $\hat{b} = \mathbb{1}[\hat{x} < 0]$ at the destination. QPSK is defined in Section 4.5. Both are normalized to unit average symbol energy, $E_s = 1$, so they differ in bits per symbol rather than in transmit power: $k = 1$ and 2 respectively, giving $E_b = E_s/k$. Higher-order constellations as a relay design question are future work; 16-QAM appears only as a link-adaptation rung.

4.1.2 Hop Model

In the half-duplex two-hop model studied in this thesis, the relay cannot transmit and receive simultaneously, and there is no direct source–destination link. The communication proceeds in two time slots. On the canonical Rayleigh fast-fading channel each hop applies an i.i.d. complex fading coefficient followed by complex AWGN:

Slot 1 (Source → Relay), listening phase:

$$y_R[i] = h_1[i]\, x[i] + n_1[i], \quad h_1[i] \sim \mathcal{CN}(0, 1), \quad n_1[i] \sim \mathcal{CN}(0, \sigma^2) \tag{4.3}$$

Slot 2 (Relay → Destination), transmission phase:

$$y_D[i] = h_2[i]\, x_R[i] + n_2[i], \quad h_2[i] \sim \mathcal{CN}(0, 1), \quad n_2[i] \sim \mathcal{CN}(0, \sigma^2) \tag{4.4}$$

where $x[i]$ is the complex symbol transmitted by the source at time i (QPSK on this canonical channel; the real BPSK alphabet $\{-1, +1\}$ never rides this complex channel, and is used in the real AWGN special case below and in the Chapter 6 experiments), $y_R[i]$ is the signal received at the relay, $x_R[i]$ is the signal transmitted by the relay, and $y_D[i]$ is the signal received at the destination. The relay output is written generally as

$$x_R[i] = f_\theta(\mathcal{Y}_{R,i}) \,,$$

where $\mathcal{Y}_{R,i}$ denotes the observation supplied to the relay-processing function; for AF and symbol-wise DF, $\mathcal{Y}_{R,i} = y_R[i]$.

Each hop applies a channel function followed by optional equalization (Equation 4.5):

$$y = \mathcal{H}(x, \text{SNR}) + n \tag{4.5}$$

where $\mathcal{H}(\cdot)$ denotes the canonical Rayleigh fast-fading channel operator (the AWGN special case, $|h| \equiv 1$, is retained because the closed forms of Chapter 2 are stated in it). The calligraphic symbol distinguishes the channel *operator* from the scalar fading *coefficient* h used throughout the rest of this thesis.

Both hops use the same channel, canonically. $\mathcal{H}(\cdot)$ applies identically to both hops there, so a configuration is named by the pair (constellation, channel). The unknown-channel study is the exception, pairing an impaired Hop 1 with an unrelated Hop 2 – AWGN in the fixed-ISI, QPSK-ISI, nonlinear-bias and coded-memory studies, compensated Rayleigh in the flat, composite, pilot and blind studies (Table 9.7).

One-tap equalization: which statistic is actually formed. Two one-tap operations are commonly conflated, and they are not interchangeable for anything downstream of a hard decision. Phase compensation, $y \cdot h^*/|h|$, leaves a real channel with a random gain and *unchanged* noise variance; zero forcing, y/h, removes the gain as well and leaves a *conditional* noise variance inflated by $1/|h|^2$. Hard decisions cannot tell them apart, since the two differ by a positive scale factor, but soft relay inputs, posterior probabilities, AF power normalization and BCJR metrics all can.

The canonical simulator (`relaynet/channels/fading.py`) applies **zero forcing**, so the observation each receiver acts on is

$$\tilde{y}_R[i] = x[i] + \frac{n_1[i]}{h_1[i]}, \qquad \tilde{y}_D[i] = x_R[i] + \frac{n_2[i]}{h_2[i]}, \tag{4.6}$$

with conditional complex-noise power $\sigma^2/|h[i]|^2$ given $h[i]$, hence variance $\sigma^2/(2|h[i]|^2)$ on

each real decision axis — a random, heavy-tailed quantity, not the constant per-axis variance $\sigma^2/2$ that phase compensation would give. The unknown-channel study of Chapter 6 does not use this convention throughout, and the difference is worth naming once here because the same symbol $\tilde{y}$ denotes three different observations across this thesis. Its core unknown-ISI experiment has no fading at all: the hop is the deterministic filter plus noise, $\tilde{y}[i] = (h * x)[i] + n[i]$ at constant σ^2, which is what makes a trellis built from h a genuine genie-CSI detector there. Its QPSK subsection adds a per-symbol Rayleigh magnitude on top of that filter and retains it rather than dividing it out, $\tilde{y}[i] = g[i]\,(h * x)[i] + n[i]$ with $g[i] = |h[i]|$, again at constant σ^2 — phase compensation rather than zero forcing. All three are legitimate receivers; they are simply different observations, and each chapter's derivations refer to the one its own simulator forms. Where it matters — soft relay inputs, posterior means, and the branch metrics of a genie detector — the distinction is stated at the point of use rather than assumed. Because the channel is complex, the canonical constellation is two-dimensional: QPSK places one bit on each axis, so both dimensions the channel acts in are used and the relay processes the pair; the real reduction above then applies per axis rather than discarding one. BPSK is not transmitted over the complex canonical fading channel here — a choice of experimental design, not of validity, since coherent BPSK over complex flat Rayleigh is perfectly well posed and loses nothing by projecting onto the real decision axis (Section 4.1.3). Besides its use as the transmitted alphabet of the unknown-channel study (Chapter 6), it appears only in the real AWGN special case below and the closed-form expressions of Chapter 2. The *AWGN limit* referenced in those closed forms is the special case $|h[i]| \equiv 1$, i.e. $y_R[i] = x[i] + n_1[i]$ and $y_D[i] = x_R[i] + n_2[i]$.

The complex noise terms are independent and identically distributed with total power

$$\sigma^2 = \frac{P_s}{\gamma} \tag{4.7}$$

where $P_s = \mathbb{E}[|x|^2]$ is the transmit symbol power (unit for every constellation studied, per the normalization of Section 4.1.1) and γ denotes the per-hop linear signal-to-noise ratio (SNR), equal on both hops.

4.1.3 Evaluated Configurations

This thesis evaluates the primary (constellation, channel) configurations listed in Table 4.1, plus the QPSK/unknown-ISI generalization of Section 6.2.2 noted there.

Coherent BPSK over complex flat Rayleigh fading is theoretically valid and is not excluded on any principled grounds. With $y = hx + n$ and $x \in \{-1, +1\}$, and h known at the receiver, $z = \Re\{h^* y\}$ (or any positive scaling of it) is a sufficient statistic: once the channel phase is removed the entire BPSK component lies on the real decision axis, and the orthogonal component carries no information about x. Discarding it therefore discards neither signal energy nor half

of anything useful.

QPSK is the canonical constellation here for reasons of experimental design rather than validity. It exercises a genuinely two-dimensional complex constellation, carries two bits per symbol, and matches the complex-signal neural processing this thesis sets out to study: the fading coefficient $h \sim \mathcal{CN}(0, 1)$ rotates as well as scales, and a QPSK constellation has a phase for it to act on. BPSK is retained in Chapter 6 for the opposite reason, to keep the ISI trellis exact and the controlled comparisons compact. A complex constellation on AWGN is likewise not wrong, but it exercises none of what makes the complex channel interesting, since there is no phase rotation to undo, and it is not reported here.

Table 4.1: The primary evaluated (constellation, channel) pairings. The canonical and coded rows use the same channel on both hops; the unknown-channel row's Hop-1 filter is fixed throughout (not redrawn per block) and unknown to AF/DF/MLP, though the MLP is trained on it, matching the genie-CSI/pilot-aided Viterbi baselines; Hop-2 is AWGN (Chapter 6). A QPSK generalization of the unknown-ISI study, with the same Hop-1/Hop-2 split, is reported separately in Section 6.2.2

Constellation	Channel	Role	Why this pairing
QPSK (complex)	Rayleigh (complex)	Canonical	The complex case, and the operating point this thesis draws its conclusions on. QPSK places one bit on each axis, so both dimensions the fading coefficient acts in carry information and none is discarded.
BPSK (real)	Unknown 3-tap ISI	Extension	Used only in Chapter 6, where the impairment under study is channel *memory* rather than fading, and the one-dimensional alphabet keeps the trellis small enough for an exact Viterbi benchmark.
16-QAM (complex)	Rayleigh (complex)	Coded MCS rung	Used only as one rung of the modulation-and-coding grid of the coded study (Section 5.5), where the object of study is the rate-and-modulation choice rather than the relay's ability to process a denser constellation.

Beyond its role as the noise term of every channel model and its use in the closed-form expressions quoted as analytical background (Chapter 2), AWGN also appears as the Hop-2 channel of several unknown-channel studies (Table 9.7) and on both hops of one channel of the size sweep (Table 9.3).

A consequence worth stating plainly. The channel functions set N_0 from the average *symbol* energy, so the reported SNR axis is E_s/N_0. For real BPSK, $E_s = E_b$; for QPSK, $E_b = E_s/2$, so its bit-energy label is 3.01 dB below the plotted value. The real branches use variance $N_0/2$ and the complex branches split the same N_0 equally across I and Q. Where a BER figure is read against a BPSK closed form, this conversion is stated explicitly at the point of use.

Power Normalization. All relay strategies normalize their output power to ensure a common empirical transmit-power constraint:

$$x_R \leftarrow x_R \cdot \sqrt{\frac{P_{\text{target}}}{P_{\text{current}}}} \tag{4.8}$$

Here P_{current} is estimated over the complete simulated relay-output block. The normalization is therefore block-adaptive and noncausal at the symbol level; it is appropriate to the block-buffered simulation but should not be read as an online gain known before the block has been observed. For AF on the canonical Rayleigh channel this matters more than for the other relays. AF forwards the zero-forced observation $y/h = x + n/h$, and the implementation applies Eq. (4.8) to it directly, with no clipping or regularization of deep fades. Since $|h|^2$ is exponentially distributed, $\mathbb{E}\left[1/|h|^2\right]$ is infinite, so the forwarded power has no finite mean: the empirical block power does not converge as the block grows, and the few most deeply faded symbols in a block set the gain applied to all the others. The AF results in this thesis therefore describe this block-normalized implementation, not a conventional finite-power AF relay, and the learned relays' advantage over AF is an advantage over this implementation.

4.1.4 Relay Processing

On Hop 1 the relay's single receive antenna observes, as shown in Equation 4.9:

$$\tilde{y}_R[i] = x[i] + \frac{n_1[i]}{h_1[i]}, \quad \frac{n_1[i]}{h_1[i]} \sim \mathcal{CN}\left(0, \frac{\sigma^2}{|h_1[i]|^2}\right) \tag{4.9}$$

matching Eq. (4.6): the effective observation follows one-tap zero-forcing equalization of the Rayleigh fading coefficient (Section 4.1), which removes the gain and leaves conditional complex-noise power $\sigma^2/|h_1[i]|^2$, or variance $\sigma^2/(2|h_1[i]|^2)$ per real axis, rather than the fixed per-axis variance $\sigma^2/2$ that phase compensation would leave. Note that the per-symbol effective SNR is $|h_1[i]|^2\gamma$ (a random, exponentially distributed quantity) not a constant — the same value either convention would give, which is why the two are easy to conflate — and the AWGN limit is recovered by setting $|h_1[i]| \equiv 1$. The relay applies its processing function to a sliding window of received samples and transmits a cleaner estimate $\hat{x}_R = f_\theta(\tilde{y}_{R,i-w:i+w})$; classical AF/DF use $w = 0$. This is purely a noise-removal task: in the canonical SISO setup there is no

inter-stream interference at any stage, and the destination demodulates the Hop-2 output after that one-tap equalization with no further equalizer.

Remark 4.1 (Realizability of the symmetric window)**.** The canonical *channel* of this chapter is memoryless: conditional on current CSI, neighboring samples add no symbol information; a $w = 0$ detector with that CSI therefore loses no temporal information, and the classical AF and symbol-wise DF baselines do use $w = 0$ and are strictly causal. The learned relays nevertheless take a symmetric window $y_R[i - w : i + w]$ with $w > 0$ everywhere they appear, the canonical experiments included: the MLP uses $w = 2$ (a 5-sample window, Section 4.2) and the sequence models $w = 5$. The Transformer attends across the whole window, but the two Mamba relays scan it causally and are read at its centre, so their decision on $x[i]$ depends only on $y_R[i - 5 : i]$: their effective look-ahead is zero. This is an *architectural input convention*, not a claim about the channel. Under the canonical i.i.d. assumptions the extra samples are redundant, they carry no additional information about $x[i]$, and the convention is adopted so that the same *input format*, a symmetric window of received samples, carries over to the three-tap ISI channel of Chapter 6, where the window is what spans the memory and is no longer redundant.

What carries over is the format, not the dimensions, and it is worth being exact about this because the two chapters are otherwise easy to read as using one network. The canonical relay of this chapter takes $w = 2$ (a 5-sample window) into 24 hidden units, 169 parameters. The unknown-ISI relay of Chapter 6 takes $w = 5$ (an 11-sample window, wide enough to span the three-tap response) into 13 hidden units, 170 parameters. Both are two-layer windowed perceptrons of essentially the same size, trained with the same loss and training-SNR grid but different activations, learning rates, epochs and sample counts (Chapter 6), and neither is tuned per channel; but the window had to widen for the memory, and the hidden layer was traded down to keep the parameter budget comparable. The constant held across the two settings is therefore the *class* of relay and its order of magnitude in parameters, not a single fixed network.

Two consequences follow, and both apply to the canonical experiments rather than only to Chapter 6. First, a symmetric window is not causal: it reads w samples ahead. It is admissible in the half-duplex protocol only because the relay buffers the whole listening-phase block before transmitting, and it costs a fixed w-symbol processing latency. Second, that non-causality is a limitation of the windowed MLP, VAE and Transformer relays, recorded as such rather than justified: a streaming deployment needs past-only processing. Past-only processing was evaluated only through the two simplified Mamba relays, whose centre readout uses $y_R[i-5{:}i]$; a past-only window for the other learned relays was not evaluated. Because the window is redundant on the canonical channel, none of the canonical results depend on the look-ahead; the cost is real only where the window is doing work, which is Chapter 6.

4.1.5 The Relay-Design Problem and Its Training Surrogates

The background of Chapter 2 motivates relay processing as local estimation of the transmitted symbol, whereas every experiment evaluates the bit error rate at the destination, after power normalization and a second noisy hop. A local estimator is not in general the relay mapping that minimizes that error rate, so the problem the relays are compared on is stated here, together with the sense in which their training losses stand in for it.

The problem. For a block of N symbols the relay applies a local function f to each observation, $u[i] = f\big(\mathcal{I}_R[i]\big)$, and transmits $x_R[i] = \alpha(\mathbf{u})\, u[i]$, $i = 1, \dots, N$, where $\alpha(\mathbf{u}) = \sqrt{P \big/ \frac{1}{N}\sum_j |u[j]|^2}$ is the block normalization of Eq. (4.8). The local function sees only its own window, but α depends on every $u[j]$, so each transmitted symbol depends on the whole block: the relay map $g = \alpha \cdot f$ is a block map, not a symbol map, and its output is final only when the block ends. For hard-output relays $|u[i]|$ is constant and α is a fixed scale. The elements are as follows.

- *Observations and channel information* $\mathcal{I}_R[i]$. In the canonical study, the window $\tilde{y}_R[i - w{:}i + w]$ of zero-forced samples (Eq. (4.9)): h_1 is used by the front end, but $|h_1[i]|$ is not passed to the learned relays; the matched-metric block-DF decoder of Section 5.5.1 uses $|h_1[i]|^2$ as a reliability weight. In Chapter 6, the window of raw samples y_R from a channel whose taps the relay is not told; the classical comparators there receive either the true taps (genie) or a pilot estimate.
- *Look-ahead.* The decision on $x[i]$ may use samples up to $i + w$, with $w = 0$ for AF and symbol-wise DF and $w > 0$ for the windowed learned relays other than the two causal Mamba relays (Remark 4.1); trellis detectors look ahead by their traceback depth. Because the protocol is half-duplex, the whole listening-phase block is available before transmission begins (Remark 4.1).
- *Power constraint.* $\frac{1}{N}\sum_i |x_R[i]|^2 = P$, which holds by construction of α: it constrains the block map, not f.
- *Destination decision rule.* Fixed and not jointly designed with the relay: one-tap zero forcing followed by a symbol-wise slicer per axis in the uncoded studies, and the soft-decision Viterbi decoder of Section 5.5 in the coded ones. The coded studies are reported with the unweighted Euclidean metric of the published pipeline and, in Section 5.5.1, also with the matched metric, which weights each residual by $|h_2[i]|^2$ (by the end-to-end variance for AF).
- *Objective.* The destination bit error probability $P_b(g) = \Pr\big(\hat{b}_D \neq b\big)$, averaged over bits, fading and noise; in the coded studies, the information bit error probability after decoding.

The relay-design problem is therefore $\min_f P_b(\alpha \cdot f)$ over local functions that use only $\mathcal{I}_R$ within the permitted look-ahead. No relay in this thesis solves it directly. The obstacle is not differentiability as such. The empirical hard-decision error count that estimates P_b is piecewise constant in the relay's parameters, so its gradient vanishes almost everywhere, which is a real training

difficulty. The expected P_b, averaged over the Hop-2 noise and fading, is instead a smooth function of the forwarded values, since for the Gaussian second hop and slicer used here each conditional error probability is a Q-function of $x_R[i]$, and it could be differentiated through α and f. Doing so requires the second hop and the destination rule in the training loop, which a relay trained on Hop 1 alone does not see.

The surrogates. The learned relays are trained on Hop 1 alone, with one of two losses. Mean squared error is minimized by the posterior mean $\mathbb{E}[X \mid \mathcal{I}_R]$ (Eq. (2.5)); cross-entropy over the constellation is minimized by the posterior probabilities $p(x \mid \mathcal{I}_R)$, whose arg-max is the symbol-wise MAP decision. Both optima are taken under the training distribution, which for the canonical relays is BPSK over AWGN rather than the deployment channel (Section 4.3.3). How closely these surrogates track P_b depends on what the relay forwards.

Hard-output relays forward constellation points: symbol-wise DF, the four-class QPSK classifier MLP of Sections 6.2.2 and 6.5.1, the Viterbi relays, and the scalar MAP relay. Their output power is constant, so the normalization of Eq. (4.8) is a fixed scale. Let p_1 be the relay's per-axis bit error rate and p_2 the second hop's. Hop 2's fading and noise are independent of Hop 1 and the constellation is symmetric, so in the uncoded studies a destination bit is wrong exactly when one hop, but not both, flips it:

$$P_b = p_1 + p_2 - 2p_1p_2 = p_2 + p_1\,(1 - 2p_2) \tag{4.10}$$

Since $p_2 < 1/2$, P_b increases with p_1. Minimizing the relay's own bit error rate therefore minimizes the destination's. The BER-optimal target is bit-wise MAP. For BPSK, or QPSK with a posterior that factorizes across its two bit axes, this coincides with symbol-wise MAP. A joint four-class QPSK arg-max minimizes symbol error, not generally BER; bit-wise MAP instead marginalizes the class probabilities for each bit. Cross-entropy learns those probabilities, but does not make the joint arg-max BER-optimal. The Viterbi relays minimize sequence error rather than the bit-wise objective.

Soft-output relays forward a real value per axis: AF, the MSE-trained MLP of the canonical study and of Chapter 6, and the other canonical learned relays (the Hybrid only below its SNR switching threshold, above which it acts as DF). For these no such link holds. After normalization, the magnitude of each forwarded symbol sets how much of the block power it receives, so shrinking unreliable symbols towards zero moves power onto reliable ones; whether that lowers P_b depends on Hop 2 and is not decided by the MSE optimum. For binary inputs on each axis, $\operatorname{sign}\mathbb{E}[X \mid \mathcal{I}_R]$ coincides with the MAP decision, so a perfectly trained soft relay and its hard-thresholded version make the same relay-side decisions and differ only in the magnitudes they forward. The posterior mean is therefore a well-motivated local target but not a solution to the relay-design problem. The comparisons in this thesis *measure* P_b for each relay rather than establishing that any of them minimizes it.

In the coded studies Eq. (4.10) applies to the coded bits that reach the decoder. Because the

decoder uses soft values, the information error rate after decoding is not a function of p_1 alone, so the link between the relay's loss and the end-to-end objective is weaker there than in the uncoded case.

4.2 Relay Strategies

Nine relay strategies were implemented, spanning four learning paradigms. The selection of these nine strategies is designed to systematically explore the relay design space along three axes:

1. the degree of processing (from no learning to deep generative models).
2. the type of inductive bias (feedforward, recurrent, attention, generative).
3. the model capacity (from 0 to 26K parameters).

This section describes each strategy's architecture, training procedure, and the design rationale motivating each choice.

4.2.1 Selection and Classification of Relay Strategies

The selected nine relay strategies are drawn from four distinct architectural concepts:

(i) classical methods (AF and DF); (ii) supervised-learning models (MLP and hybrid MLP–DF); (iii) generative models (VAE and cGAN); and (iv) sequence models (Transformer, Mamba-S6, and Mamba-2 SSD).

1. **Classical AF:** Block-adaptive power normalization as in Equation 2.1, preserving the received signal and its noise. The CSI-assisted analytical SNR expression in Equation 2.2 is background, not an exact prediction of this empirical block-normalization implementation.

2. **Classical DF:** Demodulates, recovers bits, and re-modulates clean symbols (QPSK on the canonical channel, applied per axis). DF is the maximal-processing classical baseline. It performs full signal regeneration through symbol-wise demodulation and re-modulation. On the canonical matched channel and in the uncoded setting studied here, DF achieves the lowest BER among the classical baselines at medium-to-high SNR (see Remark 4.2). Its theoretical BER follows the error-composition formula of Equation 2.4.

3. **Supervised MLP (Minimal):** A two-layer feedforward neural network (multilayer perceptron, MLP) with 169 parameters following the relay architecture defined in Equation 2.9,

   where $\mathbf{y}_i \in \mathbb{R}^5$ is a sliding window from one real decision axis ($w = 2$ neighbors on each side), and the hidden layer has 24 neurons. Parameters: $(5\times24+24)+(24\times1+1) = 169$.

   Real network on a complex channel. The network above is real-valued, while the canon-

ical channel and its QPSK constellation are complex. The two are reconciled by operating *per axis*: the identical 169-parameter network is applied to the real and imaginary parts of the zero-forced sample, each carrying $\pm 1/\sqrt{2}$ plus noise and passed without rescaling, and the two outputs are recombined as one complex symbol before the common power-normalization step (Eq. (4.16)). The parameter count is unchanged by this — one set of weights is shared across both axes, which is what makes the two sub-problems the *same* problem — but the arithmetic cost per transmitted symbol is twice the per-window figure. Every learned relay in the canonical comparison of Chapter 5 (MLP, hybrid, VAE and the sequence models) is applied this way; AF and symbol-wise DF operate on the complex sample directly.

This is a design choice rather than the only option, and the coded study of Section 5.5 makes the opposite one for contrast: its relay interleaves the I and Q windows into a single input vector and classifies over the joint four-point QPSK alphabet, so it sees the constellation as a two-dimensional object rather than two independent axes. That choice is the main reason its parameter count is larger, and it is stated where it is used.

> **Architecture note:** The MLP relay is a standard discriminative two-layer perceptron trained with supervised learning (MSE loss on input–output pairs). In contrast, the generative models in this study are the VAE and cGAN (Section 4.2), which learn to *sample from* or *approximate* the data distribution. The MLP relay simply learns a deterministic mapping $f : \mathbb{R}^5 \to [-1, +1]$ from a noisy observation window to a denoised estimate of one axis: a classical regression task.

Design rationale: The tanh output activation naturally constrains the output to $[-1, +1]$, matching the ± 1 BPSK targets it is trained on; the per-axis outputs are rescaled by the block power normalization (and, in the real-valued unknown-channel study of Chapter 6, match the BPSK symbol range directly). The ReLU hidden layer provides the nonlinearity needed to approximate a soft threshold. The fixed-variance form $\tanh(y/\sigma^2)$ is qualitative motivation, not the exact target: under zero forcing, reliability varies with $|h_1[i]|^2$, so the CSI-conditioned posterior mean forms a family of fading-dependent thresholds. The network receives no channel coefficient of its own. The canonical pipeline as a whole is not CSI-free, however: the one-tap zero-forcing front end that produces $\tilde{y}_R = y_R/h_1$ uses h_1, so CSI enters before the learned stage rather than inside it. Only the unknown-channel study of Chapter 6 withholds a per-block estimate from the pipeline as a whole. Because $|h_1[i]|$ is not supplied explicitly to the network, the noise on $\tilde{y}_R$ is not fixed-variance AWGN but a scale mixture of Gaussians, its per-symbol variance $\sigma^2/|h_1[i]|^2$ randomized by the fading. The Bayes target on the deployment channel is therefore $f^*(\tilde{y}) = \mathbb{E}[X \mid \tilde{Y}]$ under that mixture, not the fixed-σ^2 AWGN denoiser $\tanh(y/\sigma^2)$ that Chapter 3 uses for intuition. The network is not trained toward that target, however. Every learned relay of the canonical comparison is trained once

on BPSK over real AWGN (Section 4.3.3) and applied unchanged to each axis of the zero-forced QPSK/Rayleigh observation, so what it implements is an empirical AWGN denoiser marginalized over the training SNRs, deployed on heavy-tailed noise that never appeared in its training data. With 24 hidden neurons and a 5-dimensional input, the network has approximately 34 parameters per input dimension; whether this capacity is sufficient is determined experimentally, not from the parameter count alone. Its weights are initialized with the Xavier (Glorot) uniform rule (Section 4.3.3).

Training uses MSE loss with multi-SNR training data (i.e., 5, 10, and 15 dB), 25,000 samples, and 100 epochs with a learning rate of 0.01.

4. **Hybrid:** SNR-adaptive relay that switches between its own MLP sub-network (low SNR) and DF (high SNR) by comparing an SNR estimate with a fixed threshold of 5 dB. It is intended to combine the learned relay's behavior at low SNR with DF's zero-parameter operation at high SNR. Same 169-parameter architecture as the MLP.

   Design rationale: The Hybrid relay is motivated by the prediction that the two families are strongest in different SNR regimes — hypotheses H1 and H2 of Section 3.3 state that expectation and Chapter 5 tests it. If it holds, neither family is the right choice across the whole range, and a switching threshold lets the relay select between them per operating condition. The threshold is a fixed 5 dB, not fitted to data. The relay compares it with an estimate formed from the received power of each block, separately for each axis and assuming unit signal power: $\hat{\gamma} = 1/(\hat{P}_{\text{rx}} - 1)$, set to 100 dB when $\hat{P}_{\text{rx}} \leq 1$ (`relaynet/relays/hybrid.py`). On the zero-forced Rayleigh observation this is not the operating SNR: each axis carries signal power $1/2$ rather than 1, and the heavy-tailed noise makes the block power fluctuate. Replaying the Hop-1 draws of the canonical run (`hybrid_branch_selection.py`, which reproduces the published per-trial BER of every block on which both axes chose DF) shows the MLP branch on all 20 axis-blocks at each SNR from 0 to 6 dB, on 6 of 20 at 8 dB, 2 at 10 dB and 1 at 12 dB, and on none from 14 dB upward. The switch therefore falls between 6 and 8 dB and is set by the estimator, not by the nominal threshold. The approach requires no parameters beyond those of the MLP sub-network.

   Training points for the MLP sub-network: SNR = **2, 4, 6 dB**. The sub-network is a separately trained instance of the 169-parameter MLP architecture, not the Minimal MLP relay's trained network (Section 4.3.3).

5. **Generative VAE:** Probabilistic relay with encoder $q_\phi(\mathbf{z}|\mathbf{x})$ mapping to a latent space and decoder $p_\theta(\mathbf{x}|\mathbf{z})$ reconstructing the signal. Architecture: encoder $(7 \to 32 \to 16 \to \mu, \sigma^2(8))$, decoder $(8 \to 16 \to 32 \to 1)$. Total: 1,777 parameters. Trained with β-VAE loss ($\beta = 0.1$) for 100 epochs.

   Training points at the relay node: SNR = **5, 10, 15 dB**.

Design rationale: The low $\beta = 0.1$ prioritizes reconstruction quality over latent space regularization, appropriate for a task where accurate signal recovery is paramount. The 8-dimensional latent space provides sufficient representational capacity for the BPSK signal manifold while maintaining a tractable KL divergence. The encoder window of 7 symbols provides local context.

6. **Generative cGAN (WGAN-GP)** [36, 37]**:** Adversarial relay with a generator conditioned on the noisy signal and a critic providing the training signal. Generator: $(7 + 8 \to 32 \to 32 \to 16 \to 1)$, Critic: $(1 + 7 \to 32 \to 16 \to 1)$. Total: 2,946 parameters, of which the critic's 833 are used only in training. Trained with Wasserstein loss, gradient penalty ($\lambda = 10$) and spectral normalization of the critic's layers, and L1 reconstruction loss ($\lambda_{\text{L1}} = 20$), for 200 epochs. The cGAN is implemented but, on cost grounds, not evaluated in the reported comparisons.

   Training points at the relay node: SNR = **5, 10, 15 dB**.

   Design rationale: The weight $\lambda_{\text{L1}} = 20$ makes the reconstruction loss dominate the generator's objective, with the adversarial term acting as a regularizer that encourages outputs to lie on the clean signal manifold. The noise vector $\mathbf{z} \in \mathbb{R}^8$ provides the stochastic input needed by the GAN framework. The 5:1 critic-to-generator update ratio follows the WGAN-GP recommendation for stable critic training. The doubled epoch count (200 vs. 100) compensates for the slower per-step convergence of adversarial training.

7. **Sequential models: Transformer:** Multi-head self-attention over a window of 11 symbols. Architecture: $d_{\text{model}} = 32$, 4 attention heads, 2 encoder layers, feedforward dimension 64 (twice d_{model}). Total: 17,697 parameters, counted from the instantiated model. Trained for 100 epochs with Adam optimizer (lr $= 10^{-3}$).

   Training points at the relay node: SNR = **5, 10, 15 dB**.

   Design rationale: The Transformer is included as the dominant sequence architecture in modern deep learning. It takes the same 11-symbol window as the Mamba-S6/SSD models but attends across all of it, whereas the Mamba relays use only its causal half (Remark 4.1). With $d_{\text{model}} = 32$ and 4 heads, each head operates in an 8-dimensional subspace, which is sufficient for capturing the local noise structure. The feedforward dimension of 64 is half the usual $4 \times d_{\text{model}}$ expansion.

8. **Sequential models: Mamba-S6** Selective state space model with input-dependent state transitions. Architecture: $d_{\text{model}} = 32$, $d_{\text{state}} = 16$, 2 Mamba blocks with residual connections. Total: 24,001 parameters. Each block applies LayerNorm $\to$ expand $(32 \to 64)$ with SiLU $\to$ S6 scan $\to$ contract $(64 \to 32)$ $\to$ residual. This is a simplified Mamba: it has neither the reference design's depthwise convolution nor its gating branch, its scan is driven by the mean of the 64 expanded channels, and a learned linear map, not the input-dependent C, returns the state to them. Trained for 100 epochs with Adam optimizer

(lr $= 10^{-3}$).

Training points at the relay node: SNR = **5, 10, 15 dB**.

Design rationale: The expand factor of 2 doubles the internal dimension during the S6 scan, and the state dimension $N = 16$ makes each layer a 16th-order adaptive filter, more flexible than a fixed low-order linear filter (though not a replacement for matched or Wiener filtering under the assumptions in which those are optimal). LayerNorm, residual connections and the SiLU activation are carried over from the reference implementation; its convolution and gate are not.

9. **Sequential models: Mamba-2 SSD** Structured State Space Duality model that replaces the sequential S6 recurrence with a chunk-parallel structured matrix multiply. Architecture: $d_{\text{model}} = 32$, $d_{\text{state}} = 16$, chunk size 8, 2 Mamba-2 blocks with gating and residual connections. Total: 26,179 parameters. Each block applies LayerNorm, then two parallel branches, an SSD layer on a SiLU-activated expansion ($32 \to 64$) and a sigmoid gate, whose product is contracted ($64 \to 32$) and added to the residual. This too is a simplified Mamba-2: it omits the reference design's depthwise convolution and replaces its normalized SiLU gate with a sigmoid gate. The SSD layer builds a lower-triangular causal kernel M per chunk and applies it via batched matmul, with inter-chunk state passing for continuity. Trained for 100 epochs with Adam optimizer (lr $= 10^{-3}$) and gradient clipping ($\|\nabla\| \leq 1$).

   Training points at the relay node: SNR = **5, 10, 15 dB**.

   Design rationale: The chunk size of 8 balances parallelism against the quadratic memory cost of the $L \times L$ chunk matrix, giving two chunks at the 11-symbol window. Gradient clipping at norm 1.0 and the S4D-style initialization $A_{\log} = \log(1, 2, \ldots, N)$ are carried over from the reference implementation.

Remark 4.2 (Terminology: symbol-wise DF vs. information-theoretic DF)**.** The term "decode-and-forward" is used in two distinct senses in the literature, and the distinction matters for interpreting the results of this thesis. In its *information-theoretic* sense [1, 2], DF is a *block* strategy: the source protects each message block with a strong (ideally capacity-achieving) channel code, the relay decodes the entire block, and only then re-encodes and forwards the recovered message. This block-DF is inherently *not* memoryless and, over a single link, can operate reliably at any rate below the first-hop capacity.

The canonical system studied in the core of this thesis is *uncoded*: one QPSK symbol carries two information bits, one per axis, and no outer error-correction code is applied there. Consequently, the "DF" baseline throughout the canonical comparison refers to *symbol-wise slicing*: per-symbol Gray demodulation on the coherently equalized I and Q components followed by QPSK re-modulation, with $w = 0$. This is the appropriate classical counterpart to the learned relays under the uncoded BER metric used there, but it is strictly weaker than block-DF with coding; the reported DF results should not be read as bounds on coded block-DF performance.

This caveat is measured directly, not left as a standing assumption: Section 5.5 adds a practical finite-length coded block-DF baseline (a rate-1/2 convolutional code with soft-decision Viterbi decoding, swept over constraint length on the QPSK/Rayleigh link) and reports where it does and does not beat the symbol-wise DF used elsewhere in this thesis.

4.3 Simulation Framework

4.3.1 Monte Carlo BER Estimation

BER is estimated through Monte Carlo simulation, which provides an unbiased estimate of the true BER at each SNR point. The simulation parameters are:

- **Bits per trial:** 10,000
- **Trials per SNR:** 10 (independent random seeds)
- **Total bits per SNR point:** 100,000
- **SNR range:** 0 to 20 dB (step: 2 dB), yielding 11 evaluation points
- **Random seed control:** Each trial uses a unique seed for bit generation and noise realization

BER Computation:

$$\hat{P}_e = \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}(b_i \neq \hat{b}_i) \tag{4.11}$$

where $\mathbb{1}(\cdot)$ is the indicator function and $N = 10{,}000$ is the number of bits per trial. Under independent Bernoulli errors, the per-trial variance is $P_e(1 - P_e)/N$. Window overlap and block normalization can invalidate that independence; the reported trial-level intervals instead use variation across independent simulation trials.

Confidence Intervals. The 95% confidence interval is computed from the $M = 10$ independent trial estimates as:

$$\text{CI}_{95\%} = \bar{P}_e \pm t_{0.975,M-1} \cdot \frac{s}{\sqrt{M}} \tag{4.12}$$

where $\bar{P}_e$ and s are the mean and sample standard deviation of ten trial BERs, and $t_{0.975,9} = 2.262$. Sparse errors can make the approximation unreliable. A zero-width interval or apparent curve convergence at the measurement floor does not establish stable ranking.

BER resolution and dependence. With 100,000 aggregate bits, one observed error changes the pooled estimate by 10^{-5}; an observed zero is not evidence of zero BER. For a fixed budget of independent Bernoulli errors, the exact one-sided 95% zero-error upper bound is $1-0.05^{1/n} \approx 3/n$

[38]. That assumption does not follow from independent fading alone: overlapping windows, shared I/Q reliability, coding and whole-block normalization can correlate bit errors. Use independent blocks or trials as the uncertainty unit, and separate training-seed replication. A zero-width trial interval at zero observed errors is uninformative. Historical first-error-dependent exposures in Chapter 6 require separate validation and do not inherit fixed-budget binomial coverage.

Per-experiment simulation budgets. The canonical and normalized comparisons use ten trials of 10,000 bits per SNR. The original unknown-ISI, flat-channel and QPSK studies use ten trials of 100,000 bits; the composite study uses the same unique-trial budget, with its historical aggregation caveat in Section 6.2.4. Blind and pilot studies use 50 trials of 20,000 bits. Training replication and high-budget confirmations are additional, separately declared units. The coded-memory precision sweep uses five seeds; other coded tests use 100 paired trials (30 for the matched-metric reruns of Section 5.5.1). No single trial count or significance resolution applies to every experiment. For a two-sided exact Wilcoxon test with five nonzero pairs, the smallest attainable p is 0.0625, so those five-seed measurements cannot support a $p < 0.05$ claim from that test.

4.3.2 Statistical Significance Testing

Differences between relay methods are assessed using the **Wilcoxon signed-rank test**, a non-parametric paired test. For each pair of relay strategies (A, B) at each SNR point, the test compares the $M = 10$ paired BER observations (Equation 4.13):

$$H_0 : \text{the paired-difference distribution is symmetric about } 0 \tag{4.13}$$

The Wilcoxon signed-rank results are exploratory paired comparisons. Their location interpretation requires assumptions on the paired-difference distribution, and failure to reject is not equivalence. No multiplicity correction was applied to the post-inspection comparisons, so the reported p-values are descriptive, per-comparison quantities. Original fixed-checkpoint Monte Carlo intervals do not include retraining variance. Separate three-initialization studies cover the size sweep, native and equal-budget architectures, and several unknown-channel experiments; those sections state their own aggregation rules. Initialization spread is strongly configuration-dependent, so the pooled median 0.176 dB is not a universal resolution threshold.

4.3.3 Training Protocol

The runs behind Tables 5.2 and 5.3. Both canonical tables come from runs recorded in the repository's history. Table 5.2 was produced on 17 August 2026 by `run_experiments.py` with `--exp 7.10`, `--retrain` and `--skip-relays cgan`, ten trials of 10,000 bits at

0–20 dB, from the code of commit `b4059e6` (data committed as `69a470c`), in 2,892 s for training and evaluation on a four-core CPU without CUDA; Table 5.3 by `--exp 7.8` with the same options from commit `4cf9753` (data `d27fcf7`, 746.6 s). Every learned relay was trained afresh, once, in its run, and the same trained network is evaluated at every SNR; the cGAN was not run. No code these runs execute has changed since, except that the Mamba wrappers gained an optional readout argument whose default is the configuration used. Re-evaluating the parameter-free AF and DF with today's code reproduces both tables' AF and DF per-trial BERs bit for bit (`canonical_run_provenance.py`), which pins the channel, the trial seeds and the budget that the learned columns share. Table 4.2 gives each learned relay's settings as executed.

Table 4.2: Training of the learned relays of Tables 5.2 (upper block) and 5.3 (lower block) as executed. All are trained for 100 epochs on BPSK ± 1 over real AWGN, $N/3$ samples at each training SNR with seed $42 + \mathrm{SNR}$, on interior windows only; "windows" counts the training set, with the part used for gradient steps in brackets where the sequence wrappers hold out 15%. Loss MSE (β-VAE with $\beta = 0.1$ for the VAEs). "Kept" is the checkpoint evaluated: the final epoch, or the epoch of lowest mean training loss (`results/canonical_run_provenance.json`)

Relay	Params	Win.	SNRs (dB)	Windows	Adam lr/batch	Kept
MLP	169	5	5/10/15	24,987	10^{-2}/32	final
Hybrid (MLP part)	169	5	2/4/6	24,987	10^{-2}/32	final
VAE	1,777	7	5/10/15	49,980	10^{-3}/64	final
Transformer	17,697	11	5/10/15	49,968 (42,473)	10^{-3}/64	lowest loss
Mamba-S6 / Mamba-2	24,001 / 26,179	11	5/10/15	49,968 (42,473)	10^{-3}/64	lowest loss
MLP-3K, Hybrid-3K	3,004	11	5/10/15	24,969	10^{-2}/32	final
VAE-3K	3,037	11	5/10/15	24,969	10^{-3}/64	final
Sequence models, 3K	3,004–3,027	11	5/10/15	24,969 (21,224)	10^{-3}/64	lowest loss

Scaling and evaluation. Each learned relay is applied per axis to the zero-forced observation, the real and imaginary parts separately and without rescaling, so each axis carries $\pm 1/\sqrt{2}$ against the ± 1 of training (Section 4.5.3); windows are edge-padded at block ends. The tanh output of each axis is scaled to unit mean power over the 5,000-symbol block (Eq. (4.8)). Hop 2 sets its noise from the measured power of the forwarded block, so this scale does not change the operating SNR. The Hybrid chooses between its MLP and DF per axis and per block. Each SNR point uses ten trials of 10,000 bits, with trial seeds 0–9 shared by every SNR and every relay. The sequence wrappers hold out a random 15% of their windows, used only to report accuracy, and restore the weights of the epoch with the lowest mean training loss, without early stopping; the sequence models of Section 6.4.2 use a separate loop that keeps the final epoch.

What cannot be established. PyTorch's weight initialization, the batch order and the VAE's sampling noise were not seeded, the trained weights were not saved, and the per-epoch losses and library versions of the run environment were not recorded. The learned columns therefore cannot be regenerated exactly; their training data, which are seeded, can. The multi-SNR training grid

was a design choice with no single-SNR ablation, and with training at 5, 10 and 15 dB the test points from 6 to 14 dB interpolate while the others extrapolate, nominally only: the $\pm 1/\sqrt{2}$ axis amplitude shifts the per-axis signal-to-noise ratio by 3 dB and fading spreads it further.

Weight initialization. The PyTorch relays use Xavier (Glorot) initialization [39]: the uniform form for the MLP, the Hybrid and the sequence models, the normal form for the cGAN, with zero biases; the VAE keeps PyTorch's default layer initialization. The NumPy MLPs of Chapter 6 instead draw $W \sim \mathcal{N}(0, 2/n_{\text{in}})$, the He rule [40], although their layers are tanh.

4.3.4 Reproducibility and Weight Management

Experiment drivers declare source, channel, and training seeds where replication was performed. Seed formulas differ by driver; the committed scripts and result metadata are the reproduction authority. The original sequence checkpoints did not control every training RNG, so they cannot be reconstructed exactly from a later seed declaration.

A **weight checkpoint system** saves trained model parameters to disk after training completes, enabling experiment resumption without retraining. This is particularly important for the cGAN (2-hour training time) and sequence models (24–37 minutes). It saves weights with architecture metadata, validates the architecture on resume, and keeps seed-specific directories (e.g., `trained_weights/seed_42/`).

The framework includes 347 automated tests (pytest) in 24 test modules, covering channels, modulation, coding (including the matched decoder metrics), relay strategies, weight save and load, simulation, statistics and the verification tooling. In the revision environment (Python 3.11 with PyTorch) all 347 pass; the four tests that need the Mamba-2 module are skipped when it cannot be imported.

Code and reproducibility archive. The complete simulation framework (`relaynet`), every experiment script, and the committed result files each figure and table is generated from are maintained under version control (git) in the `Gilzuk/relaynet2` repository; experiments registered in `provenance_audit.py` are linked to their generating script and committed result file, and `verify_thesis_tables.py` checks a declared subset of numerical table cells against those files and flags mismatches. Neither tool covers every number, figure or prose statement, and superseded result files are retained in the repository as provenance records alongside the files that replaced them (Appendix 9.3). The code and data of this revision are archived as the tag `thesis-2026-10` of that repository, and the canonical runs were made from commits `b4059e6` and `4cf9753` (Section 4.3.3).

4.4 Normalized Parameter Comparison

A fundamental confound in comparing AI architectures is that different models have vastly different parameter counts: from 169 (MLP) to 26,179 (Mamba-2 SSD): a ratio of 155:1. Per-

formance differences between these models could reflect either (a) the superiority of one architecture's inductive bias, or (b) the simple advantage of having more learnable parameters. To reduce this confound, all seven AI models were scaled to approximately 3,000 parameters, giving a comparison at an equal parameter budget. The control is over capacity alone: as Table 4.3 shows, meeting a common budget required changing width, depth, state dimension and, for the feedforward models, the input window size, so those factors co-vary with architecture and the comparison does not isolate architectural inductive bias from them. The MLP-3K in particular sees an 11-sample window against its canonical 5, so it is not fed the same context as the models whose windows were left unchanged.

Choice of target parameter count. The target of ~3,000 parameters was chosen as a compromise: it is large enough that all architectures can express a reasonable denoising function (avoiding underfitting), yet small enough that the normalization represents a meaningful reduction for the larger models (Transformer: 5.9× reduction, Mamba-S6: 7.9×, Mamba-2 SSD: 8.7×). The MLP model is scaled **up** from 169 to 3,004 parameters, testing whether additional capacity benefits the simple feedforward architecture.

Scaling methodology. Each architecture's hyperparameters were adjusted to reach the target while preserving its essential structure:

Table 4.3: Normalized 3K-parameter model configurations: parameter count and scaling method for each architecture

Model	Parameters	Scaling Method
MLP-3K	3,004	Increased window (5→11) and hidden (24→231)
Hybrid-3K	3,004	Same as MLP-3K + DF switching
VAE-3K	3,037	Increased window (7→11), adjusted latent/hidden
cGAN-3K	3,004	Increased window (7→11), adjusted hidden layers
Transformer-3K	3,007	Reduced d_model (32→18), heads (4→2), layers (2→1)
Mamba-S6-3K	3,027	Reduced d_model (32→16), d_state (16→6), layers (2→1)
Mamba-2-SSD-3K	3,004	Reduced d_model (32→15), d_state (16→6), layers (2→1)

All seven configurations use a window size of 11 (versus 5 for the original MLP/Hybrid and 7 for the original VAE/cGAN; the original sequence models already used 11), and every parameter

count lands within 1.3% of the 3,000 target. The cGAN-3K is listed for completeness; like the cGAN, it was not run.

Parameter counting convention. All learnable parameters are counted, including biases, normalization parameters (LayerNorm gain/bias), and projection matrices. Non-learnable buffers (e.g., positional encoding tables, fixed **A** matrices) are excluded. The counts are verified programmatically via `sum(p.numel() for p in model.parameters() if p.requires_grad)`.

Unified input window. All 3K models receive a window of 11 samples (vs. 5 for the original MLP/Hybrid, and 7 for the original VAE/cGAN), but not a common context: the two simplified Mamba relays read their causal scan at the window centre and so use only the six samples up to the current one (Remark 4.1), while the MLP, VAE and Transformer use all eleven. The common window therefore removes the difference in input length for the feedforward models and the Transformer only. It does not isolate architecture: as the parameter-normalization above records, meeting the shared budget also required adjusting width, depth, state dimension, head count and layer count together, so those degrees of freedom still co-vary with architecture. Any residual gap is therefore attributable to the remaining structural and training differences collectively, of which architecture is the most salient but not the only one.

This normalization approximately matches raw learnable-parameter count, not effective capacity, expressivity, or optimization difficulty. Convergence at that budget would establish similar measured performance for those configurations, not identify capacity rather than architecture as its cause. Persistent gaps likewise cannot be attributed to architectural inductive bias alone: width, depth, state dimension and optimization remain confounded. The normalized models share one window length, but the change from the original comparison also changes the feedforward models' context. Equal parameter counts also leave training unequal. In the 3K run every model was trained on $25{,}000$ samples at 5, 10 and 15 dB for 100 epochs, the Hybrid-3K's subnetwork included, but with family-specific optimizer settings (Table 4.2), and no architecture-specific hyperparameter tuning was performed, so a shared setting may disadvantage some architectures more than others. The conclusions of the equal-budget comparison are therefore statements about these architectures under these training protocols, not about the architectures' intrinsic inductive biases.

4.5 Modulation: QPSK

The experiments use QPSK, the canonical constellation. This section defines it and the I/Q-splitting technique that lets real-valued relays process a complex constellation.

4.5.1 QPSK: Gray-Coded Quadrature Phase-Shift Keying

Quadrature Phase-Shift Keying maps pairs of bits (b_0, b_1) to complex symbols [31]:

$$x = \frac{(1 - 2b_0) + j(1 - 2b_1)}{\sqrt{2}} \tag{4.14}$$

yielding four constellation points at $\{(\pm 1 \pm j)/\sqrt{2}\}$ with unit average power ($E_s = 1$). The Gray coding ensures that adjacent constellation points differ by exactly one bit:

Table 4.4: QPSK Gray-coded symbol mapping: bit pairs to complex symbols

Bit pair (b_0, b_1)	Symbol	Quadrant
00	$(+1 + j)/\sqrt{2}$	I
01	$(+1 - j)/\sqrt{2}$	IV
11	$(-1 - j)/\sqrt{2}$	III
10	$(-1 + j)/\sqrt{2}$	II

Demodulation applies independent sign decisions on each component: $\hat{b}_0 = \mathbb{1}(\mathrm{Re}(\hat{x}) < 0)$ and $\hat{b}_1 = \mathbb{1}(\mathrm{Im}(\hat{x}) < 0)$. The spectral efficiency is 2 bits/symbol, double that of BPSK.

Theoretical QPSK BER. For uncoded QPSK over an AWGN channel, the BER per bit equals the BPSK BER at the same E_b/N_0 because each I/Q component carries an independent BPSK stream [31]:

$$P_b^{\mathrm{QPSK}} = Q\left(\sqrt{\frac{2E_b}{N_0}}\right) = P_b^{\mathrm{BPSK}} \tag{4.15}$$

The advantage of QPSK is doubled throughput for the same BER and per-bit energy.

4.5.2 Constellation Diagram Summary

Figure 4.2 presents the constellation diagrams for the modulation schemes appearing in this thesis. The progression from BPSK (2 points on the real axis) through QPSK (4 points on the unit circle) to denser grids illustrates the fundamental trade-off between spectral efficiency and noise robustness: higher-order constellations pack more bits per symbol but reduce the minimum Euclidean distance between points, increasing the BER at a given SNR. QPSK is the constellation used in every canonical relay-architecture comparison; 16-QAM appears only as a rung of the coded link-adaptation study (Section 5.5), and BPSK in the unknown-channel study (Chapter 6), the size sweep and the training of the canonical relays. The others are shown for orientation.

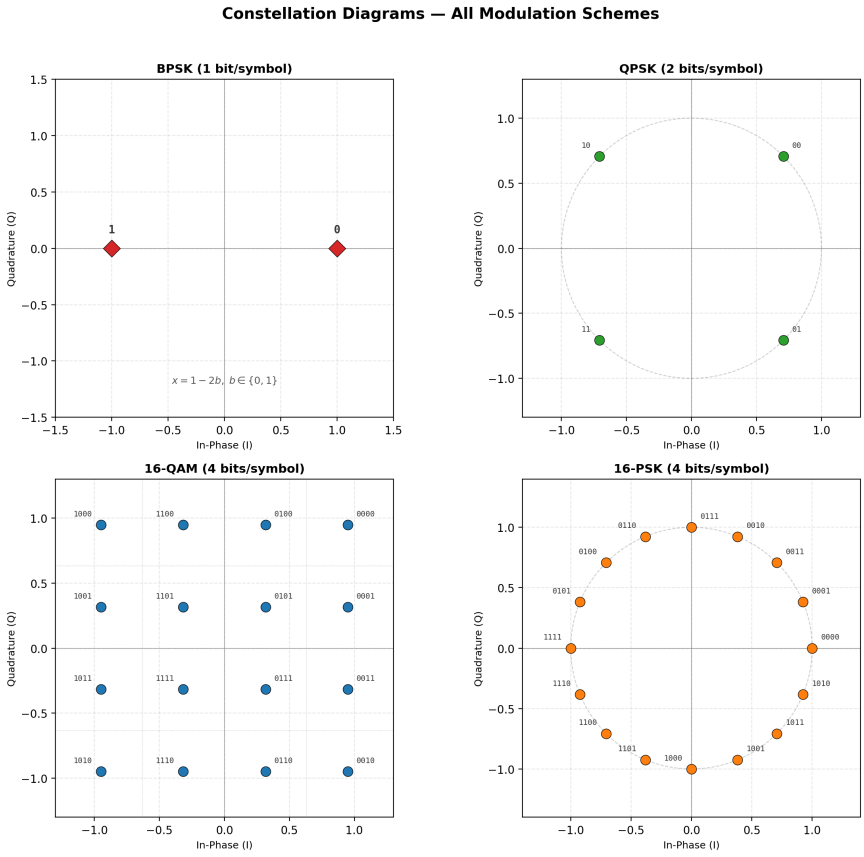


Figure 4.2: Constellation diagrams. (a) BPSK: 2 real-valued symbols at ± 1. (b) QPSK: 4 Gray-coded symbols on the unit circle. (c) 16-QAM: 16 Gray-coded symbols on a 4×4 rectangular grid with decision boundaries (dotted lines). (d) 16-PSK, shown for completeness only; it is not studied in this work. All constellations are normalized to unit average power

4.5.3 I/Q Splitting for AI Relay Processing of Complex Constellations

$$\hat{x}_R = c_I \, f_\theta\big(\mathrm{Re}(\tilde{y}_R)\big) + j \, c_Q \, f_\theta\big(\mathrm{Im}(\tilde{y}_R)\big) \tag{4.16}$$

Here c_I and c_Q scale each axis's output to the target block power of Eq. (4.8), and f_θ acts on the window of each axis. No rescaling is applied to the axes on the way in: each carries $\pm 1/\sqrt{2}$ plus noise, whereas f_θ was trained on ± 1 BPSK symbols, so the network operates on inputs a factor $\sqrt{2}$ smaller than those it was trained on.

Justification and caveat. For rectangular constellations (QPSK, QAM), the I and Q components carry independent information and are corrupted by independent noise conditional on the channel, so per-axis denoising with a shared real-valued function is a natural and parameter-efficient decomposition. It is, however, a *restriction* of the hypothesis class to axis-separable functions, not an equivalence to joint 2D processing. In particular, when fading reliability is not supplied explicitly, a joint processor could use the shared I/Q magnitude as information about that reliability. No joint-QPSK ablation was run, so zero cost from splitting is not claimed; for denser constellations it becomes a structural limitation that a joint two-dimensional relay formulation would be needed to remove, which is left to future work.

Relay-specific handling:

Table 4.5: Relay-specific complex signal handling strategies and rationale

Relay type	Complex signal processing	Rationale
AF	Amplifies complex signal directly	Power normalization ($\|y\|^2$) is valid for complex vectors
AI relays	I/Q splitting (process Re and Im separately)	Real-valued networks; independence of I/Q in rectangular constellations

Limitation beyond binary-per-axis constellations. I/Q splitting works here because QPSK places a single binary decision on each axis, which is exactly what a tanh-output relay trained on $\{\pm 1\}$ represents. Constellations with more than two levels per axis (16-QAM and denser) break that correspondence and would require either modulation-specific retraining or a joint two-dimensional relay formulation; neither is evaluated here, and both are identified as future work.

Chapter 5

Core Experiments: The Canonical Setup

5.1 Experiments summary

Two experiments constitute the core of this chapter: the canonical relay comparison (E2) and the parameter-normalization and complexity study (E3), which together answer every canonical hypothesis (H1–H4). A supplementary coded study (Section 5.5) departs from the canonical setup within this same chapter, adding a rate-1/2 convolutional code, a block-DF relay and an adaptive modulation-and-coding scheme. The principal contribution (Chapter 6) is the unknown-channel study (E6), which tests H5. The core (E2, E3), unknown-channel (E6), coded (E7) and size-sweep (E8) configurations are summarized in the table below; the coded study is detailed in Section 5.5.

Each experiment subsection reports the key results inline (BER tables and figures); supporting material and model hyperparameters are collected in the appendices (Appendices 9.1–9.2).

Table 5.1: Summary of experimental configurations (E1, E4 and E5 are studies not reported here)

Exp.	Topology	Modulation	Channel / Equalization	Focus
E2	SISO	QPSK	Rayleigh (canonical)	Canonical relay comparison (H1, H2)
E3	SISO	QPSK	Rayleigh (canonical)	Parameter normalization & complexity (H3, H4)
E6	SISO	BPSK, QPSK	Unknown (ISI; composite; blind; partial-posterior); control: Rayleigh	Extension: unknown-channel failure of AF/DF and learned mitigation (H5)

Exp.	Topology	Modulation	Channel / Equalization	Focus
E7	SISO	QPSK	Rayleigh (canonical), rate-1/2 convolutional code	Supplementary: coded block-DF against symbol-wise DF and coded-aware learned relays (Section 5.5)
E8	SISO	BPSK, QPSK	All nine families above, plus the coded scenario	Minimum relay size and the capacity hypothesis H3, with three initializations per configuration (Section 5.4)

5.2 Canonical Relay Comparison (SISO, Rayleigh, QPSK)

Goal: Evaluate the classical and AI-based relay strategies on the canonical setup of this thesis.

Configuration. SISO on both hops; QPSK; i.i.d. Rayleigh fast fading (canonical); one-tap zero-forcing equalization per hop, with no ISI equalizer (the channel is memoryless); six learned relays: supervised (MLP, Hybrid), generative (VAE) and sequence (Transformer and the simplified Mamba-S6 and Mamba-2 SSD of Section 4.2), each with a tanh output; training and run settings as executed are in Table 4.2.

Conclusion: On the canonical Rayleigh channel, AI relays match but do not surpass symbol-wise DF: DF achieves the lowest or tied-lowest BER at essentially every SNR point, including the low-SNR regime, while the best neural relays (MLP, Hybrid, VAE) track it within statistical uncertainty and clearly outperform AF. This supports H1 (some AI relays beat AF at low SNR) and, in its matching sense, H2: at ≥ 6 dB DF matches or beats every learned relay on the canonical channel, with zero trainable parameters. The AF column is the block-normalized implementation of Section 4.1.3. A conventional finite-power, variable-gain AF with coherent destination processing reaches 0.3418, 0.2450 and 0.1437 at 0, 4 and 8 dB, so against it the MLP's 0 dB margin is 0.007 rather than 0.073; a fixed-gain AF does better at 0 dB (0.3384), leaving 0.003 (Section 7.1.1).

5.2.1 Results Tables

Table 5.2: BER on the canonical setup: QPSK over i.i.d. Rayleigh fast fading, SISO on both hops, uncoded, $10 \times 10{,}000$ bits per SNR point. "DF th." is the closed-form two-hop composition of Eq. 2.27 (Section 2.5.2.2), included for direct comparison against the measured DF column. Column abbreviations: Transf. is the Transformer relay, Mamba-S6 and Mamba2 the simplified Mamba-S6 and Mamba-2 SSD relays. Run and training settings as executed: Table 4.2

SNR (dB)	AF	DF	DF th.	MLP	Hybrid	VAE	Transf.	Mamba-S6	Mamba2
0	0.4076	0.3337	0.3333	0.3349	**0.3329**	0.3359	0.4014	0.4071	0.4032
4	0.3129	**0.2219**	0.2216	0.2233	0.2220	0.2254	0.2758	0.2812	0.2772
8	0.1852	**0.1218**	0.1203	0.1229	0.1220	0.1235	0.1424	0.1448	0.1416
12	0.0822	0.0580	0.0560	0.0586	**0.0579**	0.0589	0.0628	0.0637	0.0621
16	0.0309	**0.0245**	0.0239	0.0247	**0.0245**	0.0247	0.0260	0.0259	0.0251
20	0.0110	**0.0097**	0.0098	0.0099	**0.0097**	**0.0097**	0.0101	0.0100	0.0098

Two features of Table 5.2 deserve comment. First, symbol-wise DF tracks the closed form (the "DF th." column above): reading the abscissa as E_s/N_0 and converting to E_b/N_0 for QPSK ($k = 2$, so 3.01 dB lower), the two-hop expression $2P(1-P)$ with $P = \frac{1}{2}\big(1 - \sqrt{\bar{\gamma}/(1+\bar{\gamma})}\big)$ gives agreement to within a few percent relative error at every tabulated point (1.0% at 20 dB, 2.5% at 16 dB, worst case 3.6% at 12 dB, each recomputable from the four-decimal values as printed), which provides a consistency check on the channel, equalization and hard-decision pipeline against the analysis of Section 2.5.2.2. Figure 5.1 plots this closed form alongside the measured curves, together with the single-hop floor P itself – the BER a genie that recovered hop 1 perfectly would still incur on hop 2 – which is a reference for this uncoded fixed-rate QPSK comparison; no evaluated curve crosses it.

Second, the relays divide by family rather than by paradigm. The three feedforward relays, the MLP, the Hybrid and the VAE, show no difference from DF detectable at this simulation budget from 4 dB upward, nor from each other throughout. That is not a finding of equivalence: as Section 4.3.2 states, failure to reject requires a prespecified practical margin and an equivalence test to become one, and neither was performed. The VAE reaches 0.00972 at 20 dB, matching DF exactly at the resolution of this budget. The three sequence models are the ones that separate, trailing by roughly 0.07 in BER at 0 dB before converging by 16 dB. Raw parameter count does not explain it, since the sequence models are the larger of the two groups. One reading is that a memoryless channel offers no temporal structure for their inductive bias to exploit; another is that their training protocols, which differ from those of the feedforward relays and were not tuned per architecture, suit this task less well; a third is that all were trained on BPSK over AWGN and meet $\sqrt{2}$-smaller, heavy-tailed inputs here. The equal-parameter comparison of Section 5.3 removes the difference in parameter count but cannot separate these readings.

Why QPSK and not BPSK. QPSK places one bit on each axis the Rayleigh coefficient acts in, carries two bits per symbol, and is processed by the I/Q-split relays without modification, which makes it the constellation that exercises the complex-signal processing this thesis studies.

Coherent BPSK over the same complex channel would be entirely valid (after channel-phase removal the whole BPSK component lies on the real decision axis, so projecting onto it discards no signal energy) and the choice here is one of experimental design rather than of correctness (Section 4.1.3). One consequence must be kept in view when reading these BER figures against BPSK closed forms: the SNR axis here is E_s/N_0, and QPSK carries two bits per symbol, so a given abscissa corresponds to an E_b/N_0 that is 3.01 dB lower than the same number on a BPSK axis.

5.2.2 Results Figures

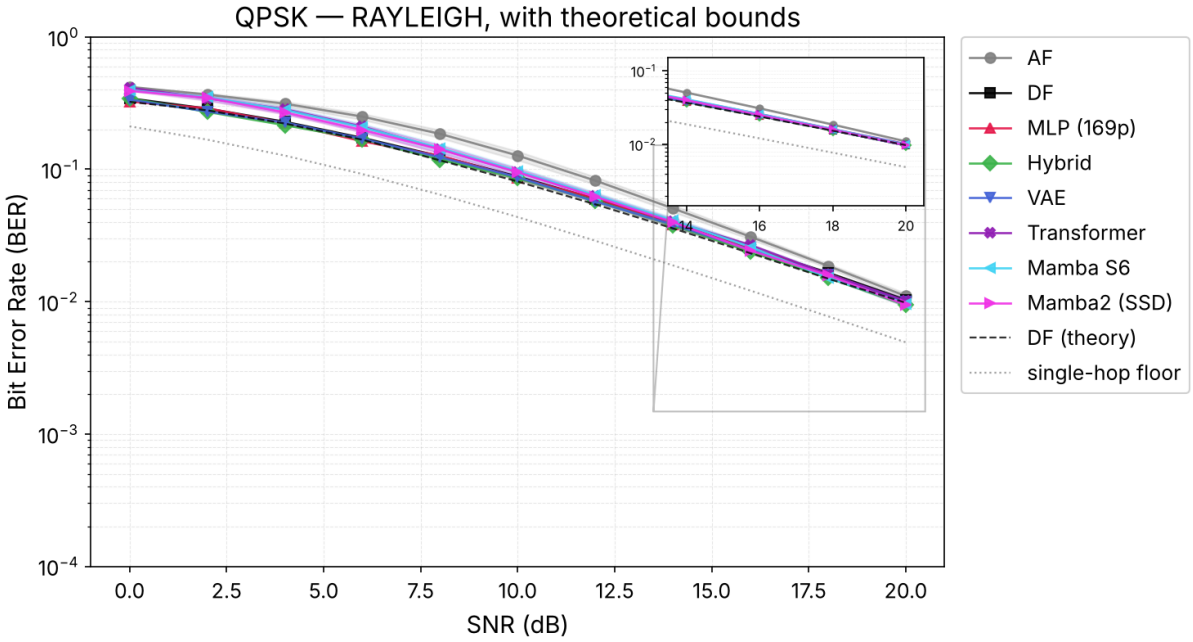


Figure 5.1: The canonical setup: QPSK over i.i.d. Rayleigh fast fading, BER vs. E_s/N_0 for the eight relays of Table 5.2, with 95% confidence intervals, plus two theoretical reference curves: "DF (theory)" (dashed), the closed-form two-hop composition, which the measured DF curve tracks closely, within a few percent relative error everywhere (worst case 3.6%, at 12 dB); and the "single-hop floor" (dotted), the BER a genie that recovered hop 1 perfectly would still incur on hop 2 – a reference floor for this fixed-rate, fixed-power relay chain. AF is uniformly worst; no difference between symbol-wise DF and the feedforward learned relays is detected from 4 dB upward at this budget; the three sequence models trail at low SNR and converge by 16 dB

5.3 Parameter Normalization & Complexity Trade-off

This study tests H3 (complexity versus BER) and H4 (architecture convergence at an equal parameter budget) on the canonical setup.

Goal: Compare the architectures at an approximately equal parameter count, removing raw parameter count as one difference between them, and characterize the complexity-performance trade-off for relay denoising. Equal counts do not isolate inductive bias: width, depth, state dimension, window and the training protocols still differ (Section 4.4).

Configuration. As in Section 5.2, with every learned relay normalized to about 3,000 parameters, alongside the original sizes (169 to 26K parameters).

Conclusion. The 169-parameter MLP is competitive with the larger models in the original canonical comparison. At approximately 3,000 parameters, the cross-model relative BER spread, (max − min)/max, falls from 17.4% at 0 dB to 4.1% at 20 dB. This does not isolate parameter count as the cause: width, depth, state dimensions, the context actually used (the Mamba relays read only the past half of their window) and optimizer settings also differ, although the run trained every model on the same 25,000 samples for 100 epochs (Table 4.2), so the result describes these architectures under these training protocols. Later seed replication shows that the apparent high-SNR convergence is not robust to every initialization (Section 7.6).

5.3.1 Results Tables

Table 5.3: Equal-parameter-budget comparison on the canonical setup (QPSK over Rayleigh). All learned columns are the ∼3,000-parameter variants; every learned model is scaled to approximately 3,000 parameters, so parameter count is removed as an explanation for the residual differences; width, depth, state dimension and window size still co-vary with architecture (Table 4.3). AF and DF are the unparameterized references. Column abbreviations: Transf. is the Transformer relay, Mamba-S6 and Mamba2 the simplified Mamba relays. Run and training settings as executed: Table 4.2

SNR (dB)	MLP	Hybrid	VAE	Transf.	Mamba-S6	Mamba2	AF	DF
0	0.3361	0.3391	0.3424	0.4068	0.4007	0.4001	0.4076	**0.3337**
4	0.2244	0.2271	0.2291	0.2818	0.2751	0.2742	0.3129	**0.2219**
8	0.1229	0.1229	0.1267	0.1458	0.1406	0.1405	0.1852	**0.1218**
12	0.0586	0.0580	0.0604	0.0640	0.0620	0.0622	0.0822	**0.0580**
16	0.0247	**0.0245**	0.0251	0.0259	0.0250	0.0251	0.0309	**0.0245**
20	0.0099	**0.0097**	0.0101	0.0100	0.0097	0.0098	0.0110	**0.0097**

5.4 Minimum Relay Size Across Channel Families

The complexity study of Section 5.3 finds no evaluated model detectably better than the 169-parameter MLP, but leaves two questions open: how far *below* 169 parameters the relay can be taken, and whether the answer is a property of the architecture or of the channel. This section answers both by sweeping relay size downward on every channel family used in this thesis.

Goal: Locate the smallest relay that costs nothing measurable against a reference relay trained on the same channel, and determine which architectural axis — window, depth or width — sets that floor.

Configuration. SISO, two hops, Hop 2 following each channel family's own convention (AWGN or coherently compensated Rayleigh); BPSK or QPSK per channel; the nine channel families of Chapters 5 and 6 plus the coded scenario of Section 5.5; comparators symbol-wise

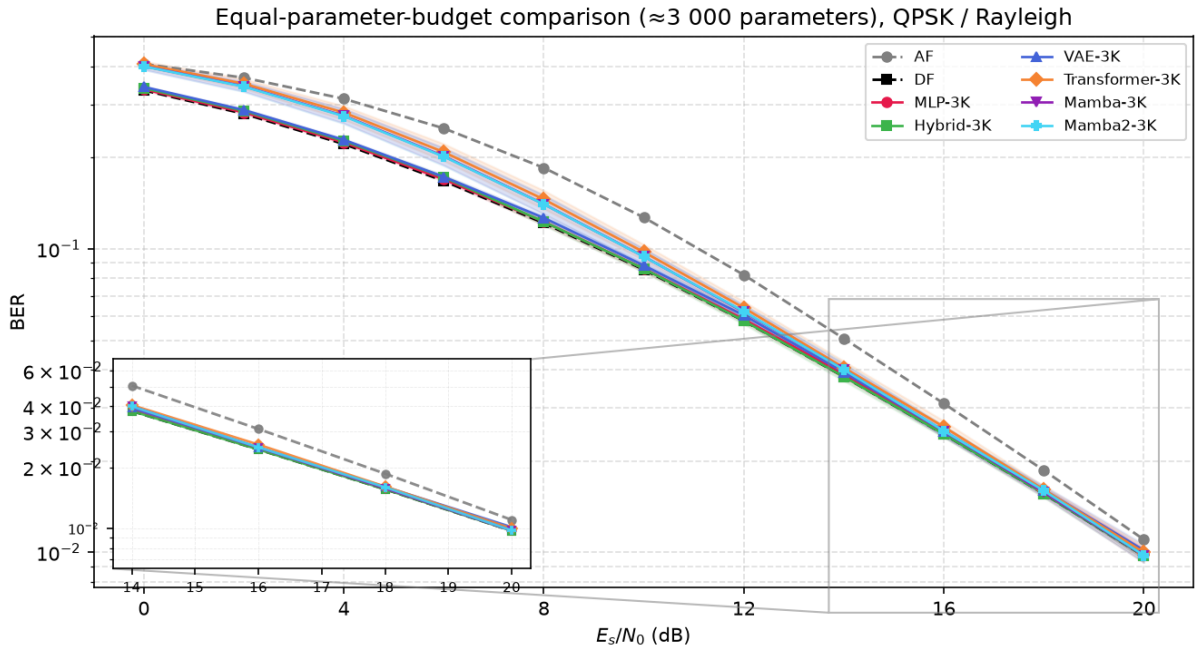


Figure 5.2: BER vs E_s/N_0 for all relay variants normalized to ≈3,000 parameters, QPSK over Rayleigh fading. AF and DF (dashed) are the unparameterized references. Shaded bands show 95% confidence intervals for the learned relays. The feedforward group (MLP-3K, Hybrid-3K, VAE-3K) tracks DF across the full range; the sequence models (Transformer-3K, Mamba-3K, Mamba2-3K) converge with them only at high SNR

DF on memoryless channels, Viterbi MLSE on channels with memory and block DF on the coded scenario, plus a reference relay (a window-5 MLP-169 retrained per channel, or MLP-756 coded); window $\in \{1, 3, 5, 7\}$, depth $\in \{1, 2, 3\}$, width 1 to 48; three initializations per configuration, each held to its worst; metric: the additional SNR needed to reach a given BER.

Conclusion. Within the tested grid and degradation criteria, observation-window length is an important resource for the selected three-tap channels. Three of the five memoryless cases meet the zero-degradation criterion with four parameters, whereas the three-tap cases use 73–145. Widening the window from 1 to 7 improves the reported metric by more than 8 dB on every three-tap channel, a lower bound because no window-1 seed reaches every target the comparator reaches; added depth does not improve the best tested configuration. These are smallest *tested* configurations under finite training and evaluation budgets, not mathematical lower bounds on architecture size. The budget-by-budget table (Table 9.4), the seed spreads and the two figures that show the window effect (Figures 9.1 and 9.2) are in Appendix 9.4.

5.5 Coded Block-DF: Measuring the Remark 4.2 Caveat

A note on constellation scope. The fixed-MCS relay-decoding comparisons here are QPSK throughout. 16-QAM appears only as a rung of the modulation-and-coding ladder in Section 5.5.5, where the object of study is the rate-and-modulation choice rather than the relay's ability to process a denser constellation; that question is left to future work (Section 4.5).

Remark 4.2 cautions that the symbol-wise DF results "should not be read as bounds on coded block-DF performance". That is asserted there and measured here, by a practical finite-length block-DF relay built from a rate-1/2 convolutional code and a soft-decision Viterbi decoder, as a supplementary study on the canonical channel rather than a revision of the core comparison.

Setup. A standard rate-1/2 convolutional code (constraint length K, 2^{K-1} trellis states, zero-tail terminated per 200-information-bit frame) is applied before QPSK modulation on the canonical Rayleigh channel; `CodedDecodeAndForwardRelay` decodes the full frame at the relay via soft-decision Viterbi, then re-encodes and re-modulates a fresh codeword before forwarding. Because post-equalization noise is heteroskedastic, the decoders weight each symbol by its instantaneous channel reliability (the matched metric); these are the results of the section (Section 5.5.1). The published pipeline used an unweighted squared-distance metric instead and is kept as a diagnostic (Section 5.5.2). Two coded-aware learned relays are trained on the same task for comparison: a windowed 756-parameter MLP classifier (window 21) and the Mamba-S6 architecture already used in Table 5.2 (24,084 parameters at this configuration), both predicting the transmitted QPSK symbol from a window of noisy coded observations with no explicit knowledge of the code. All results use this study's budget of 100 trials $\times$ 100,000 information bits per SNR point.

5.5.1 Matched Decoder Metrics: The Principal Coded Comparison

The zero-forced observation $y/h = x + n/h$ has noise variance $1/(2\gamma|h|^2)$ per real axis, which changes from symbol to symbol. The matched metric weights each squared residual by $|h|^2$, which is the same as decoding $|y - hx|^2$ on the unequalized observation, and it is the comparison this section reports as its result. `coded_matched_metrics.py` runs the comparisons of Tables 5.5, 5.6 and 5.9 with it. The relay Viterbi decoder weights its branch metrics by $|h_1|^2$; the soft relay's BCJR uses the per-symbol variance $1/(2\gamma|h_1|^2)$; and the destination weights by $|h_2|^2$, or for AF by the end-to-end variance, which requires both hops' gains. The learned relays evaluated are the hard and soft read-outs of the 756-parameter MLP, unchanged and still given no channel gain; only their destination metric changes, and both destination metrics are reported. The Mamba-coded relay of the unweighted study was not rerun with matched metrics, so every statement below concerns the two MLP read-outs. The run uses the first 30 of the 100 paired trials of Table 5.9, each of 100,000 information bits. Its unweighted rows reproduce that table's per-trial values exactly. The 30-trial budget was retained after inspecting running means, so the paired tests below are exploratory; the within-trial differences isolate the metric change.

Table 5.4: Coded relaying with reliability-weighted (matched) decoder metrics, canonical QPSK/Rayleigh, rate-1/2 $K{=}3$, 30 paired trials $\times$ 100,000 information bits. "Unweighted" repeats the published pipeline on the same draws. For the two MLP read-outs, "plain / weighted" gives the unweighted and the $|h_2|^2$-weighted destination metric; the Mamba relay was not rerun. At 16 and 20 dB, matched block DF makes 30 and 3 errors in 3×10^6 bits, and the oracle 13 and 2

SNR (dB)	Block DF, un-weighted	Block DF, matched	Soft block DF, matched	MLP hard, plain / weighted	MLP soft, plain / weighted	Oracle, matched
0	0.4207	**0.2685**	0.2845	0.4436 / 0.4080	0.4332 / 0.3960	0.1603
4	0.2231	**0.0473**	0.0553	0.2966 / 0.2341	0.2760 / 0.2131	0.0244
8	0.0638	**0.003235**	0.003502	0.1005 / 0.0787	0.0875 / 0.0619	0.001588
12	0.0154	**0.000203**	0.000206	0.0214 / 0.0224	0.0177 / 0.0154	9.5×10^{-5}
16	0.004127	$\mathbf{1.0 \times 10^{-5}}$	$\mathbf{1.0 \times 10^{-5}}$	0.004359 / 0.006797	0.003559 / 0.004311	4.3×10^{-6}
20	0.001356	$\mathbf{1.0 \times 10^{-6}}$	$\mathbf{1.0 \times 10^{-6}}$	0.001050 / 0.002377	0.000889 / 0.001449	6.7×10^{-7}

With matched metrics, block DF leads at every tested SNR. Matched hard block DF has lower BER than both MLP read-outs at all six SNRs from 0 to 20 dB, in all 30 paired trials at each point (Wilcoxon $p < 10^{-5}$), whichever destination metric the MLP is given. At 8 dB it reaches 0.0032 against the better MLP's 0.0619. At 20 dB it makes 3 errors in 3×10^6 bits (1.0×10^{-6}), against 2,667 for the soft MLP (8.9×10^{-4}), and stays within a factor of 2.4 of the clean-codeword oracle at every SNR. Soft block DF with the matched BCJR is no better than hard block DF: it is worse at 0–8 dB and not resolved from it at 12–20 dB.

The weighted destination metric does not help the MLP read-outs at high SNR. Weighting by $|h_2|^2$ is matched to the second hop only. It lowers the MLP read-outs' BER below 12 dB but raises it from 12 dB for the hard MLP and from 16 dB for both, because a wrong relay symbol received over a strong second hop is then trusted. A destination metric that also models relay errors, and a learned relay given $|h_1|$, were not evaluated. Either way the MLP read-outs remain far behind matched block DF.

Consequences for this section. The published coded tables used an unweighted metric, under which the learned relays appeared ahead of block DF at high SNR. Changing only the metric reverses that ordering at every SNR for the two MLP read-outs, so the published comparisons established an advantage over a mismatched decoding pipeline, not a disadvantage of decoding and re-encoding at the relay. They are kept below as a diagnostic (Section 5.5.2). The other coded studies of this chapter, the error-location diagnostic (Table 5.7), the constraint-length sweep, the reliable-decoding regime (Table 5.8), the equal-throughput, rate-adaptation and readiness-budget studies (Tables 5.10, 5.12 and 5.13) and the coded size sweep, were rerun with the matched metric on their published seeds (30 trials, `CODED_METRIC=matched`), and their

matched results are given first where each appears. The coded-memory study of Section 6.5.1 has no fading, so per-symbol reliability weights do not arise there.

5.5.2 Diagnostic: The Published Unweighted Pipeline

The tables of this subsection and the next are the published comparisons, with unweighted decoder metrics at the relay and the destination and the Mamba-coded relay included. They show what the metric mismatch did; the results of the section are those of Section 5.5.1.

Table 5.5: Coded block-DF vs. uncoded symbol-wise DF and coded-aware learned relays, canonical QPSK/Rayleigh, rate-1/2 $K{=}3$ code. "coded-AF" forwards the noisy codeword unmodified (the relay does not decode), isolating the code's own gain from any relay intelligence. Decoder metrics are unweighted; Table 5.4 repeats the comparison with matched metrics

SNR (dB)	uncoded DF	coded AF	coded DF	MLP-coded	Mamba-coded
0	**0.3337**	0.4902	0.4204	0.4439	0.4505
4	**0.2216**	0.4468	0.2229	0.2974	0.3214
8	0.1205	0.2722	**0.0638**	0.1010	0.1174
12	0.0560	0.0539	**0.0154**	0.0214	0.0226
16	0.0239	0.0099	0.0042	0.0045	**0.0041**
20	0.0098	0.0020	0.0014	0.0011	**0.0010**

Coded-DF is stronger over part of the tested range. At equal E_s/N_0, coded block-DF beats uncoded symbol-wise DF from 8 dB upward, by $5.7\times$ in BER at 16 dB and $7.2\times$ at 20 dB. The rate-1/2 code also reduces information throughput; Section 5.5.4 examines an approximately equal-rate comparison. At 0 and 4 dB, coded AF and coded DF are worse than uncoded DF, with the crossover between 4 and 8 dB for this code, metric and frame.

The larger coded relay does not consistently improve BER. Both learned relays trail coded-DF at 4–8 dB, and the simplified Mamba-S6 has higher measured BER than the 756-parameter MLP there (Table 5.5), while at higher SNR it can be lower.

Paired high-SNR comparisons resolve different orderings. Table 5.6 uses 100 paired trials with the same source bits and channel draws for each relay. At 20 dB both learned relays have lower BER than coded-DF in at least 99 trials, with Wilcoxon $p < 10^{-4}$. At 16 dB Mamba-coded wins 68 trials, whereas MLP-coded wins 23 and loses 77; both paired comparisons give $p < 10^{-4}$. With matched metrics the ordering reverses at both points for the MLP read-outs; the Mamba relay was not rerun (Section 5.5.1).

Table 5.6: Paired re-measurement of the two high-SNR points of Table 5.5, 100 paired trials × 100,000 information bits. "Diff" is the mean per-trial BER difference against coded-DF (negative favors the learned relay); "W/L" counts trials won/lost (one tie, MLP-coded at 20 dB); p is Wilcoxon signed-rank. Decoder metrics are unweighted; with matched metrics the ordering reverses (Table 5.4)

SNR	Relay	BER	Diff vs. coded-DF	W/L	p
16 dB	coded-DF	0.004245	—	—	—
	MLP-coded	0.004437	$+0.000192$	23/77	$< 10^{-4}$
	Mamba-coded	**0.004124**	-0.000121	68/32	$< 10^{-4}$
20 dB	coded-DF	0.001362	—	—	—
	MLP-coded	0.001066	-0.000295	99/0	$< 10^{-4}$
	Mamba-coded	**0.000967**	-0.000395	100/0	$< 10^{-4}$

The metric mismatch is sufficient to produce the ordering. Table 5.7 records fewer relay-output symbol errors for coded-DF but a lower destination repair fraction than for the learned relay, which decode–re-encode forwarding of wrong codewords could explain. The matched rows show why that explanation does not apply once the metric is matched: the matched relay's symbol error rate is 1.6×10^{-5} at 16 dB and 1.0×10^{-6} at 20 dB, so there is almost nothing for re-encoding to lock in.

Table 5.7: Where the errors go, 100 trials × 500 frames. "Relay symbol ER" counts symbols the relay put on the air that differ from the transmitted ones; "repaired" is the fraction of those that do not survive the destination's decoder. The oracle relay forwards the clean codeword and so isolates the hop-2-only floor. Rows marked "matched" use $|h|^2$-weighted relay and destination metrics, and "weighted dest." the $|h_2|^2$ destination metric for the learned relay; those rows use 30 trials (Section 5.5.1)

SNR	Relay	Relay symbol ER	Repaired	Final BER
16 dB	coded-DF	0.00505	15.7%	0.004257
	MLP-coded	0.01065	**57.9%**	0.004487
	oracle	—	—	0.002140
	coded-DF, matched	$\mathbf{1.6 \times 10^{-5}}$	34.8%	$\mathbf{1.0 \times 10^{-5}}$
	MLP-coded, weighted dest.	0.01060	36.2%	0.006768
	oracle, matched	—	—	5.0×10^{-6}
20 dB	coded-DF	0.00182	24.7%	0.001369
	MLP-coded	0.00385	**73.0%**	0.001039
	oracle	—	—	0.000668
	coded-DF, matched	$\mathbf{1.0 \times 10^{-6}}$	32.7%	$\mathbf{6.7 \times 10^{-7}}$
	MLP-coded, weighted dest.	0.00385	37.1%	0.002422
	oracle, matched	—	—	0 errors

Constraint-length sweep. With the unweighted metric, the tested $K \in \{3, 5, 7\}$ generators show higher BER for larger K across 0–8 dB, and rounded means of 0.0014, 0.0015 and 0.0012 at 20 dB (`results/coded_df_experiment.json`). With the matched metric (`results/coded_df_experiment_matched.json`, 30 trials) larger K helps from 4 dB upward, as code theory predicts: at 8 dB the BER is 0.0033, 0.00069 and 6.7×10^{-5} for $K = 3, 5, 7$, and $K = 5$ and $K = 7$ make no errors at 16 and 20 dB in 3×10^6 bits. Only at 0 dB, below the codes' thresholds, does larger K remain worse; the unweighted sweep's ordering reflects the metric, not the codes.

Higher-SNR evaluation. Table 5.8 extends the same short code to higher SNR; it does not approach the capacity-achieving code of Remark 4.2, since at 20 dB destination frame errors occur in 19.98% of coded-DF frames and 10.64% even with an oracle relay. A second learned relay is retrained on 5–30 dB.

Table 5.8: Block-DF into the reliable-decoding regime: the rate-1/2 K=3 code and canonical QP-SK/Rayleigh link of Table 5.5, extended up the SNR axis, 100 trials × 1,000 frames (20,000,000 information bits) per point. "MLP (5/10/15)" uses that table's training recipe; "MLP (5–30)" is the same architecture retrained on 5–30 dB, bracketing all but the 32 dB point, hence a different instance whose 20 dB entry need not reproduce it. Bold marks the best *relay* in each row (both marked where they tie); the oracle forwards the clean codeword and is the single-decode floor they are measured against, not a competitor

SNR (dB)	coded-DF	MLP (5/10/15)	MLP (5–30)	oracle	coded-DF FER
20	0.001378	**0.001064**	0.001092	0.000703	0.1998
24	0.000485	**0.000308**	0.000318	0.000253	0.0804
28	0.000187	**0.000105**	**0.000105**	0.000092	0.0324
32	0.000069	**0.000037**	**0.000037**	0.000034	0.0122

With matched metrics the higher-SNR gap closes. Rerun with matched metrics (`results/coded_reliable_regime_matched.json`, 30 trials of 2×10^5 bits), coded-DF makes one bit error in 6×10^6 bits at 20 dB and none at 24–32 dB, exactly as many as the matched oracle, while both MLP read-outs remain at 1.1×10^{-3} to 3.7×10^{-5} under their better destination metric. Under the unweighted metric of the table, coded-DF's BER-to-oracle ratios across 20–32 dB are 1.96, 1.92, 2.03 and 2.00, against 1.55, 1.26, 1.14 and 1.08 for the retrained MLP, whose 636 frame errors at 32 dB against the oracle's 607 in 100,000 frames are not resolved ($z = +0.8$). That gap is a property of the unweighted metric, not of block-DF.

5.5.3 Soft-Decision Relaying under the Unweighted Metrics (Diagnostic)

Table 5.9 compares hard read-out with soft constellation averages. At the classical relay, BCJR replaces Viterbi and supplies symbol weights under the implemented metric; those weights need not be calibrated under fading. For the MLP, the soft average $\sum_k p_k s_k$ replaces the hard argmax without retraining, but requires the weighted-sum computation. Its hard/soft pair shares the same

weights, isolating the implemented read-out change rather than proving that the scores are exact posteriors. The paired experiment also includes a clean-codeword oracle relay.

Table 5.9: Soft- versus hard-decision relaying, canonical QPSK/Rayleigh, rate-1/2 $K{=}3$, 100 paired trials $\times$ 100,000 information bits. The MLP hard/soft pair shares one set of trained weights and differs only in the read-out. The oracle relay forwards the clean codeword, giving the hop-2-only floor. Both block-DF variants are marked at 8 and 12 dB, where no difference is resolved. Decoder metrics are unweighted; Table 5.4 repeats the comparison with matched metrics

SNR (dB)	hard block-DF	soft block-DF	MLP hard	MLP soft	oracle
0	0.4208	**0.4150**	0.4443	0.4343	0.3022
4	0.2231	**0.2218**	0.2969	0.2761	0.1278
8	**0.0636**	**0.0636**	0.1008	0.0878	0.0327
12	**0.01537**	**0.01537**	0.02142	0.01768	0.00768
16	0.004222	0.004222	0.004403	**0.003633**	0.002105
20	0.001354	0.002348[†]	0.001056	**0.000912**	0.000682

[†]Nominal assumed noise variance, the same run as the rest of the table and as Figure 5.3; it is miscalibrated at this SNR (see text below). With the assumed variance doubled, a separate sensitivity sweep gives 0.001376, not resolved from hard block-DF's 0.001375 in that sweep. Neither variant is the matched metric of Section 5.5.1.

Soft MLP read-out improves the sampled points (Figure 5.3). Its BER is lower than the hard read-out at every measured SNR on identical weights. Here the term posterior mean denotes the softmax-weighted constellation average; calibration was not tested. At 16 dB the hard MLP loses to block-DF in 80 paired trials, wins 19 and ties one, whereas the soft MLP wins all 100. At 20 dB the soft MLP's measured BER is within 0.00023 of the clean-codeword oracle. The comparison does not identify the physical mechanism or establish optimality of this read-out for destination BER.

Soft block-DF does not rescue the implemented block-DF baseline. The relay is given a nominal noise variance, although post-equalization Rayleigh noise has variance $1/(2\gamma|h|^2)$ per real axis. Its BCJR metrics are therefore mismatched and overconfident on deep-fade symbols. Uniformly inflating the assumed variance removes the 20-dB gap, but this sensitivity sweep does not supply the required per-symbol reliability weights and should not be described as matched calibration. The same limitation applies to unweighted Viterbi distances: multiplying every branch metric by one constant leaves its decisions unchanged, whereas omitted symbol-dependent weights can change the selected path.

Soft and hard block-DF have similar reported BER at 16 dB under these decoder metrics. At 20 dB the nominal-variance soft block-DF is worse, and with the assumed variance doubled the two are not resolved (footnote to Table 5.9). This does not isolate hard quantization as the cause of the gap, nor prove a general disadvantage of decoding at a relay.

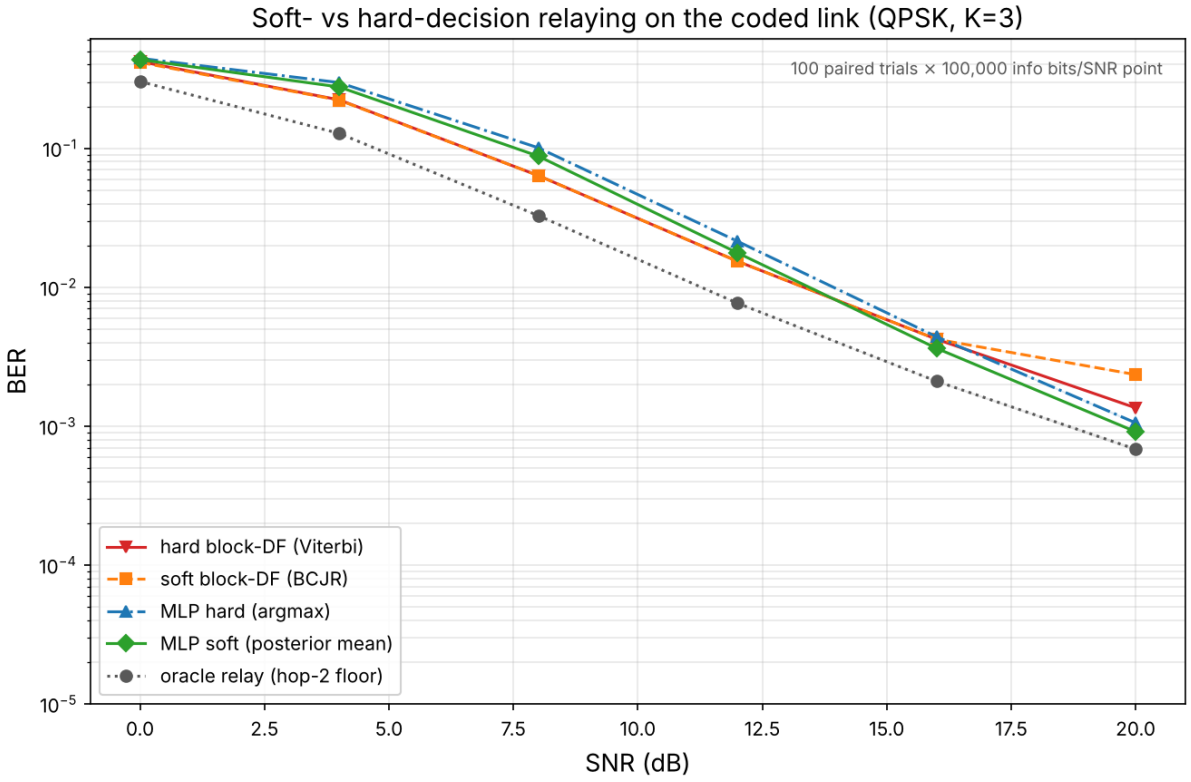


Figure 5.3: Soft- and hard-decision coded relaying under the implemented (unweighted) metrics, with 100 paired trials; Table 5.4 gives the separate 30-trial matched-metric comparison. The MLP soft constellation average has lower measured BER than its hard read-out at every sampled SNR. The soft block-DF curve uses the nominal noise variance, so its 20 dB point is the miscalibrated value discussed in the text. Neither the curves nor the repair counts isolate a causal explanation for the high-SNR ordering. The oracle forwards the clean codeword

Under the unweighted metrics the coded study therefore shows a configuration-specific high-SNR benefit for the learned relays and a benefit from soft MLP read-out; Section 5.5.1 shows that the first does not survive matched decoding.

5.5.4 Throughput and Latency: What the BER Tables Leave Out

Every table above compares relays at equal E_s/N_0. That is the correct axis for asking which relay detects better, and the wrong one for asking which relay a link should use, because it silently ignores two costs that differ by orders of magnitude across these designs.

Throughput: the coded comparison was not like-for-like. A rate-1/2 code carries half the information per channel use. Concretely, the coded QPSK link spends 202 symbols to deliver 200 information bits (0.99 bits/symbol) where uncoded QPSK spends 100 (2.00 bits/symbol). The "5.7× better at 16 dB" quoted above is therefore bought with half the data rate, and a link designer given that trade would not read it as a 5.7× improvement at all. The honest comparison holds spectral efficiency fixed, and the measurements already collected supply it: rate-1/2 *16-QAM* delivers $4 \times \frac{1}{2} \times \frac{200}{202} \approx 1.98$ information bits per symbol, matching uncoded QPSK's 2.00. Table 5.10 places those two side by side.

Table 5.10: Coded versus uncoded at equal spectral efficiency (≈2 information bits per channel symbol): uncoded QPSK against rate-1/2 K=3 16-QAM, both on the canonical Rayleigh channel. Recomputed from the data of Table 5.5 and `results/coded_df_experiment.json`; the last two columns use the matched-metric 16-QAM decoder (`results/coded_df_experiment_matched.json`, 30 trials), against the same uncoded reference, which the metric cannot change

SNR (dB)	uncoded QPSK	rate-1/2 16-QAM	Coding gain	16-QAM, matched	Gain, matched
0	**0.3337**	0.4680	0.71× (worse)	0.4496	0.74×
4	**0.2216**	0.4203	0.53× (worse)	0.3325	0.67×
8	**0.1205**	0.2726	0.44× (worse)	0.1072	1.12×
12	**0.0560**	0.1001	0.56× (worse)	0.01539	3.64×
16	**0.0239**	0.0273	0.88× (worse)	0.001954	12.3×
20	0.0098	0.0076	1.29× (better)	0.00025	39×

With the matched 16-QAM decoder the coded link wins at equal throughput from 8 dB upward: by 1.12× at 8 dB, 3.6× at 12 dB and 12.3× at 16 dB. Coding is still reliability bought with bandwidth, but on this channel the purchase pays once the decoder uses the fading reliability. Under the unweighted metric (diagnostic) the rate-1/2 code did not pay for itself below 20 dB, and gained only 1.29× there, not the 5.7–7.2× of the equal-E_s/N_0 reading.

Relay readiness: block relays buffer substantially more. The relays differ in how much they must accumulate before producing their first output. Symbol-wise AF and DF can process a current symbol — a property of the ideal symbol-wise rules with a fixed or statistically known gain; the implemented AF estimates its gain from the empirical power of the whole simulated

block (Section 4.1.3), which is block-adaptive and not symbol-causal, so the zero in Table 5.11 describes the algorithmic rule, not that implementation detail; the symmetric 21-sample learned relay needs $w = 10$ future symbols; and block-DF must receive a 202-symbol frame before decoding. The look-ahead column of Table 5.11 counts only this detector look-ahead, and two other components matter. First, power normalization: the hard MLP and hard block-DF forward exact constellation points and apply no gain, but the soft MLP and soft block-DF, like AF, scale their outputs by the power of the whole simulated block, so their forwarded symbols are not final until that block has been observed, whatever their look-ahead. An implementable streaming rule (a fixed gain, or a running power estimate) was not evaluated. Second, the 202-symbol figure belongs to the implemented full-frame decoder; a convolutional decoder with finite traceback can release decisions after a fixed delay without waiting for the frame, so it is not a requirement of every coded relay. These are relay-processing readiness delays, not complete end-to-end latency: in the half-duplex protocol every strategy must still spend channel uses on both source–relay and relay–destination transmissions.

Table 5.11: Detector look-ahead and measured compute cost. "Look-ahead" is the number of symbols the relay's detector must receive before it can decide the current symbol: the $w = 10$ future symbols of the 21-sample learned relays, or the whole 202-symbol frame that the implemented block-DF decoder needs before its first decision. It omits normalization buffering: the soft MLP and soft block DF, like the implemented AF, scale their outputs by the power of the whole simulated block, so their outputs are not final until that block ends. It also excludes the two half-duplex transmission phases and is not end-to-end latency. The AF/DF zero assumes an ideal fixed or statistically known gain; the implemented AF normalizes by block-empirical power and is not symbol-causal. Compute is the median of 7 runs over 49,894 symbols after a discarded warm-up, single-threaded CPU. The identical benchmark was run on two different machines, both reported here: neither is more correct than the other, and the pair shows that absolute figures and ratios involving sub-microsecond MLP timings are machine-dependent (`results/coded_latency_compute_machines.json`)

		μs/symbol		
Relay	Look-ahead (symbols)	Machine A	Machine B	Relative to Viterbi
AF / symbol-wise DF	**0**	—	—	—
MLP hard (756p)	10	0.30	0.34	50–63× faster
MLP soft (756p)	10	0.22	0.35	61–68× faster
hard block-DF (Viterbi)	202	15.07	21.23	1×
soft block-DF (BCJR)	202	29.30	41.21	1.94× slower

The two-machine timings are implementation measurements. BCJR takes approximately 1.94 times the Viterbi time on each machine (1.944 and 1.941, not equal to three decimals). Absolute decoder times differ by about 40%. The MLP timings are sub-microsecond, so the last column's ranges are coarse; operation counts and algorithmic readiness are reported separately.

Table 5.10 limits the coding benefit at approximately equal spectral efficiency for this grid, while Table 5.11 documents different detector look-ahead and reference-program timings. These are

separate comparisons: measured timings are not arithmetic counts, and neither yields a general hardware or deployment conclusion.

5.5.5 Adaptive Modulation and Coding: Choosing the Rate, Not Just the Relay

Every comparison above, including the equal-throughput one, holds the modulation-and-coding scheme (MCS) fixed and asks which relay achieves the lower BER. A real link does not hold the MCS fixed; it adapts. The question a link actually faces is not "which relay wins at rate R" but "what is the highest *useful* rate this link can sustain at this SNR", and answering it requires more than one code rate to choose from. The mother code of Section 5.5 is extended here with the standard punctured rates $2/3$ and $3/4$ (deleting a periodic subset of the rate-1/2 output, re-inserted as soft zeros at the decoder, so the same Viterbi decoder runs unmodified at every rate), and evaluated via bit-interleaved coded modulation across the full grid $\{\text{QPSK}, \text{16-QAM}\} \times \{\text{uncoded}, 1/2, 2/3, 3/4\}$.

The objective is an idealized one-shot *goodput*, $G = \eta\,(1 - \text{FER})$ (Eq. 2.30), assuming perfect frame-error detection and discarding failed frames without accounting for feedback or retransmission overhead. G and η are both measured per channel symbol transmitted on the active hop, matching the ergodic-capacity reference C introduced below; because the relay is half-duplex, expressing goodput instead as a fraction of the total two-slot cycle (source-to-relay plus relay-to-destination) would halve every figure in this section without changing which MCS or relay strategy is preferred, since that factor of $\frac{1}{2}$ applies identically to both. Maximizing G over the MCS grid at each SNR traces the link-adaptation envelope, compared here for two implemented classical forwarding strategies: *block-DF*, which decodes the code at the relay and re-encodes its message decision before transmitting, against *denoise-only*, which hard-decides each bit and re-modulates without ever invoking the code, retaining the original transmitted code constraints for destination decoding and requiring no training and no code knowledge at the relay.

Table 5.12: Link-adaptation envelope: the MCS maximizing goodput at each SNR, block-DF versus denoise-only relaying, canonical Rayleigh channel, 100 trials × 200 frames × 200 information bits per point. Columns are SNR in dB, the goodput-maximizing MCS and its goodput G for each relay, and C, the ergodic Rayleigh (Shannon) capacity at that SNR, $C = \log_2(e)\, e^{1/\gamma} E_1(1/\gamma)$ – an unconstrained-input upper bound no achievable goodput may exceed, included as a sanity ceiling rather than a target: the gap to it here is dominated by the finite constellation and coding loss, not by relay strategy. The capacity column is a *single-link* reference evaluated per active-hop channel symbol; it is not a capacity bound for the half-duplex two-hop relay system studied here, whose own bound would additionally account for the two-slot cycle and the relay constraint. The last column gives block-DF's best MCS and goodput with the matched decoder metric (`results/coded_matched_envelopes.json`, 30 trials)

SNR	MCS (block-DF)	G	MCS (denoise)	G	C	Block-DF, matched
8	QPSK $\frac{1}{2}$	**0.0039**	QPSK $\frac{1}{2}$	0.0003	2.40	QPSK $\frac{1}{2}$, 0.6625
12	QPSK $\frac{1}{2}$	**0.1708**	QPSK $\frac{1}{2}$	0.1342	3.45	QPSK $\frac{2}{3}$, 0.9914

SNR	MCS (block-DF)	G	MCS (denoise)	G	C	Block-DF, matched
16	QPSK $\frac{1}{2}$	0.5406	QPSK $\frac{1}{2}$	**0.6046**	4.63	16-QAM $\frac{1}{2}$, 1.5158
20	QPSK $\frac{3}{4}$	0.9545	QPSK $\frac{3}{4}$	**0.9635**	5.88	16-QAM $\frac{1}{2}$, 1.8997
24	16-QAM $\frac{2}{3}$	**1.5790**	16-QAM $\frac{2}{3}$	1.4739	7.17	16-QAM $\frac{3}{4}$, 2.3536
28	16-QAM $\frac{3}{4}$	**2.3003**	16-QAM $\frac{3}{4}$	2.2625	8.48	16-QAM $\frac{3}{4}$, 2.7254
32	16-QAM $\frac{3}{4}$	**2.6807**	16-QAM $\frac{3}{4}$	2.6726	9.80	16-QAM $\frac{3}{4}$, 2.8642
36	16-QAM uncoded	3.3920	16-QAM uncoded	3.3920	11.13	16-QAM uncoded, 3.4020
40	16-QAM uncoded	3.7336	16-QAM uncoded	3.7336	12.46	16-QAM uncoded, 3.7327

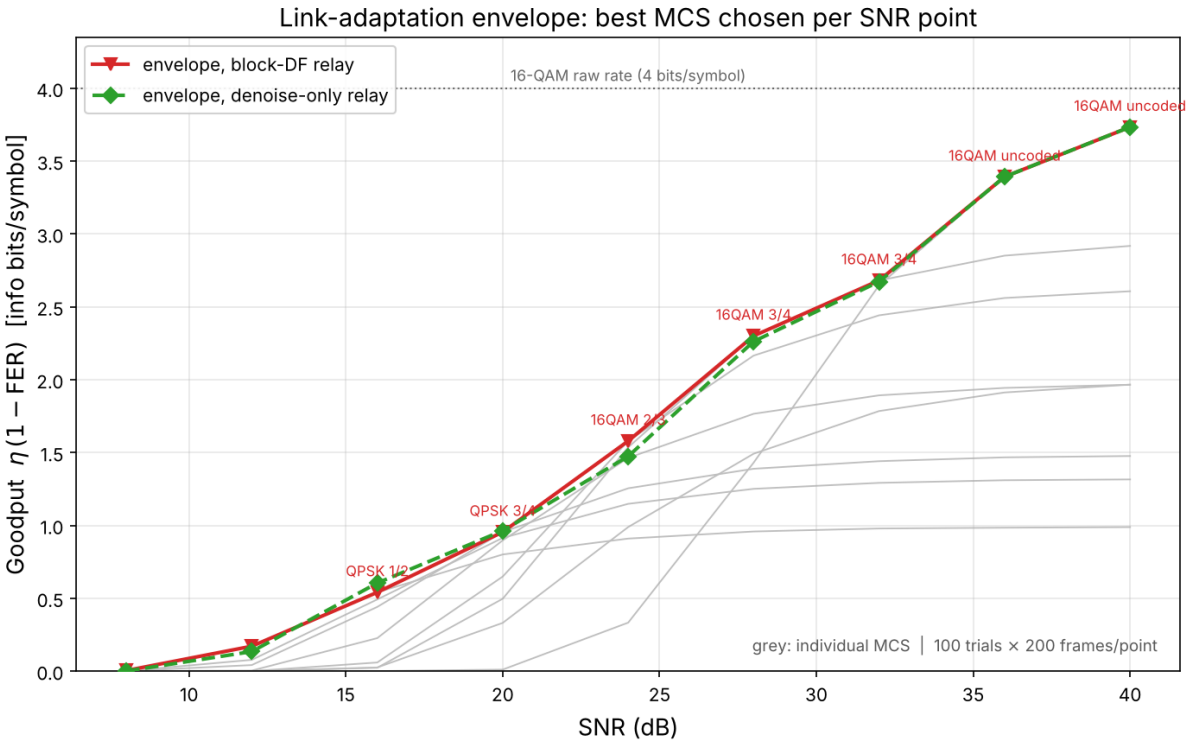


Figure 5.4: The envelope of Table 5.12 (colored) against every individual fixed MCS on the block-DF relay (grey), which is the visual case for adaptation: the envelope tracks the upper hull of the grey curves rather than any single one of them, switching MCS as SNR crosses each rate's operating threshold. The curves use the unweighted decoder metric, under which the two relay strategies' envelopes are visually almost the same curve; with the matched metric, block-DF's envelope is far higher (last column of Table 5.12)

Two observations follow within this finite modulation-and-coding grid.

The envelope is a real staircase, and its top step is "turn the code off". Both relay strategies climb QPSK 1/2 → QPSK 3/4 → 16-QAM 2/3 → 16-QAM 3/4 → *16-QAM uncoded* as SNR rises from 8 to 40 dB, each switch occurring once the next rate's threshold has cleared. The optimum at high SNR is no code at all: once the channel is good enough, coding redundancy is pure overhead, and the highest-goodput choice is the raw 4 bits/symbol of 16-QAM with zero coding loss. This is the same lesson as the equal-throughput correction above, now reached by direct optimization over the grid rather than by hand-picking one comparison. Table 5.12's C column confirms the envelope never approaches the information-theoretic ceiling: at 40 dB the

achieved 3.73 sits at 30% of the 12.46-bit Shannon capacity there, a gap set by the fixed 16-QAM constellation (itself capped at 4 bits/symbol) and residual FER, not by anything a smarter relay strategy could close.

With the matched metric, decoding at the relay matters. Block-DF's matched envelope (last column of Table 5.12) is far above both published envelopes at every SNR where a code is used: 0.66 against 0.0003 information bits/symbol at 8 dB, 0.99 against 0.13 at 12 dB, 1.52 against 0.60 at 16 dB, 1.90 against 0.96 at 20 dB and 2.35 against 1.47 at 24 dB, and it reaches each higher-rate MCS at a lower SNR. The denoise relay is taken here with the unweighted destination metric, which serves it better than the $|h_2|^2$ weight because its hard decisions carry relay errors that weight does not model. Above 32 dB both strategies select 16-QAM uncoded. Neither comparison is neural versus classical.

With the unweighted metric (diagnostic), relay strategy barely mattered. The two published envelopes in Table 5.12 differ by at most 0.105 information bits/symbol (at 24 dB, under 7% of the envelope value) and are identical from 36 dB, where both select 16-QAM uncoded, whereas the MCS choice moves goodput by 1.25 at 24 dB and by more than 1.7 from 32 dB. That finding is a property of the unweighted decoder.

5.5.6 Goodput Under an Idealized Buffer-Readiness Budget

The coded relay of Section 5.5 decodes a whole frame before forwarding, and every comparison so far has left the frame length free. A link with a deadline cannot: the deadline bounds the blocklength and with it the redundancy the relay has to spend. The achievable rate once blocklength is fixed is a standard question — the finite-blocklength capacity of Polyanskiy, Poor and Verdú [41] and the delay-limited capacity of Hanly and Tse [42] for fading channels.

The following calculation uses an idealized *buffer-readiness budget*, not physical round-trip latency. It counts one destination frame buffer for both strategies and an additional relay frame buffer for block-DF, while assigning only the learned relay's look-ahead to denoise-only relaying. It omits the channel-use duration of the two half-duplex transmission phases, which both strategies incur. Re-deriving the goodput envelope of Table 5.12 under this accounting illustrates the effect of relay buffering, but does not establish performance under a complete end-to-end deadline.

Table 5.13: Link-adaptation envelope re-derived under the idealized 150-symbol buffer-readiness budget, same layout and data source as Table 5.12, including the matched block-DF column

SNR (dB)	MCS, block DF	G	MCS, denoise	G	Block DF, matched
8	QPSK uncoded	0.0000	QPSK uncoded	0.0000	QPSK uncoded, 0.0000
12	QPSK uncoded	0.0000	QPSK 3/4	**0.0105**	QPSK uncoded, 0.0000
16	QPSK uncoded	0.0251	QPSK 3/4	**0.3582**	16-QAM 3/4, 0.2637
20	16-QAM 3/4	0.4966	QPSK 3/4	**0.9635**	16-QAM 3/4, 1.4538

SNR (dB)	MCS, block DF	G	MCS, denoise	G	Block DF, matched
24	16-QAM 3/4	**1.5357**	16-QAM 2/3	1.4739	16-QAM 3/4, 2.3536
28	16-QAM 3/4	**2.3003**	16-QAM 3/4	2.2625	16-QAM 3/4, 2.7254
32	16-QAM 3/4	**2.6807**	16-QAM 3/4	2.6726	16-QAM 3/4, 2.8642
36	16-QAM uncoded	3.3920	16-QAM uncoded	3.3920	16-QAM uncoded, 3.4020
40	16-QAM uncoded	3.7336	16-QAM uncoded	3.7336	16-QAM uncoded, 3.7327

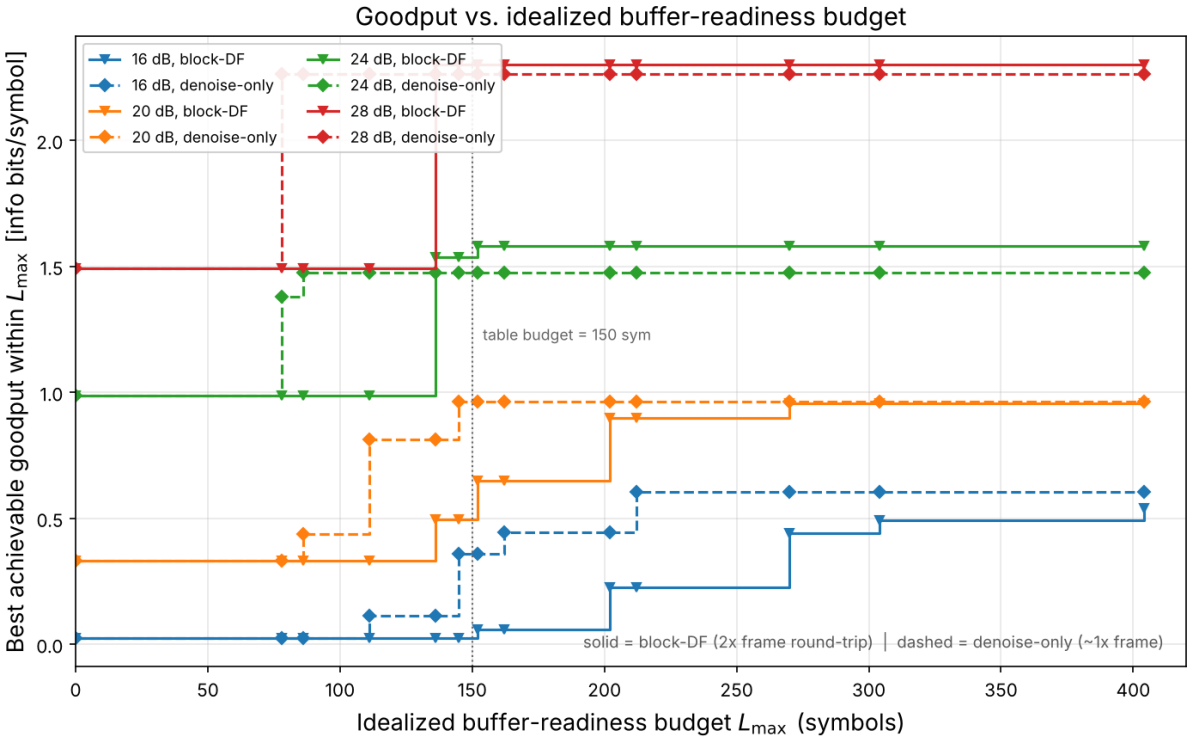


Figure 5.5: Best achievable goodput as a function of the idealized buffer-readiness budget $L_{\max}$, at four SNRs (16, 20, 24, 28 dB). Each curve is a step function under this accounting. The omitted two-hop transmission time prevents interpreting this axis as end-to-end latency. The curves use the unweighted decoder metric; the matched block-DF values at a 150-symbol budget are in the last column of Table 5.13

Under this idealized readiness budget, block-DF can be locked out of coding. At $L_{\max} = 150$ symbols (Table 5.13), block-DF's cheapest coded option barely clears the budget while its next-cheapest does not. With the matched metric (last column), that option, 16-QAM 3/4, carries 0.2637 information bits/symbol at 16 dB and 1.4538 at 20 dB, so denoise-only leads by $1.36\times$ at 16 dB and block-DF leads by $1.51\times$ at 20 dB and at every higher SNR where a code is used. Under the unweighted metric (diagnostic) the 16-dB entries differed by $14.3\times$ (0.3582 against 0.0251) and the 20-dB entries by $1.9\times$. These figures illustrate buffering sensitivity only; they are not end-to-end latency results.

Tighten the budget further and the reversal spreads to SNRs where block-DF led unconstrained. At $L_{\max} = 100$ symbols, block-DF has *no* coded option left at any of these four SNRs – every one of its coded MCS choices exceeds the budget – so it is pinned to uncoded transmission

throughout, while the denoise-only relay still reaches 16-QAM 2/3 or 3/4 at 20, 24 and 28 dB. At 24 and 28 dB, this is a direct reversal: Table 5.12's unconstrained envelope has block-DF *ahead* at both points, but under the 100-symbol budget denoise-only leads by $1.46\times$ and $1.54\times$ respectively, because the MCS that gave block-DF its unconstrained edge is no longer reachable in time. This structural lock-out does not depend on the decoder metric: with matched metrics block-DF is still pinned to uncoded transmission at 100 symbols, and denoise-only leads at 24 and 28 dB by $1.46\times$ and $1.52\times$.

The read-out comparison of Section 5.5.3 does not establish a decoding-location mechanism. The finite-grid results show sensitivity to rate selection and idealized buffering. A hard-deadline deployment recommendation requires a slot-level model of both half-duplex transmissions, processing time and reliability; that analysis is not performed here.

Chapter 6

Learned Relaying under Unknown and Mismatched Channels

This chapter tests learned relaying on selected channels with memory and model mismatch, following the matched memoryless controls of Chapter 5. The main comparison is with the implemented AF and zero-threshold DF baselines; matched sequence detection provides a stronger reference where the required channel information is supplied.

On the fixed three-tap BPSK channel, a 170-parameter windowed MLP substantially reduces destination BER relative to AF and zero-threshold DF. Those two mappings approach a 0.25 high-SNR floor for the selected taps; the MLP's fixed-budget measurements continue to fall through 16 dB. This does not establish a limitation of all memoryless detectors, nor generalization to an unknown impairment family. The MLP is trained on this fixed filter. Later random-channel studies test unseen draws from a family represented during training.

No statistically resolved advantage over matched Viterbi MLSE is demonstrated in the BPSK comparison. Exact taps at inference and finite training samples are different information resources. MLSE minimizes sequence error, not BER; the QPSK BCJR control in Section 6.2.3 measures bit errors at the relay output only. End-to-end BCJR performance is not measured. Cost comparisons are architecture- and implementation-specific: the small BPSK trellis is cheaper in nominal arithmetic than the MLP.

The following studies locate useful operating regimes for the tested learned relays, without claiming an exhaustive deployment map. The channel models, redraw intervals, pilots, window boundaries, DBPSK forwarding and the definition of the single-number SNR penalties are specified in Appendix 9.6.

A conceptual ladder of assumptions. Table 6.1 organizes separate experiments by channel memory and available channel information. Modulation, fading, training, and evaluation budgets also differ between studies. The ladder is therefore an explanatory organization, not a

controlled ablation that changes only one assumption at a time.

Table 6.1: Four groups of experiments and representative measured outcomes. The rows organize the argument; they do not hold every experimental condition fixed. Classical entries identify the strongest implemented comparator in each group

Layer	Assumptions	Strongest classical	Outcome
1	Memoryless, known channel, perfect CSI (Chapter 5)	Symbol-wise DF, zero trainable parameters	**Classical wins.** On the canonical QPSK/Rayleigh setup itself (Table 5.2), DF 0.1218 against the MLP's 0.1229 at 8 dB: the learned relay only matches DF, at 169 parameters against none (H2).
2	+ channel memory (3-tap ISI), CSI still perfect (Section 6.2)	Viterbi MLSE with exact taps	**Split.** The memoryless relays break outright, DF rising from 0.1802 at 8 dB to 0.2457 at 20 dB. The MLP restores the link (0.0065 at 8 dB), but genie-CSI Viterbi is ahead of it at every point where both are resolved: 0.001354 against that 0.0065 at 8 dB, and nominally below 3.0×10^{-8} at 16 dB against the MLP's 1.20×10^{-7}. The advantage widens with SNR rather than holding at a fixed margin — in required transmit power, $+0.26$ dB at a 10^{-1} target, $+1.18$ at 10^{-2}, $+1.88$ at 10^{-3} and $+2.53$ at 10^{-4} (Table 6.3). The fixed filter is supplied exactly to the trellis and represented through finite training samples for the MLP; these information resources are not equivalent.

Layer	Assumptions	Strongest classical	Outcome
3	+ CSI uncertainty: a finite pilot budget (Section 6.3)	Pilot-aided LS estimate then MLSE	**Crossover.** At 10 dB, Viterbi-est holds 0.0283 on 200 pilots and 0.0335 on 20, both beating the pilot-free MLP's 0.0486; by 10 pilots it has degraded to 0.0545, not resolved from the MLP, which leads at 5 pilots. The classical advantage is real but purchased with pilots.
4	+ unseen realization from the trained family, redrawn per block (Sections 6.2.4, 6.4)	CMA blind equalization	**Learned relay leads to 12 dB.** At 8 dB the MLP reaches 0.0955 against CMA's 0.1182; the lead is resolved over 50 paired draws from 0 to 12 dB, not from 14 dB. Decision-directed blind MLSE is unstable, non-monotonic and worse at 20 dB (0.0498) than at 16 dB (0.0374). The BER margin is modest; the margin in per-block cost is not.

The comparator changes from a symbol-wise relay in the memoryless control to a sequence detector in the ISI study. Viterbi's sequence-ML guarantee applies to its specified channel likelihood, not to a mismatched estimated model. The learned relays instead use fixed weights trained on the specified impairment distribution.

The tested baselines are not exhaustive. Pilot-based and blind alternatives trade estimation effort and overhead against payload BER. The learned relay requires no test-time pilot estimate or online adaptation, but its training distribution remains part of its channel knowledge. Generalization outside that family is not established.

Relationship to the canonical model. The canonical setup of this thesis (Chapter 5) assumes a memoryless two-hop relay channel with white noise, perfect receiver-side CSI, and uncoded transmission. The present chapter deliberately departs from those assumptions by introducing impairments that are not part of the canonical model, including channel memory (3-tap intersymbol interference), amplifier nonlinearity, unknown phase, and channel-model mismatch. Consequently, sequence-detection and equalization techniques such as Viterbi MLSE, pilot-aided channel estimation, and blind equalization become relevant only in this chapter. The goal is not to replace the canonical study, but to evaluate whether the conclusions established under

matched memoryless conditions remain valid when the underlying channel model no longer represents the true channel.

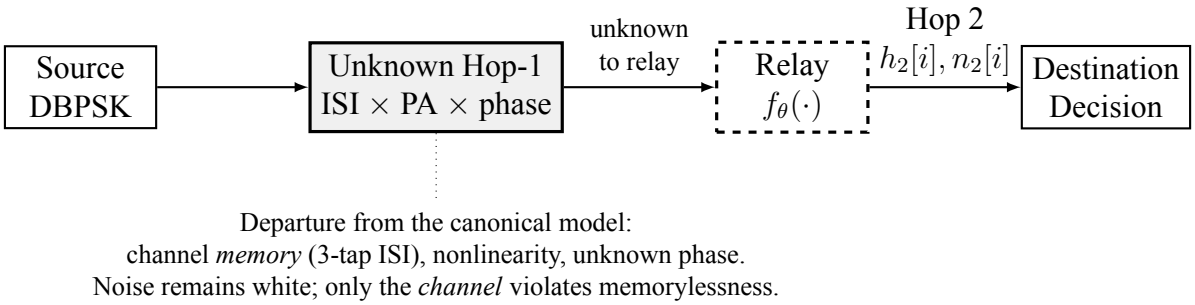


Figure 6.1: Schematic of the composite impairment study. Hop 1 includes memory, amplifier nonlinearity, and block phase; Hop 2 is compensated Rayleigh fading with AWGN. The learned relay is trained on the impairment family, while the pilot-aided classical baseline estimates a linear channel. Individual studies use the configurations stated in their methods

6.1 Layer 1 — Matched Memoryless Channel: The Control

The matched memoryless comparison is reported in Sections 5.2 and 5.4. It provides a reference setting, not a causal isolation of the reasons for performance differences in later, differently configured studies.

6.2 Layer 2 — Channel Memory with Perfect CSI: Classical Failure and Learned Mitigation

Goal: Test whether the learned relay's value changes qualitatively when the channel violates the assumptions under which AF and DF are (near-)optimal: do the *memoryless* classical relays fail on an unknown channel while the minimal 170-parameter MLP mitigates by learning it from data (H5)?

Unlike the canonical experiments, where each transmitted symbol can be processed independently, the 3-tap ISI channel couples adjacent symbols in time. The sufficient statistic for detection is therefore no longer a single received sample, making sequence detection the relevant classical benchmark. To make the classical side as strong as possible, the study benchmarks against the optimal sequence detector: Viterbi MLSE applied to the 3-tap FIR/ISI channel [32] with genie CSI, and a practical variant with a 200-pilot least-squares estimate.

MLSE is introduced here solely because the channel now contains memory. In the canonical memoryless setting of Chapter 5, per-symbol detection is sufficient and sequence detection provides no additional benefit. The need for MLSE therefore arises from the deliberate introduction of ISI in this chapter rather than from the system model studied elsewhere in the thesis.

Relation to data-driven detection. Learning channel-dependent detector computations is established prior work: ViterbiNet [24] combines learned likelihoods with a Viterbi recursion. The present contribution is a comparison of small feedforward relays and the specified classical baselines under the particular channel, pilot, and computation budgets reported here.

The flat-channel controls, pilot-budget sweep, and blind comparison delimit that empirical claim. They do not establish that prior work lacks such boundary studies, or that three taps become unidentifiable at five pilots. The pilot results concern estimation quality in the tested finite-sample systems.

Trials.

- **Topology:** SISO (both hops)
- **Modulation:** BPSK (DBPSK for the unknown-phase control and the composite cascade)
- **Hop-1 channel (not given to AF, DF or the MLP; genie Viterbi receives its taps); Hop-2 (equal SNR per hop):**
  - 3-tap ISI, $h = [1.0, 0.7, 0.5]/\|h\|$ (Hop 2: AWGN)
  - Flat unknown channels (phase, gain, asymmetry; Section 6.2.1) (Hop 2: Rayleigh)
  - Composite cascade of ISI, amplifier nonlinearity, and unknown phase (Section 6.2.4) (Hop 2: Rayleigh)
- **Relays:**
  - AF
  - Symbol-wise DF
  - MLP (window 11, one hidden layer of 13 tanh units, 170 parameters; trained at SNR $\in \{5, 10, 15\}$ dB with MSE loss, as in the canonical study, but not with its full protocol: tanh rather than ReLU hidden units, Adam at learning rate 3×10^{-3} for 25 epochs on 120,000 samples, against the canonical MLP's Adam at 0.01 for 100 epochs on 25,000)
  - For ISI setups only: Viterbi MLSE applied to the 3-tap FIR/ISI channel, evaluated with either genie CSI or a 3-tap least-squares channel estimate obtained from 200 pilot symbols
- **Evaluation:** 10 trials $\times$ 100,000 bits per SNR point, SNR range 0–20 dB. To rule out the influence of the particular random initialization and training data used to fit the MLP, the network is retrained independently with three different random seeds, giving 30 BER samples per cell (3 training instances $\times$ 10 inference trials). The mean is their average. The confidence interval is *not*: the ten trials inside one seed share a trained network, so they resample Monte Carlo noise but not training variance and are not 30 independent

draws from the population the claim concerns. Intervals are therefore computed hierarchically — average within each seed, then a Student-t interval on the three seed means, $t_{2,0.975} = 4.303$ rather than 1.96, deliberately wide because three points say little about a spread. Pooling all 30 as i.i.d. divides by $\sqrt{30}$ where the effective sample size is nearer 3; on the MLP rows here that understates the interval by a factor of about seven to eight (at 8 dB, $\pm 1.3 \times 10^{-3}$ against a pooled $\pm 1.8 \times 10^{-4}$). AF and DF are barely affected, as expected, since neither has a trained network to vary across seeds.

The ISI channel admits a sharp analytic prediction, and it is not confined to the asymptote. With BPSK symbols i.i.d. and equiprobable, the relay observes $y[n] = h_0 x[n] + h_1 x[n-1] + h_2 x[n-2] + v[n]$ and a memoryless slicer returns $\mathrm{sign}(y[n])$. Conditioned on $x[n] = +1$ the two interfering symbols take one of $2^2 = 4$ equiprobable sign patterns, each fixing an effective amplitude $A(s_1, s_2) = h_0 + s_1 h_1 + s_2 h_2$, so

$$P_e(\gamma) \;=\; \tfrac{1}{4} \sum_{s_1, s_2 \in \{\pm 1\}} \Phi\!\left(\frac{-A(s_1, s_2)}{\sigma} \right), \qquad \sigma^2 = \frac{1}{2\gamma}, \quad \gamma = 10^{\mathrm{SNR_{dB}}/10}. \tag{6.1}$$

For $h = [1.0, 0.7, 0.5]$ normalized the four amplitudes are 1.6678, 0.9097, 0.6065 and -0.1516. As $\sigma \to 0$ each term tends to 0 where $A > 0$ and to 1 where $A < 0$, so the limit is the fraction of interference patterns that *flip the symbol sign* — here exactly one of four, giving the floor of 0.25, approached from below.

AF reaches the same floor, but this needs its own argument rather than an appeal to inheritance, since AF performs no hard decision. The relay forwards $g(A + n_1)$ with g the power-normalization gain and the destination slices $g(A + n_1) + n_2$; a positive scale cannot change a sign, so the effective statistic is $A + n_1 + n_2/g$. That is the same four-amplitude structure with an inflated variance $\sigma^2(1 + g^{-2})$, $g^2 = 1/(\mathbb{E}[A^2] + \sigma^2)$, so the $\sigma \to 0$ limit is again the fraction of sign-flipping patterns. The floor belongs to the four amplitudes, not to the decision rule. The two relays approach it differently, and AF is the better of the two at high SNR *because* of its extra noise: on the one pattern whose amplitude is negative, DF errs with probability tending to one, while AF's additional noise still occasionally rescues the decision.

Both closed forms predict the measured curves across the sweep, not the asymptote alone. The measured rows below come from the three-seed run (ISI on Hop 1, AWGN on Hop 2) behind the 8 dB column of Table 6.3 (its 12 and 16 dB AF and DF cells are fixed-budget reruns), read at six of its eleven SNR points (`isi_slicer_floor.py` for the closed forms, `e6_sim_ported.py` for the measurements):

SNR (dB)	6	8	12	16	18	20
DF closed form, Eq. (6.1)	0.1786	0.1803	0.2009	0.2280	0.2389	0.2460
DF measured	0.1802	0.1802	0.2009	0.2296	0.2391	0.2457
AF closed form	0.1939	0.1814	0.1835	0.2075	0.2213	0.2338
AF measured	0.1943	0.1813	0.1834	0.2067	0.2213	0.2349

The closed forms reproduce the broad high-SNR trend and the different minima of AF and DF. The DF expression is a relay-slicer calculation and omits second-hop errors; the AF expression includes both hops with an ensemble power-normalization approximation. Differences of order 10^{-3} in the displayed values preclude claiming agreement to three or four decimal places. This analytic check does not validate the stopping rule or confidence coverage of the historical rare-event simulations.

The 0.25 limit is specific to the chosen taps and zero-threshold rule. In general, average $\mathbf{1}\{A < 0\} + \frac{1}{2}\mathbf{1}\{A = 0\}$ over the four equiprobable interference patterns; zero-amplitude ties contribute one half. If $h_0 > |h_1| + |h_2|$, this floor is zero. For the selected taps, the noiseless observations conditioned on the current bit form disjoint sets, even though one amplitude has the opposite sign. A non-monotonic symbol-wise MAP rule can exploit those sets, and it was measured under the matched protocol (`e6_scalar_map.py`), told the taps and noise variance as genie-CSI Viterbi is but given no temporal context. At 6 dB and below its relay-output errors are identical to zero-threshold DF's. Above that it removes the floor: its destination BER is 0.170 at 8 dB, 0.099 at 12 dB, 0.033 at 16 dB and 0.0043 at 20 dB, while zero-threshold DF rises to 0.246; its relay-output error matches the numerically integrated Bayes error to within 1% at every SNR. It nonetheless stays far above the windowed MLPs of the same run (Table 6.3 uses other checkpoints), at 0.099 against 1.09×10^{-4} at 12 dB. A better scalar decision rule therefore accounts for the absence of a high-SNR floor, but not for the learned relay's practical gain, which comes from the temporal context its window spans. The result demonstrates a benefit over the evaluated memoryless mappings, not an impossibility theorem for all memoryless detectors.

6.2.1 Control: Flat (Memoryless) Unknown Channels

Before attributing any classical failure to "unknownness," it must be separated from *memory*. This subsection isolates unknown channels that are *flat* (memoryless): the receiver chain does not know the channel parameters, but the channel introduces no intersymbol coupling. Three physically motivated cases are tested, each with the parameter redrawn per block and unknown to all relays: (F1) an unknown carrier phase $\theta \sim \mathcal{U}[0, 2\pi)$ (oscillator/synchronization offset), with a DBPSK source and conventional differential detection as the classical relay; (F2) an unknown positive gain $g \sim \mathcal{U}[0.3, 2.0]$ (path loss / AGC error), coherent BPSK; and (F3) an unknown per-branch amplitude asymmetry $a_+ \neq a_-$ (per-symbol saturating-PA gain), coherent BPSK. The learned relay is trained identically to before. Its size differs slightly by setup, because the phase case consumes complex I/Q pairs (22 inputs into 7 hidden units, 169 parameters) while the two real-valued cases take an 11-sample window into 13 (170); the two are the same architecture class at the same budget, and are labeled by their exact counts in the table.

These flat-channel controls preserve sample-wise channel mappings, although the differential receiver uses adjacent samples. They test three specified forms of parameter uncertainty; they do not cover all unknown memoryless channels.

Conclusion. In these three controls, the MLP and classical relay have similar measured BER from 8 dB upward: the largest difference is 0.0013 across the tabulated points. Below 8 dB the unknown-phase MLP trails differential detection, by up to 0.0106 at 0 dB, which lies below its lowest training SNR; in the gain and asymmetry cases the MLP is level with DF or slightly ahead throughout. Differential detection accommodates the block phase, and a positive gain preserves the noiseless BPSK sign. These observations show that the tested uncertainty alone need not defeat the chosen classical receiver; they do not prove equivalence or exclude benefits on other flat-channel families.

Table 6.2: Flat-channel controls at 8/12/16/20 dB. Learned-relay means use three training seeds and ten inference trials per seed. Similar measured BER is not a statistical equivalence claim. MLP-169 uses complex I/Q input; MLP-170 uses real input. Hop-2 noise follows the common convention (Appendix 9.6). At 16 and 20 dB every entry is set by the Hop-2 Rayleigh error rate, common to all relays

Flat channel	Relay	8 dB	12 dB	16/20 dB
Unknown phase (DBPSK)	DF (diff.)	0.0363	0.0148	0.0061 / 0.0025
	MLP-169	0.0376	0.0148	0.0061 / 0.0025
Unknown gain	DF	0.0453	0.0160	0.0061 / 0.0025
	MLP-170	0.0456	0.0162	0.0061 / 0.0025
Branch asymmetry	DF	0.0379	0.0149	0.0061 / 0.0025
	MLP-170	0.0381	0.0149	0.0061 / 0.0025

A memoryless channel on which the classical relays do fail. On the asymmetric channel $y = \tanh(1.5x) + 0.5 + n$ of Table 9.7 (setup S3 of `e6_sim_ported.py`, AWGN on Hop 2) the bias moves the decision boundary away from zero: at 8 dB AF, zero-threshold DF and the 170-parameter MLP reach 0.0820, 0.0377 and 0.00108 (three seeds, ten trials of 10^5 bits).

Conclusion (H5, within the tested scope). The fixed-filter BPSK study shows a large MLP improvement over AF and zero-threshold DF. Matched Viterbi gives lower measured BER over the resolved comparison range: at 8 dB the genie and 200-pilot LS values are 0.001354 and 0.001460, respectively, versus 0.0065 for the MLP. There is no 8-dB crossover. At 0 dB the MLP and genie means are close and the study does not resolve their ordering. Interpolated required-SNR penalties relative to genie MLSE vary with the target: 0.26, 1.18, 1.88, and 2.53 dB at BER 10^{-1} through 10^{-4}. They are not a universal 1–1.5 dB offset. The fixed-budget checkpoint results support the separation through 16 dB, not an extrapolation beyond that range.

Note on simulation validity. The observed AF/DF rise is specific to the evaluated relay mappings and tap vector; it is not a theorem about every memoryless relay or every ISI channel. The fixed-budget measurements at 12, 14, and 16 dB support a large MLP improvement over those memoryless comparators. The historical 18–20 dB entries use exposure determined by the first error and are retained only as exploratory records. They do not establish the precise BER, a con-

fidence bound valid under correlated errors, or monotonicity beyond the fixed-budget points; separately declared fixed-budget reruns are required before making those claims.

Status of the rare-event estimates. Extending the exposure to $10N_1$ after the first error does not produce an ordinary fixed-budget estimate. The historical 16 dB entry (16 errors in 334,002,040 bits) is retained only in the provenance record; it is replaced here by a predeclared fixed-budget confirmation: three independently trained checkpoints, each evaluated over 10^8 bits in 1000 blocks. This run records 36 MLP errors in 3×10^8 bits and yields a mean BER of 1.20×10^{-7}; the wide across-checkpoint interval is descriptive rather than an i.i.d.-bit confidence claim. The 12 and 14 dB points likewise use three predeclared 10^7-bit runs. The historical 18–20 dB values remain exploratory, are omitted from the MLP curve in Figure 6.2, and must not support precision or monotonicity claims pending the same validation.

Table 6.3: Destination BER for fixed three-tap BPSK ISI followed by AWGN. At 8 dB, AF/DF/MLP use the original three-seed study; the Viterbi rows use the matched-protocol run. AF/DF/MLP at 12 and 16 dB use respectively 10M and 100M bits per checkpoint (three checkpoints). Viterbi entries with $<$ are nominal independent-Bernoulli rule-of-three bounds from zero observed errors: 1M bits at 12 dB, 100M at 16 and 20 dB. They indicate insufficient data to estimate a positive BER, not zero true BER; correlated-error coverage is not established

Relay	8 dB	12 dB	16 dB	20 dB
AF	0.1813	0.1835	0.2075	0.2349
DF	0.1802	0.2009	0.2280	0.2457
MLP (170p)	**0.0065**	1.456×10^{-4}	$1.20 \times 10^{-7\dagger}$	not validated
Viterbi (genie CSI)	0.001354	$< 3.0 \times 10^{-6}$	$< 3.0 \times 10^{-8}$	$< 3.0 \times 10^{-8}$
Viterbi (200-pilot LS)	0.001460	$< 3.0 \times 10^{-6}$	$< 3.0 \times 10^{-8}$	$< 3.0 \times 10^{-8}$

† At 16 dB: 36 MLP errors in 300M bits. The 14-dB fixed-budget mean is 9.2×10^{-6}. Historical adaptive 18–20 dB MLP estimates are excluded from quantitative conclusions.

6.2.2 Does the Result Generalize to QPSK?

The unknown-ISI study above uses BPSK throughout. The canonical setup of Chapter 5 is QPSK, so this subsection asks whether the failure mode survives the move to that constellation. It is not a clean constellation control and is not offered as one: as Eq. (6.2) below records, the QPSK study also puts an independent per-symbol fade on top of the fixed filter, so the two studies differ in channel as well as in modulation. Isolating the constellation would need plain-QPSK ISI with these relays; Tables 6.6 and 6.7 use plain QPSK ISI, but with other relays. The 0.25 floor derived in Section 6.2 is a property of the *binary zero-threshold* slicer: it counts the $2^2 = 4$ equiprobable interfering-symbol patterns that flip a ± 1 decision. It does not imply a QPSK symbol-error floor or a fixed QPSK BER, because QPSK decisions use two axes and this

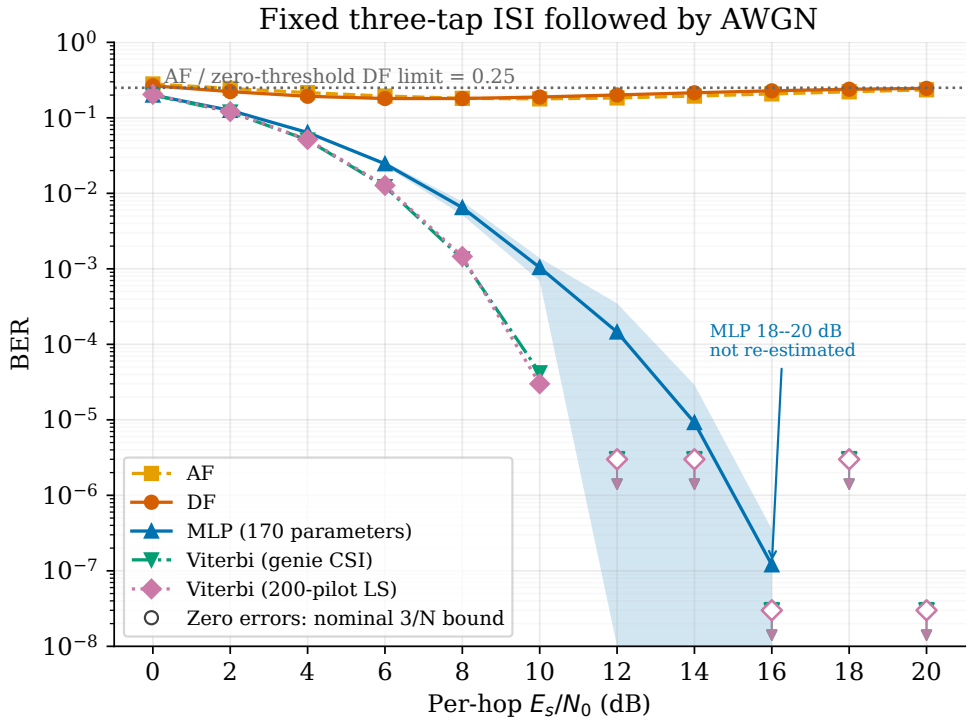


Figure 6.2: Destination BER versus per-hop E_s/N_0. The Viterbi curves use the same matched-protocol data as Table 6.3. MLP points above 16 dB are omitted. Open Viterbi markers show zero-error nominal $3/N$ bounds at the actual bit budgets, with downward arrows; these are not measured BERs. AF/DF/MLP bands use the stored across-checkpoint uncertainty at 12–16 dB and the historical intervals below 12 dB

experiment also changes the fading model. The QPSK result is therefore a separate measured extension, not a direct theorem-level generalization. This subsection repeats the relay comparison with Gray-coded QPSK in place of BPSK, using the same nominal per-hop SNR axis ($\gamma = 10^{\mathrm{SNR_{dB}}/10}$) and the same relay families generalized to the complex alphabet: AF, symbol-wise DF (hard per-axis decision), a 193-parameter 4-class MLP-QPSK classifier (window 11, softmax cross-entropy, trained at SNR $\in \{5, 10, 15\}$ dB), and Viterbi MLSE generalized to the complex QPSK trellis ($4^2 = 16$ states, complex branch metrics).

One difference from the BPSK study matters more than it first appears. Hop 1 here is not the plain ISI channel of Section 6.2 but

$$y[n] \;=\; g[n]\,(h * x)[n] \;+\; v[n], \qquad g[n] = |\,\mathcal{CN}(0,1)\,| \text{ i.i.d.}, \tag{6.2}$$

an independent Rayleigh magnitude on every symbol on top of the fixed filter. A trellis built from h alone is therefore *not* a genie-CSI detector on this channel: its branch metrics compare $y[n]$ against unfaded expected values, leaving a residual $A^2(g[n] - 1)^2$ that does not shrink as the noise does. Both detectors are reported below — the taps-only trellis, and a genie-CSI trellis given $g[n]$ as well as h, which scales each branch's expected observation accordingly. Evaluation uses 10 trials $\times$ 100,000 bits per SNR point over the same 0–20 dB range. No closed-form floor is derived for QPSK, so the curves below are reported as measured, not fit to

a predicted asymptote. Because the QPSK relays are applied jointly to the complex signal rather than decomposed into two independent BPSK problems, and the per-relay power normalization is computed on total (I+Q) power, the resulting AF/DF operating points are not required to, and empirically do not, numerically match the BPSK figures of Table 6.3 at the same SNR label; no such match is claimed.

Table 6.4: QPSK unknown-channel BER (mean over 10 trials) on the per-symbol faded ISI channel of Eq. (6.2). AF and DF plateau at an elevated, non-monotonic BER across the SNR range tabulated, and the 193-parameter MLP-QPSK restores monotonically decreasing BER. The last two rows are the same trellis given different information: with the taps alone it is model-mismatched here and the MLP passes it above about 2 dB; given the per-symbol fading gains as well, it is the genie-CSI detector for this channel and leads the MLP everywhere.

Setup	Relay	8 dB	12 dB	16 dB	20 dB
QPSK: ISI → AWGN	AF	0.2447	0.2116	0.2019	0.2100
	DF	0.2197	0.2024	0.2074	0.2211
	MLP-QPSK (193p)	**0.1292**	**0.0827**	**0.0601**	**0.0508**
	Viterbi (taps only)	0.1307	0.0852	0.0670	0.0618
	Viterbi (genie CSI)	**0.0739**	**0.0155**	**0.0018**	**0.0001**

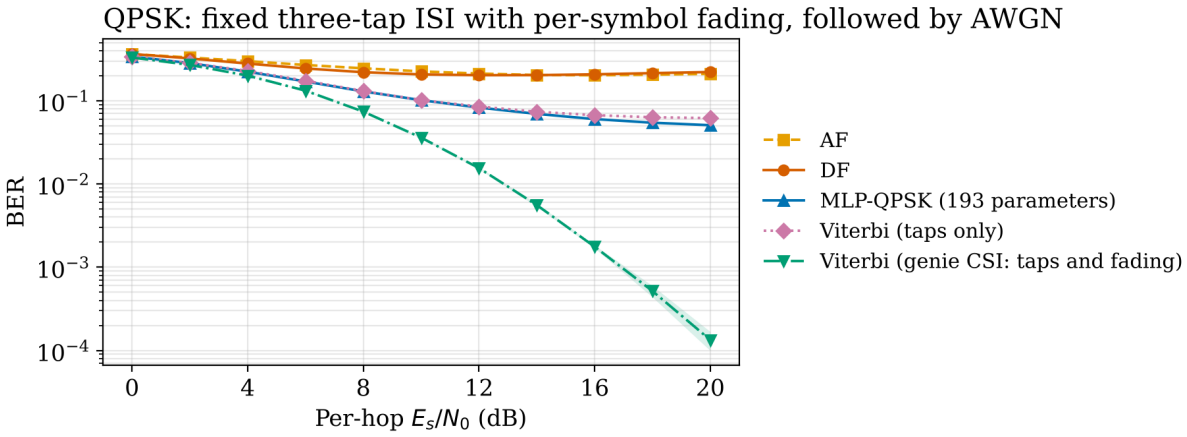


Figure 6.3: QPSK unknown-channel relaying on the per-symbol faded ISI channel of Eq. (6.2), BER vs. per-hop E_s/N_0 (10 trials, 100,000 bits/point). As with BPSK, AF and DF plateau far above the MLP and Viterbi curves at high SNR. The MLP-QPSK curve lies just below the *taps-only* trellis from about 2 dB, and the gap opens above 12 dB; the correctly informed genie-CSI trellis (given the per-symbol fading gains as well as the taps) leads the MLP at every SNR, by a gap that widens with SNR

Conclusion (QPSK with ISI and additional fading): The evaluated QPSK experiment changes both the modulation and the channel, adding per-symbol Rayleigh magnitudes to the fixed ISI filter. Its results establish a benefit over the evaluated memoryless relays under that combined change. Isolating modulation order would need a plain-QPSK-ISI control with these relays and consistent E_b/N_0 accounting.

Two trellis detectors have to be distinguished on this channel, and Eq. (6.2) is why. A trellis built from h alone, on a channel that also fades per symbol, is model-mismatched rather than genie-informed, and its error rate flattens for that reason. Three controls on the same trellis (`qpsk_trellis_controls.py`, run under the exact configuration of Table 6.4) separate the two explanations at 20 dB: given the taps on the faded channel it reaches a symbol error rate of 0.113; on the same channel with the fading removed, 0.000; and given the fading gains as well, 0.0003. The trellis is not at fault (it is verified never to return a path less likely than the transmitted sequence, and to reduce exactly to the nearest-symbol slicer when the ISI is removed) only its channel model was.

With the genie-CSI detector, the rounded destination BERs range from 0.3300 against the MLP's 0.3389 at 0 dB to 0.0001 against 0.0508 at 20 dB. The measured ordering agrees with the BPSK study, but the simultaneous modulation and channel changes prevent attributing that agreement to either factor alone.

The ordering is not an artifact of the error criterion either. A decomposition of the QPSK error events (`qpsk_error_decomposition.py`) separates the detectors on symbols as well as on bits: measured against the taps-only detector, the MLP is ahead on *symbol* error rate as well from about 8 dB up (0.0939 against 0.1126 at 20 dB), and the two lose almost the same number of bits per symbol error (1.073 against 1.090), so the Gray map is not the route either. It cannot be, for a structural reason stronger than that measurement. The taps are real and the fading gain is a real magnitude, so $y[n] = g[n](h * x)[n] + v[n]$ separates into two independent real ISI channels carrying the in-phase and quadrature bits; conditional on the genie-supplied gains and taps, the symbol posterior therefore factorizes into its I/Q bit marginals, and bit-wise MAP and symbol-MAP are the same detector on this channel, not merely close ones (`tests/test_bcjr_qpsk.py`, agreement to 5.6×10^{-16}).

At QPSK, the supplied-taps-and-gains Viterbi receiver substantially outperforms the 193-parameter MLP: the rounded destination BERs at 20 dB are 0.0001 and 0.0508. A precise multiplicative ratio should not be inferred from the rounded classical value. The supported benefit is relative to the tested memoryless baselines, not near-optimal BER. The relay-output BCJR control below addresses a different measurement endpoint.

6.2.3 Relay-Output Control: BCJR/APP Against Viterbi MLSE

Viterbi MLSE minimizes sequence error, whereas bit-MAP decisions minimize expected bit error under the supplied model [33]. The control in `qpsk_bcjr_benchmark.py` compares these two decisions with both taps and per-symbol fading gains supplied. It measures errors *at the relay output before Hop 2*; Table 6.4 measures destination errors. Thus this is not an end-to-end BCJR-versus-MLP experiment.

Two controls check the detector implementation. Removing fading while retaining ISI does not require MLSE and bit-MAP to agree: at 8 dB their BERs are 0.021764 and 0.021424, with

paired difference $+0.00034 \pm 0.00016$, a resolved difference at the reported level. At 20 dB both record zero errors in 10^6 bits; 3.0×10^{-6} is a nominal independent-bit reference bound, not a proof of zero BER. Removing ISI is the appropriate exact-agreement control: BCJR reduces to the per-axis slicer on all $10{,}000$ tested symbols.

Table 6.5: Genie-CSI Viterbi MLSE against the BER-optimal BCJR/APP detector on the same trellis, relay output, 10 trials of 100,000 bits. The interval is on the paired per-trial difference, $t(9, 0.975) = 2.262$

SNR (dB)	Viterbi BER	BCJR BER	Relative gain	Paired 95% CI
0	0.250177	0.239113	4.42%	resolved
4	0.162702	0.155861	4.20%	resolved
8	0.068918	0.067245	2.43%	resolved
12	0.015075	0.014857	1.45%	resolved
16	0.001726	0.001725	0.06%	includes zero
20	0.000127	0.000126	0.79%	includes zero

Conclusion. BCJR has no higher empirical relay-output BER at the sampled points; its paired advantage is not resolved at 16 and 20 dB. Bit-MAP optimality concerns expected loss under the model and does not guarantee ordering in every finite Monte Carlo sample. These controls support the local detector implementation. They neither establish an end-to-end BCJR margin against the MLP nor close the unmeasured BPSK bit-MAP comparison.

6.2.4 A Fixed Composite Channel: Learned and Estimated-Model Relays

This study cascades fixed three-tap ISI, a fixed saturating amplifier, and a randomly drawn block phase before additive noise, using DBPSK. It tests a combined impairment, not a factorial isolation of each cause. The implemented comparators are AF, differential detection, and pilot-LS Viterbi followed by differential decoding. The latter estimates a linear FIR model and is mismatched to the nonlinear amplifier; no claim that it is the strongest possible classical receiver is made. The 169- and 1,153-parameter MLPs are trained on this cascade family without receiving its analytic parameters at inference.

Evaluation uses 10 unique trials of 100,000 source bits per SNR and three trained MLP instances. Repeated classical seeds and the training/trial hierarchy prevent interpreting the historical plotting bands as independent-trial 95% confidence intervals. They are retained only as descriptive bands. The composite AF path originally added half the intended complex Hop 2 noise power. Its curve was rerun with the corrected variance, original ten unique trial seeds and bit budgets, recording errors, exposures and Student-t intervals in `e6_unknown_channel_results/codex_composite_af_validation.json`. The other relays emit real signals and do not use the affected complex-noise branch; their stored point estimates are unchanged. Overlapping bands do not establish equivalence.

Conclusion within this cascade. AF and differential detection remain near BER 0.25 at high

SNR; this is an observation, not a derived universal floor for nonlinear channels. The 169-parameter MLP has much lower measured BER, reaching 2.5×10^{-3} at 20 dB. Pilot-LS Viterbi has lower BER at 8 dB (0.0674 vs. 0.1073) and both round to 0.0025 at 20 dB. Equal rounded values do not establish equivalent reliability. The larger MLP is also close at the last point, ending at a rounded 0.0025 at 20 dB. Its mid-SNR means are lower than the small MLP's (0.1016 versus 0.1073 at 8 dB; 0.0482 versus 0.0542 at 10 dB), but the historical intervals do not support a formal capacity-equivalence or paired-significance conclusion. These results support learned processing on this trained cascade family, not arbitrary impairment stacks, unseen channel families, or superiority over an optimally matched receiver.

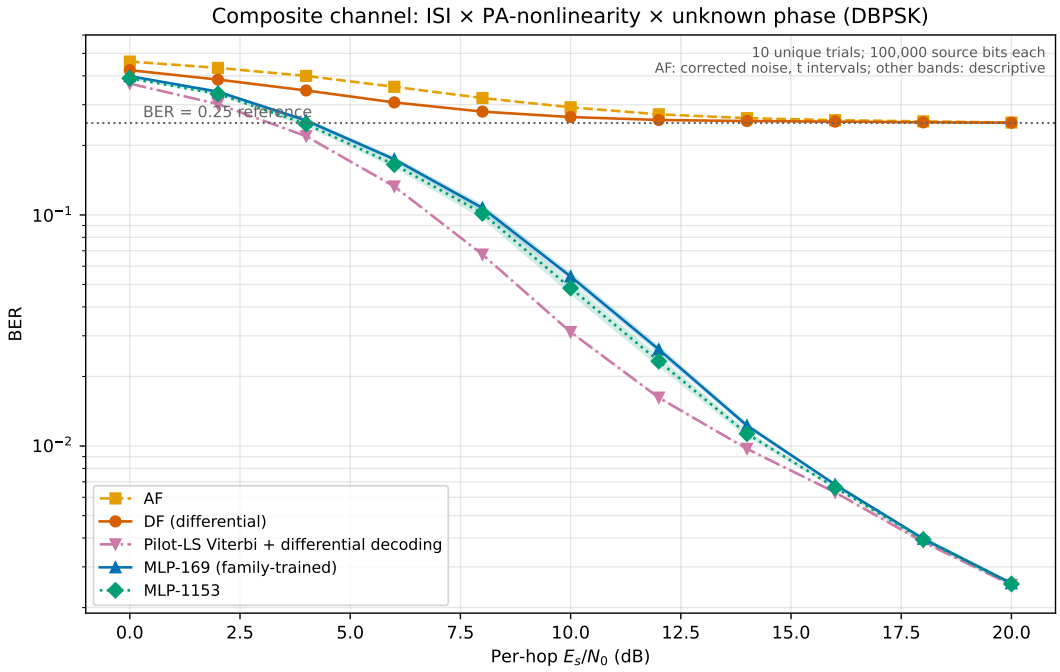


Figure 6.4: Fixed ISI and amplifier with random block phase, DBPSK. The corrected AF curve uses ten unique fixed-budget trials and Student-t intervals. Other curves retain historical point estimates and descriptive bands, which are not calibrated equivalence tests. Both MLPs were trained on this cascade family. Similar high-SNR rounded values do not demonstrate equal BER

6.3 Layer 3 — CSI Uncertainty: The Pilot-Budget Crossover

This is layer 3 of the ladder in Table 6.1: the pilot count is varied within the specified random composite-channel protocol, which differs from the preceding fixed-composite experiment. The classical receiver substitutes a single least-squares linear-channel estimate into Viterbi MLSE. It neither computes a channel posterior nor marginalizes channel uncertainty, and its linear model is mismatched to the nonlinear cascade. The pilot-free MLP supplies a fixed reference after training on the channel family; its outputs have not been checked for calibration. The sweep measures where the BER ordering changes at 10 dB, rather than a Bayesian-optimal information boundary. **Evaluation:** 50 trials × 20,000 bits per operating point (1,000,000 bits), matching Section 6.4;

the channel is redrawn per block. In panel (b), each trial is subdivided into independent blocks of the swept length, retaining the same total bit budget across block lengths.

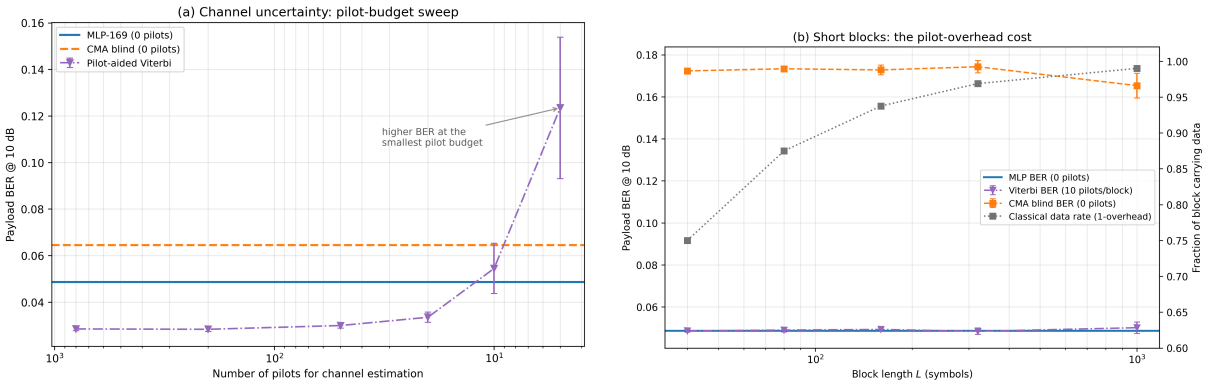
Conclusion (a measured pilot-budget crossover at 10 dB): Pilot-aided Viterbi reaches payload BER 0.0283 at 200 pilots and 0.0335 at 20 pilots, below the MLP's pilot-free 0.0486. At 10 pilots its BER is 0.0545, with a 95% interval of ± 0.0108 that contains the MLP's value, and at 5 pilots it reaches 0.1235. The reported 95% interval at five pilots is 0.1235 ± 0.0304, against ± 0.0011 at 50 pilots. Five observations for three complex taps do not by themselves establish rank deficiency; identifiability depends on the design matrix. Under a correctly specified linear model, estimation covariance depends on noise power and pilot geometry [43]. Here nonlinear model mismatch is also present, and the experiment does not isolate estimation variance as the cause of degradation. The MLP reference remains at 0.0486 because it uses no test-time pilots. The ordering is thus resolved at 20 pilots and at 5, not at 10; for all three trained MLP instances the point estimates cross between 20 and 10. No sweep at another SNR establishes how that crossover moves.

The practical force of this depends on block length, because a fixed pilot *count* is a growing pilot *fraction* as blocks shorten. Panel (b) sweeps block length at the same operating point with the classical receiver spending its minimum viable ten pilots per block, and adds blind CMA as the second pilot-free reference. The payload-BER confidence intervals overlap at the evaluated block lengths, both estimates being approximately 0.049; the study has not established equivalence. At a 1000-symbol block the ten pilots cost 1% of the block, whereas at a 40-symbol block, representative of low-latency or fast-varying links, they cost 25%, so the classical receiver carries data on only 75% of the block against the learned relay's 100% with no payload-BER difference detected at this budget.

Among the tested pilot-free methods, the fixed MLP has lower BER on these short blocks than CMA: CMA records 0.1723 at $L = 40$ and 0.1653 at $L = 1000$, versus its 0.0645 long-block reference. This is consistent with limited adaptation time, but convergence failure is not isolated from other algorithmic settings. The MLP and pilot-aided receiver have no resolved difference at this budget; that is not proof of equal reliability. The conclusion is limited to the implemented baselines and block lengths.

6.4 Layer 4 — Unknown Per-Block Realization: The Blind Regime

This is layer 4, the bottom of the ladder. Per-block CSI and pilots are removed, and a fresh realization is drawn from the same impairment family used during MLP training. The comparison is against CMA, which assumes constant modulus and adapts an equalizer from the received block, and decision-directed MLSE, which bootstraps a channel estimate from its own decisions. Thus no method receives per-block CSI, but neither is assumption-free: the MLP knows the training



(a) pilot-budget sweep at 10 dB

(b) block-length sweep at 10 pilots

Figure 6.5: Finite-pilot regime; both panels are means over 50 channel draws at 20,000 bits/point. **(a)** Pilot-aided Viterbi has lower BER than the pilot-free MLP at 20 pilots per block and more; at 10 pilots the difference is not resolved (0.0545 ± 0.0108 against the MLP's 0.0486), and at 5 pilots it collapses when the least-squares estimate becomes too noisy to construct a reliable trellis (0.1235 ± 0.0304); the MLP is flat across the sweep, operating with a family-trained discriminative estimator and requiring no pilots. **(b)** With 10 pilots per block, pilot-aided Viterbi and the MLP have no detected payload-BER difference at the evaluated block lengths (both ≈ 0.049), but the classical receiver's pilot overhead grows from approximately 1% at $L = 1000$ to 25% at $L = 40$. The tested CMA implementation has higher BER throughout. The MLP uses no pilots and has no resolved BER difference from the pilot-aided receiver at this budget; equivalence and superiority to untested pilot-free methods are not established

distribution and CMA knows signal/equalizer structure. **Evaluation:** 50 independent channel draws $\times$ 20,000 bits per SNR point. Classical-method intervals use those 50 draws; neural uncertainty is assessed across trained instances rather than by treating repeated inference under one network as independent training evidence.

The blind equalizer. The evaluated CMA normalizes its update by input-segment energy, because a fixed-step update overflowed on longer blocks. This is an implementation choice, not a proof that normalized CMA always converges or improves on fixed-step CMA; the results concern this implementation and its stated settings.

Conclusion (learning leads CMA up to 12 dB and avoids per-block adaptation): The corrected CMA reaches BER 3.3×10^{-3} at 20 dB, and the family-trained MLP reaches 2.6×10^{-3}. The margin is largest at mid SNR, the MLP at 0.0955 against CMA's 0.1182 at 8 dB, while decision-directed blind MLSE is non-monotonic (0.0374 at 16 dB and 0.0498 at 20 dB). Whether a margin is resolved is a question the intervals have to answer rather than the point estimates. Classical methods have no training seed, so their intervals use the 50 independent channel draws, and the three trained MLP instances are treated hierarchically (`e6_blind_intervals.py`): the decision-directed MLSE interval at 8 dB is ± 0.0361, against the MLP's ± 0.0046 and CMA's ± 0.0140. Paired over the 50 shared draws, the MLP's lead over CMA is resolved from 0 to 12 dB (Wilcoxon $p \leq 0.003$) but not from 14 to 20 dB ($p \geq 0.40$); at 20 dB the MLP is better on only 19 of the 50 draws, and its lower mean comes from a few much worse CMA draws.

The observed instability belongs to the decision-directed MLSE implementation. CMA instead carries an online adaptation cost and requires enough samples to converge, whereas the MLP uses one fixed forward pass after offline training. The supported conclusion is therefore narrow: on unseen per-block realizations from its training family, the MLP outperforms the evaluated CMA implementation up to 12 dB while avoiding per-block adaptation, and it avoids the non-monotonic behavior of the evaluated decision-directed MLSE.

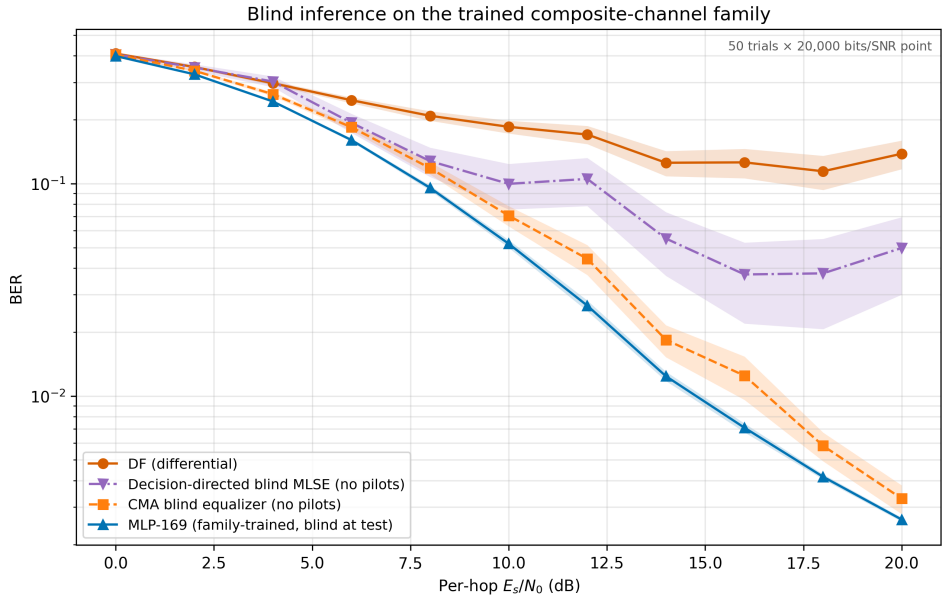


Figure 6.6: No pilots or per-block CSI; fresh realizations from the training family. The evaluated CMA and MLP means are close, while decision-directed MLSE has a non-monotonic high-SNR tail. Historical bands are descriptive; the neural training hierarchy prevents treating all pooled trials as independent confidence evidence. The MLP avoids test-time adaptation but uses family knowledge acquired in training. No claim against all blind equalizers is made

6.4.1 A Low-Complexity Linear Baseline: MMSE Equalization

The comparisons above set the learned relay against two classical relays, and neither is matched to it in cost. The evaluated zero-threshold DF has a high-SNR error floor on the chosen ISI taps. Outperforming it alone does not establish a benefit from temporal context: the scalar MAP control removes that floor without a window, while the windowed MLP achieves substantially lower measured BER (Section 6.2). Viterbi MLSE is the optimal sequence detector for the channel model it is given, but its cost per symbol is exponential in the channel memory (M^L branch metrics for the joint trellis, $2M^{L/2}$ after I/Q factorization of a square constellation), which is precisely the expense Section 6.5 invokes on the learned relay's behalf. The case between them was not tested: an N-tap MMSE linear equalizer is a natural low-complexity remedy an engineer might reach for before a trellis. If it recovered most of what the learned relay recovers, the contribution of Chapter 6 would be a narrower claim about a nonlinearity rather than about learned processing.

It does not. Table 6.6 gives the equalizer's SNR penalty against MLSE for equalizer lengths

from 3 to 11 taps, measured on the same two-hop chains, with the exact channel taps supplied and the weights redesigned at every SNR point — the same genie CSI granted to MLSE, and a longer window than the learned relay is given.

Table 6.6: SNR penalty against Viterbi MLSE for an MMSE linear equalizer of N taps, with the best configuration of the size sweep on the same channel for comparison (window 7; 433, 433 and 289 parameters; best of three initializations; `results/mlp_min_size_all_channels.json`). Negative values beat MLSE. Channels as in Table 9.3: "ISI (QPSK)" has no fading, and on "Composite" (noise $1/\gamma$) the linear MLSE is mismatched to the amplifier. The reader should observe that the learned relay leads the equalizer by 1.8 to 2.4 dB on every channel (1.4 to 2.3 dB at the median initialization), despite the equalizer being given exact channel knowledge, per-SNR weight redesign, and up to 11 taps against the relay's window of 7. Penalties follow the definition in Appendix 9.6

Channel	3 taps	5 taps	7 taps	11 taps	Learned relay
ISI (BPSK)	$+6.58$	$+4.40$	$+4.23$	$+4.04$	$+\mathbf{1.67}$
ISI (QPSK)	$+7.81$	$+9.11$	$+5.02$	$+4.84$	$+\mathbf{3.03}$
Composite	-0.33	-0.37	-0.37	-0.44	$-\mathbf{2.26}$

The learned relay leads the tested linear-MMSE equalizers in the reported required-SNR metric. This is not a comparison against every cost-matched classical method; decision-feedback and reduced-state detectors were not evaluated. The MMSE decreases with filter length, as expected for nested linear estimators. BER and interpolated SNR penalties need not decrease with MSE, however. Consequently the non-monotonic QPSK penalty cannot be dismissed as an interpolation artifact without an additional check. The four-dB sampling grid is a further limitation on the precision of these penalties.

6.4.2 Does a Sequence Architecture Help Where a Window Does?

Chapter 5 finds that Transformer, Mamba-S6 and Mamba-2 buy nothing over a feedforward relay on the canonical channel, and offers one reading of it, that the channel rather than the architectures is responsible: a memoryless channel offers no temporal structure for their inductive bias to exploit. That reading is a prediction about what would happen given such structure, and until now it was untested here, no sequence model having been evaluated anywhere in this chapter. The minimum-size study sharpens the prediction, finding the relay's *window* to be the binding resource on channels with memory, worth more than 8 dB between window 1 and window 7 while width and depth buy almost nothing. A window is a blunt way of granting a feedforward network temporal context; sequence models are built for it.

The prediction fails, at least for the Transformer, the one sequence architecture here that sees the whole window. Table 6.7 compares the four architectures at an equal budget of approximately 3,000 parameters, each trained on the channel under test rather than on a memoryless surrogate, with identical hyperparameters and three initializations apiece.

For this equal-budget, shared-hyperparameter protocol, the feedforward relay has the best reported initialization on all three channels, by 1.55 and 3.14 dB on the two ISI cases and 0.32 dB

Table 6.7: SNR penalty against Viterbi MLSE at an equal budget of ≈3,000 parameters, with the across-initialization spread in brackets. Each entry is the *best* of three initializations, not their mean, so the values are optimistic by construction; the spreads did not reproduce in a rerun (see text), so the ordering of the three sequence architectures should not be read as resolved. Negative values beat MLSE. The reader should observe that the feedforward relay leads every sequence architecture on every channel. Every initialization is held to the targets MLSE reaches; †one Mamba-2 initialization does not reach 10^{-3} by 20 dB, so its penalty is only bounded below ($> +9.11$) and the spread is a lower bound; dropping that target, as the earlier rule did, scored it at $+4.69$. Channels as in Table 9.3; the simplified Mamba relays see only the causal half of their window (Remark 4.1). Penalties follow the definition in Appendix 9.6

Channel	MLP-3K	Transformer-3K	Mamba-S6-3K	Mamba2-3K
ISI (BPSK)	$+$**1.97** (0.39)	$+3.52$ (0.33)	$+4.30$ (1.15)	$+4.55$ (0.14)
ISI (QPSK)	$+$**2.58** (0.07)	$+5.96$ (0.45)	$+5.72$ (1.33)	$+5.83$ (>3.28)†
Composite	$-$**2.08** (0.03)	-1.76 (0.03)	-1.56 (0.08)	-1.60 (0.06)

on the composite. Its observed training times are 35–66 seconds per initialization versus 217–609 for the sequence models. These results favor the tested windowed implementation; they do not establish that channel memory never benefits a tuned sequence architecture.

The bracketed spreads do not support a ranking by reproducibility. A rerun of the whole table on another machine, with the same code, seeds and training data, reproduced eleven of the twelve best values within 0.25 dB (not the Transformer's at QPSK) but not the spreads: on the ISI channels Mamba-S6 spread 0.12 and 0.19 dB instead of 1.15 and 1.33, and the Transformer 0.90 and 1.31 instead of 0.33 and 0.45. Seeded training repeats bit for bit on one machine, but the same Transformer seed moved by 0.89 dB between two machines running identical software, so three-seed spreads on these channels vary between machines by as much as they differ between architectures. Of the three architectures retested, only the feedforward relay retrained identically on both machines.

One limitation is carried. All four architectures receive identical hyperparameters, matching the equal-budget protocol of Section 5.3; no per-architecture tuning was attempted, and a tuned sequence model might close part of the gap. A second limitation is structural: the two Mamba relays read a causal scan at the window centre (Remark 4.1), so they never see $y[n+1]$ and $y[n+2]$, which carry 42.5% of the energy of $x[n]$ on the ISI taps and 34% on the composite. Their rows do not test whether a sequence model can exploit the window; the Transformer, which sees all of it, does, and it also trails. Giving the Mamba relays the whole window at the same parameter count, by reading the scan after its last sample or by averaging forward and reversed scans at the centre, made them worse under this protocol: the first failed to train (BER about 0.1 at 20 dB), and the second trailed the centre readout at every target on every channel.

6.5 The Cost Side: Decision Delay and Arithmetic

The cost comparison distinguishes nominal arithmetic from measured implementation time. The BER tradeoff is configuration-dependent, and is not uniformly small, particularly in the QPSK study.

Per-symbol cost and scaling. MLSE maintains M^{L-1} trellis states for a constellation of size M and channel length L (memory $L-1$), performing an add–compare–select over M^L branch transitions per symbol; its cost therefore grows *exponentially* in channel memory for fixed M, and polynomially in M for fixed L. The 170-parameter MLP performs one fixed forward pass, about 330 floating-point operations per symbol. That figure is a constant of the *deployed* architecture, not a quantity independent of the channel: the input window has to span the memory, so its length grows with L, and the output dimension grows with the constellation, from a single real output at $M = 2$ to the 193-parameter QPSK relay used earlier in this chapter, and further still for denser constellations. Once those are fixed, the per-symbol cost is the same for every symbol and every channel realization, which is the property the comparison below relies on. The contrast with MLSE is therefore one of *growth rate*, not of a constant against a variable: the network's cost rises roughly linearly in window length and output dimension, the trellis count is M^L. That count is for the joint trellis. With real taps and circular noise, a square M-QAM trellis separates exactly into two $\sqrt{M}$-ary trellises (Section 2.6), with $2M^{L/2}$ branch metrics in total: 2^{L+1} for QPSK, and 2,048 rather than about 10^6 for 16-QAM at $L = 5$. The factorized count still grows exponentially in L, but with half the exponent, so the growth-rate contrast holds while the crossover with a fixed network moves to longer channels. For BPSK no such reduction exists.

For the studied BPSK three-tap channel, the nominal Viterbi count is about 64 operations per symbol — $M^L = 8$ branch metrics, each computed from scratch at about eight operations (the three-tap convolution for the expected sample, the difference, the square, and the path-metric addition), a deliberately broader convention than the multiply-only MAC count used in Section 6.5.1 — below the MLP's approximately 330 FLOPs (excluding activation-function cost). Full-state trellis counts grow rapidly for larger M and L. A fixed network's arithmetic does not depend on the channel realization, but maintaining accuracy as channel complexity increases may require changing its window, width, or output dimension. Extrapolated operation counts for untested architectures are not evidence of retained BER or hardware speedup. Reduced-state sequence detection [44] and decision-feedback methods are unmeasured alternatives.

Measured inference time. For BPSK with $L = 3$, the vectorized NumPy MLP is 58–82× faster than the interpreted Python Viterbi implementation on the measured workload (single-threaded; the ratio depends on threading and machine), despite Viterbi's lower nominal operation count. The timed network is the 169-parameter I/Q relay of the composite study, whose per-symbol arithmetic matches the 170-parameter relay's. This includes implementation overhead. Optimized kernels and target-hardware throughput were not measured; no universal latency or hardware advantage is inferred.

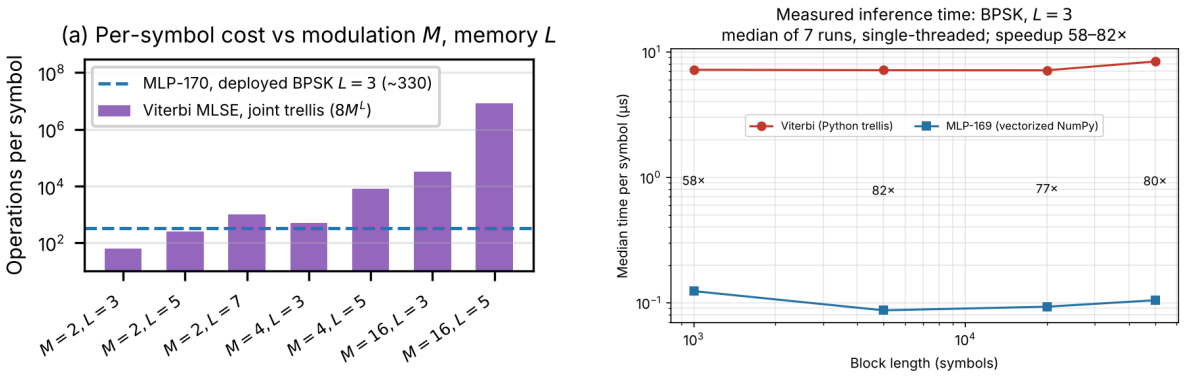


Figure 6.7: Complexity of Viterbi MLSE against a fixed MLP relay. **(a)** Per-symbol computational cost versus modulation order M and channel memory L: the plotted Viterbi count is that of the joint trellis and grows as M^L, while the network's cost grows only linearly in window length and output dimension; the dashed line marks the deployed 170-parameter BPSK, L=3 relay (∼330 flops). For $M \geq 4$ and real taps, exact I/Q factorization (Section 2.6) reduces the trellis to $2M^{L/2}$ branch metrics, a factor $M^{L/2}/2$ below the plotted bars (512 at M=16, L=5), so the panel overstates matched classical cost at those points. On the small studied channel (BPSK, L=3) Viterbi is actually cheaper; the MLP wins on *scaling*, not on the small case. **(b)** Measured time per symbol for BPSK with $L = 3$ (single-threaded, median of seven runs; `e6_complexity_ported.py`), timing the 169-parameter I/Q relay of the composite study, whose ∼330 flops per symbol match the 170-parameter relay's: the MLP is 58–82× faster as a vectorized matmul than Viterbi's interpreted sequential trellis scan

6.5.1 A Channel with Memory Under a Latency Constraint

This subsection measures accuracy, structural decision delay and arithmetic within one coded-memory system, including block DF. It complements the earlier information-limited studies, but uses a different channel configuration and endpoint. It therefore does not measure an intersection with the pilot-budget crossover or a quantitative distance between the two studies' resource boundaries.

Every scheme is placed inside one identical coded system so that the comparison is between relays and nothing else: a rate-$1/2$, $K = 3$ convolutional code on QPSK, 200 information bits per frame (202 symbols after tail), with soft Viterbi decoding of the code at the *destination*. All schemes therefore deliver an information BER at the same information rate. Hop 1 carries the unknown L-tap ISI response ($h_k = 0.7^k$, unit-energy normalized) plus complex AWGN and hop 2 is AWGN; 150 frames per trial and three seeds per point.

Each relay is then labeled with the structural decision delay it forces, in symbols: zero for AF and symbol-wise DF; $\lfloor W/2 \rfloor$ for a relay of window W, which must wait for the later half of its own window; D for an MLSE with traceback depth D; and the full frame, 202 symbols, for block DF, which cannot begin until the last symbol of a codeword has arrived. These are detector delays. The window relay and the MLSE relays here forward exact constellation points and apply no power normalization, so their delays are also their readiness delays; AF's zero is the algorithmic value, since the implemented AF normalizes by the power of the whole simu-

lated block (Section 4.1.3). The 202-symbol figure describes the implemented full-frame block decoder, not a lower bound for coded relaying: a finite-traceback convolutional decoder could release decisions sooner. Protocol scheduling and end-to-end delay are not measured in either case.

No latency separation was resolved on the tested short-memory channel. Table 6.8 gives the result at $L = 3$. The BER at 12 dB is unchanged at the reported precision from traceback depths $D = 2$ through $D = 15$. This finite-budget observation does not prove that all survivor paths merge by three symbols or that full-asymptotic accuracy is reached. Among the tested settings, MLSE at $D = 2$ has less delay than the window-11 relay's five symbols and lower measured BER. Block DF spends 202 symbols without a resolved accuracy gain in this experiment; with an equalizer it reproduces the reported MLSE value but adds frame buffering.

Table 6.8: Coded information BER at 12 dB on the three-tap channel, against the structural decision delay each relay forces and its arithmetic per symbol. The delay column is detector look-ahead only: it excludes the whole-block power normalization of AF (the other relays here forward exact constellation points and apply none), protocol scheduling and the two half-duplex transmission phases, so it is not end-to-end latency. MLSE arithmetic is that of the exactly factorized detector, with the implemented joint trellis in brackets. Mean of three seeds. No BER difference among the tested MLSE traceback depths is resolved at the reported precision; this is not an equivalence or survivor-merger test

Relay	Delay (symbols)	MACs/symbol	BER at 12 dB
AF	0	2	0.0609
DF-hard (symbol-wise)	0	0	0.3027
MLP $W = 5$	2	112	0.0010
MLP $W = 11$	5	208	0.000189
MLP $W = 21$	10	368	0.000278
MLSE $D = 2$	2	16 (128)	**0.000033**
MLSE $D = 3$	3	16 (128)	**0.000033**
MLSE $D = 15$	15	16 (128)	**0.000033**
block DF	202	16	0.0411
block DF + MLSE	205	32 (144)	0.000033

The symbol-wise relays confirm the chapter's Layer 2 finding inside the coded chain: AF and hard DF do not merely lose, they fail, DF's BER *rising* with SNR from 0.2913 at 8 dB to 0.4253 at 20 dB as the interference rather than the noise comes to dominate its decisions.

The cost convention. The tables count real multiplications per transmitted symbol, labeled MACs for the network's multiply–accumulate operations; standalone additions, comparisons, activations and traceback are excluded. The network count is $2WH + 4H$. Real taps and circular AWGN permit exact separation of the QPSK trellis into two binary channel trellises (Section 2.6), whose combined residual-metric count, 2^{L+1}, is the classical count this section compares against. The implemented joint QPSK trellis, with M^{L-1} states, M^L branches and two

real multiplications per complex squared residual, costs $2M^L$; it is kept as a diagnostic of that implementation. Both counts apply one rule: one real multiplication per real squared residual, so a complex residual costs two. The two binary trellises have 2^L branches each, giving 2^{L+1} per QPSK symbol. Charging two multiplications per *real* residual instead would double the factorized count to 2^{L+2} (32, 128 and 512 at three, five and seven taps), place the arithmetic crossover with the 208-multiplication network near 5.70 taps rather than 6.70, and give a seven-tap ratio of about 2.46 rather than 1.23; it would also double the joint count, so it is not the convention used here. The joint-trellis count is therefore not a lower bound on matched classical detection cost.

Arithmetic at a fixed architecture. With hidden width H fixed, increasing the input window W increases the network count linearly. No experiment establishes that fixed H or a rule $W = O(L)$ preserves accuracy for arbitrary longer channels. Equating the network count with the factorized count gives the following arithmetic crossover, not a performance boundary:

$$L^\star = \log_2(2WH + 4H) - 1 = 6.70 \text{ taps (MAC-count crossover)} \tag{6.3}$$

for $W = 11$ and $H = 8$, so the first integer tap count above the crossover is seven. The factorized counts are 16 at three taps and 256 at seven, against the relay's 208, a ratio of about 1.23 at seven taps. For the implemented joint trellis (diagnostic) the counts are 128 and 32,768, the crossover falls at $\log_4 104 = 3.35$ taps (2.90 at $W = 5$, 3.76 at $W = 21$), and the seven-tap ratio would be 158. These are analytical counts, not new BER or runtime measurements; the historical tables retain the joint implementation's measurements. Equal counts imply neither equal accuracy nor a hardware speedup.

In the separately measured three-, five-, and seven-tap settings, the architecture uses 208 MACs per symbol and is retrained for each channel length. Its BER ranges from 1.91×10^{-4} to 2.36×10^{-4}; overlapping intervals do not establish constant error or equivalence. The factorized MLSE count rises from 16 to 256 over the same range and stays below the network's until six taps; the joint implementation's count rises by a factor of 256 and is a diagnostic of that implementation, not a necessary cost of matched detection.

The crossovers concern selected arithmetic counts, not accuracy or measured deadline feasibility. The classical detector has lower measured BER throughout, and exact factorization weakens the computational case for the learned relay on this channel. At three taps, both classical counts are below the network's count. The earlier comparisons therefore establish benefits over the evaluated AF and zero-threshold DF mappings, not that matched MLSE is computationally unavailable.

Taken together, the studies show benefits of the tested learned relays over specified mismatched or memoryless comparators, with limitations from training-family dependence, finite budgets, and untested classical alternatives. They do not overturn the matched memoryless controls or

Table 6.9: Coded information BER at 12 dB and arithmetic against channel memory: MLSE MACs for the exactly factorized detector, 2^{L+1}, with the implemented joint trellis, $2M^L$ with M^{L-1} states, as a diagnostic; the measured BERs are those of the joint implementation, which factorization leaves unchanged. MLSE has genie CSI and traceback $5L$; the 220-parameter window-11 architecture is retrained for each tap count. Measurements at $L = 3, 5, 7$ use 1.5M information bits and five seeds; rows $L = 4, 6$ contain arithmetic only. Bracketed nominal 95% Wilson intervals inherit the adjacent BER exponent; coverage under coded error correlations is not established. MLSE has lower measured BER throughout

L	MLSE MACs, factorized	Joint (diagnostic)	Relay MACs	MLSE BER (nominal CI)	Relay BER (nominal CI)
3	16	128	208	3.20×10^{-5} [2.41, 4.24]	2.36×10^{-4} [2.13, 2.62]
4	32	512	208	—	—
5	64	2,048	208	4.20×10^{-5} [3.28, 5.37]	1.96×10^{-4} [1.75, 2.20]
6	128	8,192	208	—	—
7	256	32,768	208	2.47×10^{-5} [1.79, 3.40]	1.91×10^{-4} [1.70, 2.14]

establish a general superiority of learned relaying.

Chapter 7

Discussion and Conclusions

The canonical, coded, and unknown-channel studies answer related but distinct questions. Their channel models, architectures, endpoints and uncertainty units must not be combined into a single causal experiment. This chapter interprets their findings; Chapter 8 records the hypothesis outcomes.

7.1 Interpretation of Results

7.1.1 Low-SNR Advantage over AF (H1)

The evaluated feedforward relays have lower canonical BER than AF at 0–4 dB, supporting H1. At 0 dB the Hybrid, MLP and VAE means are 0.3329, 0.3349, and 0.3359, compared with DF's 0.3337 and AF's 0.4076 (Table 5.2). The sequence models are closer to AF at that point. These are configuration-specific comparisons; model size alone does not explain them. This gap is large, but the implemented AF normalizes the zero-forced observation by the power of the whole block (Section 4.1.3), and its BER depends on block length: at 0 dB it is 0.387 for 50-symbol blocks, 0.407 at the canonical 5,000 and 0.413 at 50,000 (`af_block_and_conventional.py`). A conventional finite-power AF relay, which forwards the unequalized signal with variable gain $\sqrt{P/(|h_1|^2 + N_0)}$ and is detected coherently at the destination, reaches 0.342 at 0 dB, 0.245 at 4 dB and 0.144 at 8 dB, agreeing with its closed form to within 0.001 at every SNR. A fixed-gain AF (average rather than instantaneous $|h_1|^2$) does better at 0 and 2 dB (0.338 and 0.294) and worse from 4 dB. Against the variable-gain reference the MLP's margin at 0 dB falls from 0.073 to 0.007, still outside the MLP's 95% interval, and is about 0.02 at 4–8 dB; against the fixed-gain one it is 0.003 at 0 dB, just outside that interval, which omits training variance, and inside the VAE's. The sequence relays are worse than either conventional AF at 0 and 4 dB (0.401–0.407 and 0.276–0.281). The feedforward learned relays track symbol-wise DF throughout, so their remaining advantage over conventional AF is the classical advantage of regeneration over amplification, not a property of

learning. H1 is supported as stated, but the implemented AF overstated its margin.

A nonlinear soft mapping is a plausible interpretation, not an isolated mechanism. For unit-energy BPSK in real AWGN, the posterior mean is $\tanh(y/\sigma^2)$, or $\tanh(2y)$ at $E_b/N_0 = 0$ dB. It minimizes local mean-squared error, not necessarily destination BER after relay normalization and Hop 2. The canonical QPSK/Rayleigh estimator also depends on fading reliability: after equalization, each real-axis variance is $N_0/(2|h|^2)$. For a relay function that is not supplied $|h|$, the Bayes target averages over that uncertainty. The canonical relays are not trained towards that target, however: they are trained on BPSK over AWGN only (Section 4.3.3), so they apply an empirical AWGN denoiser to the zero-forced fading observation; H1–H4 measure transfer from that training distribution, and the low-SNR advantage is measured, not implied by that target.

The analytical CSI-assisted variable-gain AF expression $\gamma_1\gamma_2/(\gamma_1+\gamma_2+1)$ does not predict the block-normalized AF curve implemented here. Only the qualitative fact that linear amplification preserves noise alongside signal carries over.

7.1.2 Medium-to-High-SNR Ordering (H2)

DF matches or exceeds the evaluated learned relays from 6 dB upward. Coherent component-wise slicing is a local MAP symbol decision for the canonical channel, but that does not prove optimality among all power-constrained relay mappings for destination BER. Nor does tanh saturation isolate the cause of the learned models' finite-SNR penalties.

7.1.3 Robustness Across the Sampled Ensemble

Training uses BPSK over AWGN at 5, 10 and 15 dB (Section 4.3.3); testing uses the specified Rayleigh distribution at the sampled SNRs, without online adaptation. This establishes performance over those draws, not every possible fade or an untested fading law. The equalization front end uses CSI, though the learned relay receives no explicit channel coefficient.

7.2 Capacity, Context and Training Variation

The original fixed-checkpoint comparison finds the 169-parameter MLP competitive with much larger evaluated models; it cannot separate architecture from initialization and training. The replicated size sweep supports small configurations under its declared degradation criteria, but neither study measures the train/test decomposition needed to establish overfitting.

7.2.1 Reading the Replicated Size Sweep

The three-initialization sweep does not demonstrate the capacity-induced BER upturn proposed by H3. Its median across-initialization spread is 0.176 dB over 148 configurations, but varies from 0.008 dB on the canonical channel to 1.231 dB on nonlinear bias. The pooled median is

not a universal significance threshold. Two clusters of training outcomes do not by themselves identify optimization attractors or prove a mechanism.

Four-parameter configurations satisfy the zero-degradation criterion on three memoryless cases; selected three-tap cases use 73–145 parameters. These are empirical grid results, not architecture lower bounds. The coded size sweep spans 18–1,076 parameters with penalties of $+1.24$–$+1.92$ dB under its hard read-out against unweighted block DF, and at least $+9.75$ dB against matched block DF. This motivates testing interfaces and metrics, but does not prove an interface-limited regime.

7.2.2 Practical Implications

The canonical 169-parameter model occupies 676 bytes of float32 weights. The BPSK fixed-ISI network instead has 170 parameters, and the coded-QPSK memory-sweep network has 220 parameters and 208 MACs per symbol. Parameter storage omits activations, input buffers, software and memory traffic; no embedded implementation, energy benefit, or sustained-throughput claim follows from it.

7.3 Candidates for Further Deployment Evaluation

Table 7.1: Candidates suggested by the simulations, not hardware-validated recommendations

Tested regime	Candidate	Evidence boundary
Canonical 0–4 dB	MLP/Hybrid and DF	Feedforward relays improve on AF; no general advantage over DF.
Canonical ≥ 6 dB	Symbol-wise DF	Lowest or tied-lowest evaluated BER; no training.
Selected ISI or uncertain-channel studies	Windowed MLP or model-based detector	Choice depends on the stated information, error target and cost. Family shift is untested.

Hybrid switching compares a received-power SNR estimate with a fixed 5 dB threshold. On the zero-forced Rayleigh observation that estimate is not the operating SNR, so in the canonical run the switch fell between 6 and 8 dB (Section 4.2). A calibrated estimator, sensitivity to the threshold, quantization, synchronization and deployment robustness remain unmeasured.

7.3.1 Latency and Adaptivity

Protocol scheduling, readiness and computation are separate costs. A symmetric window needs future samples; block decoding needs a frame; half-duplex forwarding spends channel uses on

both hops. Readiness may overlap the listening phase, but no end-to-end latency saving was measured.

In Section 6.5.1, no BER difference is resolved among the tested traceback settings. A two-symbol MLSE setting has less look-ahead than the five-symbol windowed relay and lower measured BER. The exactly factorized trellis needs fewer multiplications than the network up to six taps and 1.23 times as many at seven, and MLSE remains more accurate (the implemented joint trellis crosses it between three and five taps). Neither comparison establishes a hardware latency advantage.

Neither the learned relay nor the canonical experiments demonstrate online adaptation. AF's gain is recalculated from block power, and DF requires no training; neither is a fixed-SNR baseline.

7.4 Limitations

1. **Experimental scope.** One two-hop SISO relay and no direct path are modeled. The canonical comparison uses uncoded QPSK; BPSK, DBPSK, faded-ISI QPSK and coded modulation studies are separate. The QPSK ISI re-check changes both modulation and fading. The 16-QAM coded grid is not a learned-relay architecture study.
2. **Model information.** Canonical receivers have perfect CSI. Pilot-limited and blind studies relax this within specified families. Training supplies family knowledge even when no analytic parameters or pilots are given at inference.
3. **Finite data and replication.** Fixed checkpoints, three-seed studies, paired trial tests and pooled-bit bounds have different uncertainty units. Nominal independent-bit intervals need not have their stated coverage under correlated decoding errors. Zero errors mean insufficient data for a positive BER estimate, not zero true BER. MLP fixed-budget ISI validation stops at 16 dB.
4. **Comparator scope.** Matched Viterbi optimizes sequence error. The QPSK bit-MAP control measures only relay-output BER, not destination BER. Reduced-state MLSE and decision feedback are unmeasured alternatives, as is a coded destination metric that also models relay decision errors; for relays that make decisions, the matched destination metric of Section 5.5.1 weights by the second hop only.
5. **Coded metrics.** The original coded comparisons used Viterbi and BCJR metrics that omit symbol-dependent post-equalization reliability. Rerun with matched metrics (Section 5.5.1), block DF leads both MLP read-outs at every tested SNR, so their high-SNR advantage in the original tables reflects the metric mismatch, not a disadvantage of relay decoding; the coded Mamba relay was not rerun. The secondary coded studies (error location, constraint length, reliable-decoding regime, equal throughput, rate adaptation,

readiness budget and the coded size sweep) were rerun with the matched metric and are reported beside the published values; matched block DF is stronger in every one, and only the structural readiness lock-out at a 100-symbol budget survives unchanged.

6. **Generalization and hardware.** Memory is swept only through seven taps. New impairment families, correlated Doppler fading, other topologies, complete deadline models, quantized execution, energy and actual device timing are outside the evidence.

7.5 Future Work

The most direct next controls are an end-to-end bit-MAP comparison with consistent information and power normalization, a destination metric that models relay errors for hard-output relays, and prospective training-seed replication. BPSK bit-MAP, and plain-QPSK ISI under the protocol of Section 6.2.2, would isolate questions left open by the present endpoint and modulation changes. Larger constellations require an appropriate multilevel output; a joint constellation classifier is one option, not a theoretical necessity for all separable QAM channels.

Other extensions include correlated time-selective fading, online adaptation, additional pilot SNRs, longer memories, reduced-state classical comparators, other codes and frame lengths, multi-relay/MIMO topologies, joint end-to-end learning, and measured hardware operation. These are proposals, not demonstrated capabilities.

7.6 Hypotheses Revisited

H1 and H2 are supported within the canonical comparison. H3's overfitting explanation is not established. H4 does not hold across SNR: the low-SNR BER gap between the model families persists, and the high-SNR convergence is not robust across all measured seeds. In the equal-budget replication, Mamba spreads are 0.019 and 0.041 dB, whereas the Transformer's is at least 0.89 dB. These are three-seed observations, not universal architecture variances (Table 9.6).

Among eight Transformer initializations, one high-loss run has a much worse BER penalty. The all-run correlation is driven by that outlier. It motivates studying optimization failure, but does not establish a failure probability or validated screening rule; all runs remain in Appendix 9.4.

H5 is supported in the evaluated settings against AF and zero-threshold DF, which the chosen taps defeat analytically; the informative evidence is against model-based detectors: matched BPSK MLSE keeps penalties of 0.26–2.53 dB, depending on the target, and the MLP passes LS-plus-MLSE below ten pilots and leads blind CMA up to 12 dB. Similar rounded BERs and overlapping intervals elsewhere do not establish equivalence. The coherent conclusion is bounded: small trained relays can help in particular mismatched pipelines, while model information, temporal context, decoder metrics and computational budgets determine the comparisons actually measured.

Chapter 8

Summary

This work examined classical and neural network–based relay strategies for *two-hop relay communication* using a unified simulation framework. The core comparison uses one canonical setup (two-hop SISO, i.i.d. Rayleigh fast fading, QPSK, uncoded) and varies only the relay function; the coded and unknown-channel studies depart from it deliberately. The study showed the following:

1. The best tested learned relays outperform the implemented block-normalized AF at the evaluated low-SNR points (H1). Against conventional finite-power AF the MLP's 0 dB margin is only 0.003–0.007, depending on the gain rule, and the sequence relays fall behind; the feedforward relays track DF, so the advantage is that of regeneration rather than of learning.
2. Classical *decode-and-forward (DF)* matched or outperformed all neural approaches at **medium-to-high SNR** while requiring no trainable parameters; no learned relay was detectably better, which supports H2 in this matching sense.
3. Increasing model size did not produce a **systematic BER improvement**, and the replicated size sweep found no robust upward BER trend with model capacity (H3).
4. The 169-parameter MLP is competitive with larger models in the original canonical comparison. This is not a universal minimum-size result, and no train/test-gap study establishes overfitting as the explanation.
5. The replicated size sweep identifies the smallest tested configurations meeting each degradation criterion: four parameters suffice for three memoryless cases at zero degradation, while the selected three-tap cases use 73–145. Window length matters in this grid, but the experiment does not prove architecture lower bounds.
6. At an equal parameter budget, feedforward and sequence relays remained separated at low SNR. The apparent high-SNR convergence was smaller than the Transformer run-to-run spread and is not seed-robust; equalizing parameters also left depth, width, state

dimension, window, and optimization confounded (H4).

7. Within the simulated benchmark, a `hybrid relay` combining neural processing at low SNR with DF at high SNR tracked the best evaluated relay; its uncalibrated received-power SNR estimate put the switch between 6 and 8 dB rather than at the nominal 5 dB. Threshold robustness, quantization, energy use, synchronization and hardware behavior were not evaluated.

Findings 1–7 concern the canonical comparison (the size sweep of finding 5 spans several channel families); findings 8–11 summarize Chapter 6, and the coded study is summarized after them.

8. On the selected BPSK three-tap filter, AF and zero-threshold DF approach the derived 0.25 floor, while the 170-parameter MLP substantially reduces BER through the fixed-budget 16-dB point. Matched genie MLSE leads over the resolved range; the interpolated required-SNR penalties are 0.26, 1.18, 1.88, and 2.53 dB at BER 10^{-1} through 10^{-4}. These are not a constant offset. In the separate blind study, the family-trained MLP leads the evaluated CMA implementation up to 12 dB without test-time adaptation; from 14 dB the two are not resolved.

9. The pilot-aided LS-plus-MLSE sweep at 10 dB found the classical detector ahead at 20 pilots and the MLP at 5, with no resolved difference at 10; its SNR dependence was not measured. This plug-in detector does not marginalize channel uncertainty, and the experiment does not isolate estimation variance from nonlinear model mismatch. The result is specific to the tested estimator, channel family, block lengths, and SNR.

10. The separate coded-QPSK memory sweep uses a window-11, eight-hidden-unit, four-output MLP with 220 parameters and 208 MACs per symbol, retrained for each channel length. Exact I/Q factorization of the real-tap channel gives a classical count of 2^{L+1}, fewer multiplications than the network up to six taps and 256, 1.23 times its count, at seven; the implemented joint trellis's $2M^L$, 32,768 at seven taps, is a diagnostic of that implementation. These analytical counts are not measurements; MLSE retains lower measured BER, and a fixed architecture does not establish unchanged accuracy for longer memories.

11. **Latency did not separate the learned relay from sequence detection at the available resolution** on the tested channel. No BER difference was resolved among the tested traceback depths; a two-symbol MLSE setting had fewer symbols of decision delay than the window-11 relay's five and lower measured BER. The block relay required a full frame before producing output and showed no resolved accuracy gain in this experiment. This coded-memory study and the pilot-budget study address different resources in different configurations. They do not measure a common boundary or a quantitative distance between boundaries.

Read against the two research questions of Section 3.1.1, these findings give the following answers. **O1 (no budget):** on the evaluated channels with memory, the learned relay complements the classical method where the classical relay lacks the channel model — against memoryless AF and zero-threshold DF, and against the plug-in LS-plus-MLSE detector once pilots are scarce — and does not where matched MLSE is available, which retains the lower error rate (findings 8 and 9). **O2 (latency and arithmetic budget):** on the tested coded-memory channel, latency did not separate the learned relay from sequence detection (finding 11), and the arithmetic argument depends on how the trellis is implemented: the exactly factorized detector needs fewer multiplications than the 208-MAC network up to six taps and 1.23 times as many at seven (finding 10). Within the tested configurations, neither budget therefore excludes the classical method.

Table 8.1: Research hypotheses, their statements, and experimental outcomes

Hypothesis	Statement	Result
H1	Some AI relays outperform AF at low SNR	**Confirmed**: the best neural relays beat AF at 0–4 dB; against conventional AF the MLP's 0 dB margin is 0.003–0.007, not 0.073, and the feedforward relays track DF: the advantage is regeneration, not learning
H2	DF achieves the lowest BER at ≥ 6 dB, outperforming all AI methods	**Supported as matching**: DF matches or beats all AI methods at ≥ 6 dB with 0 parameters; none is detectably better
H3	An error minimum at intermediate capacity	**Not supported in the tested range**. The replicated size sweep finds no robust BER upturn with capacity. No train/test-gap analysis establishes an overfitting mechanism; uncertainty is configuration-dependent

Hypothesis	Statement	Result
H4	Equal parameter count makes architecture-dependent differences negligible	**Not supported across SNR**. Low-SNR differences remain; the apparent high-SNR convergence is not robust to all measured seeds. Width, depth, windows, state dimensions and optimization remain confounded
H5	A small learned relay improves on AF and zero-threshold DF under the selected model mismatches	**Supported within the evaluated configurations** (impairment family, implementations, budgets), whose taps defeat AF and zero-threshold DF analytically. Matched MLSE keeps penalties of 0.26–2.53 dB at BPSK and a substantial lead at QPSK; the MLP passes LS-plus-MLSE below ten pilots and leads blind CMA up to 12 dB

The evidence supports targeted use of learned relays, not a general learned-versus-classical verdict. The measured advantages depend on model mismatch, available channel information, architecture, and budget. Reduced-state sequence estimation, decision-feedback equalization, and a coded destination metric that also models relay decision errors remain unmeasured alternatives.

The supplementary coded study first found the learned relays ahead of block DF at high SNR, but its decoder metrics ignored symbol-dependent post-equalization reliability. With matched metrics at the relay and the destination, block DF has lower BER than both MLP read-outs (the coded Mamba relay was not rerun) at every tested SNR, in every paired trial; at 20 dB it makes 3 errors in 3×10^6 bits against $2{,}667$ for the better of them. The original high-SNR advantage was therefore produced by the metric mismatch. The matched relay decoder uses $|h_1|^2$, which the learned relays are not given; a learned relay with that input was not evaluated. Under rate adaptation with matched metrics, decoding at the relay gives the higher goodput at every SNR where a code is used, except under tight idealized buffering budgets (at 150 symbols below 20 dB, and at 100 symbols throughout), where block-DF's frame buffering excludes the codes it needs; neither is a deployment guarantee (Section 7.4).

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Chapter 9

Appendices

9.1 Model Architectures and Hyperparameters

Relay	Network Topology	Train Loss (LR)	Total Params	Implementation
MLP (Minimal)	Feedforward MLP, 5-symbol sliding window	MSE, Adam lr=0.01, 100 epochs, trained at 5/10/15 dB	169	PyTorch (CPU)
Hybrid	Own MLP sub-network + DF, switched per block and axis at a fixed 5 dB threshold on a received-power SNR estimate	MSE (MLP only), Adam lr=0.01, trained at 2/4/6 dB	169	PyTorch (CPU)
VAE	Encoder–decoder latent model (7→8→1)	β-VAE ($\beta = 0.1$), Adam lr=10^{-3}	1,777	PyTorch (CPU)
cGAN (WGAN-GP)	Conditional GAN (generator + critic)	WGAN-GP, Adam lr=10^{-4}	2,946	PyTorch (CUDA)
Transformer	2-layer encoder, 4 heads, $d_{\text{model}} = 32$	MSE, Adam lr=10^{-3}; 15% held out, lowest-training-loss epoch kept	17,697	PyTorch (CUDA/CPU)
Mamba-S6	Selective SSM, 2 S6 layers	MSE, Adam lr=10^{-3}; 15% held out, lowest-training-loss epoch kept	24,001	PyTorch (CUDA/CPU)
Mamba-2 SSD	Selective SSM with SSD kernel (2 layers)	MSE, Adam lr=10^{-3}, clip=1.0; 15% held out, lowest-training-loss epoch kept	26,179	PyTorch (CUDA)

Table 9.1: Comparison of relay models, training objectives, parameter counts, and implementations, as configured for the canonical comparison (`run_experiments.py`; Section 4.3.3), from which the cGAN was excluded on cost grounds

9.2 Reproducibility

9.2.1 Testing

The simulation framework carries 347 automated pytest tests in 24 test modules, covering channels, modulation, coding, relay strategies, weight save and load, simulation, statistics and the verification tooling. In the revision environment (Python 3.11.17, PyTorch 2.14) the full suite passed. The four tests that need the Mamba-2 module are skipped when it cannot be imported.

9.2.2 Reproducibility

Evaluation seeds are controlled for source-bit and noise generation. The canonical training never seeds PyTorch's weight initialization (Section 4.3.3), so the original checkpoints cannot be retrained exactly; the later replicated sweeps report training-seed variation explicitly.

Provenance of the reported results. Each reported result is associated with an experiment artifact in the source-code repository, using the experiment-specific budgets documented in the relevant section and ledger. Registered experiments are linked to their generating scripts and committed result files by `provenance_audit.py`, and `verify_thesis_tables.py` checks a declared subset of numerical table cells against those files; neither covers every number, figure or prose statement (Appendix 9.3). Independent audit requires that repository at the archived revision identifier; the ledger below documents the claimed provenance but cannot by itself reproduce every table cell.

9.3 Master Experiment Ledger

This ledger links major studies to their experiment artifacts. The table verifier checks a declared subset of numerical cells, not every number or figure. Historical `e6_*.npy` paths are relative to `e6_unknown_channel_results/`; the central ISI figure and table additionally use the matched-protocol and fixed-budget companion files listed below.

Table 9.2: Master experiment ledger: modulation, channel, and result-file provenance for the principal studies

Study	Modulation	Channel	Tbl/Fig	Result source
Canonical benchmark	QPSK	Rayleigh	Tab. 5.2, Fig. 5.1	`results/modulation/qpsk_rayleigh.json`
Normalized 3K-param. comparison	QPSK	Rayleigh	Tab. 5.3	`results/normalized_3k/3k_rayleigh.json`
Coded block-DF, K-sweep	QPSK	Rayleigh	Tab. 5.5	`results/coded_df_experiment.json`
Coded high-SNR paired re-test	QPSK	Rayleigh	Tab. 5.6	`results/coded_high_budget_test.json`
Coded relay-error mechanism	QPSK	Rayleigh	Tab. 5.7	`results/coded_error_mechanism.json`
Coded soft-decision relaying	QPSK	Rayleigh	Tab. 5.9	`results/coded_soft_decision.json`
Equal-throughput & latency	QPSK, 16-QAM	Rayleigh	Tabs. 5.10, 5.11	`results/coded_latency_throughput.json`
Adaptive modulation/coding (AMC)	QPSK, 16-QAM	Rayleigh	Tab. 5.12	`results/coded_rate_adaptation.json`
Capacity under a latency budget	QPSK, 16-QAM	Rayleigh	Tab. 5.13	`results/coded_latency_capacity.json`
Unknown ISI (layer 2)	BPSK	AWGN (Hop 2)	Tab. 6.3	Historical AF/DF/MLP below 12 dB: `e6_sim_ported_results.npy`; fixed-budget companion `codex_isi_fixed_budget_validation.json`; Viterbi: `results/e6_matched_protocol.json` and `results/e6_matched_highsnr.json`.
Flat-channel control	BPSK; DBPSK (phase)	Flat unknown Hop 1; comp. Rayleigh Hop 2	Tab. 6.2	`e6_flat_ported_results.npy`
Unknown ISI, QPSK generalization	QPSK	ISI + Rayleigh Hop 1; AWGN Hop 2	Tab. 6.4, Fig. 6.3	`e6_qpsk_unknown_channel_results.npy`

Study	Modulation	Channel	Tbl/Fig	Result source
Composite impairment cascade	DBPSK	Fixed composite Hop 1; comp. Rayleigh Hop 2	Fig. 6.4	`e6_composite_ported_results.npy`; AF replaced by `codex_composite_af_validation.json`.
Partial posterior (pilot sweep)	DBPSK	Random composite Hop 1; comp. Rayleigh Hop 2	Fig. 6.5	`e6_partial_ported_results.npy`
Blind channel (layer 4)	DBPSK	Random composite Hop 1; comp. Rayleigh Hop 2	Fig. 6.6	`e6_blind_ported_results.npy`
Four-layer summary	mixed (above)	mixed (above)	Tab. 6.1	combination of the above
AF block length and conventional AF	QPSK	Rayleigh	Sec. 7.1.1	`results/af_block_and_conventional.json`
Scalar MAP relay on unknown ISI	BPSK	ISI Hop 1; AWGN Hop 2	Sec. 6.2	`results/e6_scalar_map.json`
ISI required-SNR penalties	BPSK	ISI Hop 1; AWGN Hop 2	Tab. 6.1	`results/e6_isi_penalty.json` (from `results/e6_matched_protocol.json`)
Factorized QPSK MLSE equivalence	QPSK	Real-tap ISI	Sec. 2.6	`results/factorized_mlse_equivalence.json`
Coded relaying, matched metrics	QPSK, 16-QAM	Rayleigh	Tabs. 5.4, 5.7, 5.10, 5.12, 5.13	`results/coded_matched_metrics.json`; `results/*_matched.json`; `results/coded_matched_envelopes.json`

The result files above are themselves generated by named, committed scripts (e.g. `coded_rate_adaptation.py` produces `coded_rate_adaptation.json`); the correspondence between script and output file is one-to-one and is not separately tabulated here since the filenames make it explicit.

9.4 Minimum Relay Size and Initialization Variance

Supporting material for the minimum-size study of Section 5.4 and the initialization-variance measurements discussed in Section 7.2. The findings themselves are stated in those sections;

what follows is the underlying tabulation and the figures, collected here to keep the experiment and discussion chapters to their conclusions.

Sweep configuration. Every channel is run by `mlp_min_size_all_channels.py` (width and window grid) and `mlp_min_size_bisect.py` (width bisection at depths 1–3), with the models of Table 9.3. Each configuration is trained three times, from initialization seeds 0, 1 and 2, on $60{,}000$ symbols drawn from its own Hop-1 channel, $20{,}000$ at each of 5, 10 and 15 dB (both axes of each QPSK symbol are used), for 25 epochs with batch size 256 and learning rate 3×10^{-3}; the network after the final epoch is evaluated. Evaluation uses 20 trials of $20{,}000$ bits at 0–20 dB in 4-dB steps, with the same bits and channel draws for every relay. Windows are completed with zeros at the trial edges; QPSK channels are processed per axis. Width runs from 1 to 48 hidden units and the window from 1 to 7 samples. Penalties follow Appendix 9.6: a candidate's worst-of-seeds BER at each SNR against its reference relay's mean-of-seeds curve.

Table 9.3: Channels of the minimum-size sweep (Table 9.4). Hop-1 laws are those of Table 9.7 unless stated; "comp. Rayleigh" is coherently compensated Rayleigh fading with AWGN

Channel	Mod.	Hop 1	Hop 2	Comparator
awgn	BPSK	AWGN	AWGN	DF
rayleigh	QPSK	Zero-forced Rayleigh (canonical)	Same as Hop 1	DF
flat gain	BPSK	$g \sim \mathcal{U}[0.3, 2.0]$ per trial	comp. Rayleigh	DF
branch asym.	BPSK	$a_{\pm} \sim \mathcal{U}[0.6, 1.4]$ per trial	comp. Rayleigh	DF
nonlinear bias	BPSK	$\tanh(1.5x) + 0.5$ plus AWGN	AWGN	DF
ISI	BPSK	Fixed taps $\propto [1, 0.7, 0.5]$	AWGN	MLSE, exact taps
ISI (complex)	QPSK	The same taps on complex symbols	AWGN	QPSK MLSE, exact taps
ISI + Rayleigh	QPSK	The same taps, then a compensated Rayleigh magnitude per symbol	AWGN	QPSK MLSE given the taps only, not the fading gains, so not genie CSI

Channel	Mod.	Hop 1	Hop 2	Comparator
composite	BPSK	Fixed taps $\propto [1, 0.6, 0.4]$, then the amplifier of Appendix 9.6; with a real BPSK input the block phase is not applied and the noise variance is $1/\gamma$, twice the convention	comp. Rayleigh	MLSE, exact taps
coded	QPSK	The coded link of Section 5.5	Zero-forced Rayleigh	Block DF (unweighted metric)

Table 9.4: Smallest relay meeting a given degradation budget against the window-5, 169-parameter MLP retrained on each channel (the published MLP-756 for the coded row), written as parameters and window. Most memoryless rows settle at window 1, but flat gain and nonlinear bias require wider windows at the strictest budget; every three-tap row except the composite requires window 5 or 7 until a full decibel is conceded. A dash indicates that no smaller configuration meets that budget; one that misses a target the reference reaches meets none, which removes the coded row's earlier 372p and 244p entries. Penalties follow the definition in Appendix 9.6

Channel	Taps	≤0 dB	≤0.1 dB	≤0.25 dB	≤0.5 dB	≤1 dB	≤2 dB
awgn	1	4p w1	4p w1	4p w1	4p w1	4p w1	4p w1
rayleigh	1	4p w1	4p w1	4p w1	4p w1	4p w1	4p w1
flat gain	1	21p w3	21p w3	21p w3	4p w1	4p w1	4p w1
branch asym.	1	4p w1	4p w1	4p w1	4p w1	4p w1	4p w1
nonlinear bias	1	29p w5	4p w1	4p w1	4p w1	4p w1	4p w1
ISI	3	73p w7	73p w7	57p w5	57p w5	37p w7	21p w3
ISI (complex)	3	73p w7	73p w7	73p w7	73p w7	57p w5	21p w3
ISI + Rayleigh	3	145p w7	73p w7	73p w7	73p w7	37p w7	21p w3
composite	3	145p w7	73p w7	57p w5	21p w3	21p w3	6p w3
coded	code	—	—	—	18p w1	18p w1	18p w1

9.4.1 Results Figures

Eight Transformer initializations are shown in the following figures: seven cluster and one has both higher final loss and worse high-SNR BER. A sample of three cannot reliably characterize such failures. If three runs were sampled without replacement from these eight, the probability of missing the one outlier would be $\binom{7}{3}/\binom{8}{3} = 5/8$; this finite-set calculation is not an estimate of the population failure rate. The association does not validate training loss as a general screening rule.

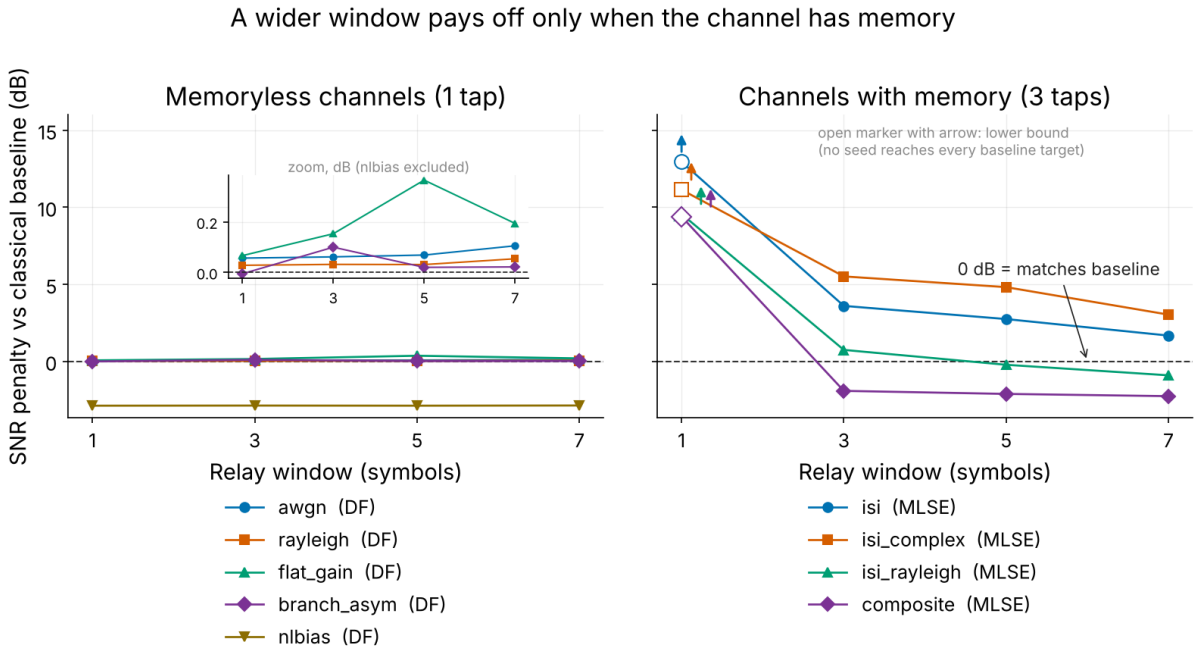


Figure 9.1: SNR penalty against each channel's classical comparator as the relay window widens, for the best configuration at each width. The reader should observe that the two panels behave oppositely: on the memoryless channels the penalty is flat to within a fifth of a decibel across every window (see inset), so the window buys nothing, while on the three-tap channels it falls by more than 8 dB between window 1 and window 7. Window-1 points there are lower bounds (open markers). This is the sense in which the relay's window must be matched to the channel's impulse response

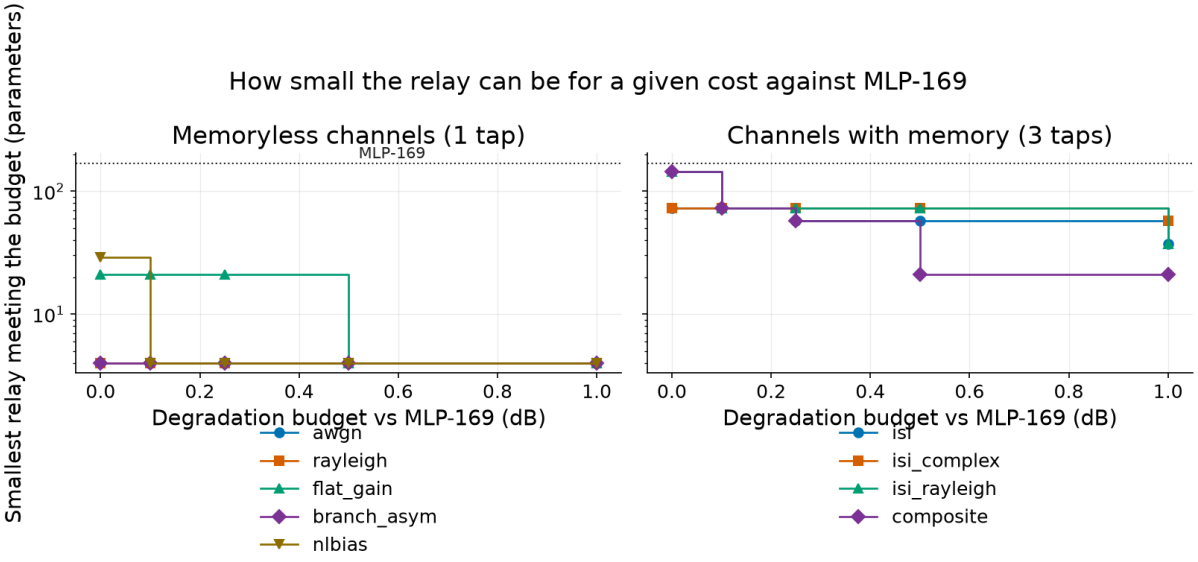


Figure 9.2: Smallest relay meeting a given degradation budget against MLP-169, drawn as a step function because the horizontal axis is a budget the designer chooses rather than a measured quantity. Most memoryless cases reach four parameters once a small degradation is allowed, while strict zero-degradation budgets require larger models for flat gain and nonlinear bias; the three-tap channels need between 6 and 145 depending on the permitted degradation

Table 9.5: Spread in SNR penalty between the best and worst of three initializations at an identical configuration, summarized over all configurations swept on each channel. The reader should observe that on five of the nine channels the median spread is comparable to or larger than the entire effect of varying model size, so a single-initialization size comparison on those channels measures the draw rather than the architecture. Configurations in which no seed reaches every target the comparator reaches have no defined spread and are not counted; three ISI (complex) spreads are lower bounds, and so are the statistics marked $\geq$ (the earlier rule gave that row a maximum of 3.182 dB). Penalties follow the definition in Appendix 9.6

Channel	Configs	Median (dB)	90th pct. (dB)	Max (dB)
rayleigh	17	0.008	0.019	0.028
composite	16	0.036	0.165	0.195
awgn	17	0.055	0.116	0.138
isi_rayleigh	16	0.177	0.588	0.795
isi	16	0.189	0.738	0.996
branch_asym	17	0.388	0.892	1.013
isi_complex	15	0.404	$\geq$1.110	$\geq$1.555
flat_gain	17	0.622	1.000	1.250
nlbias	17	1.231	1.242	1.245
All	148	0.176	$\geq$1.226	$\geq$1.555

Table 9.6: Spread in SNR penalty against symbol-wise DF between the best and worst of three initializations, for the equal-budget architectures on the canonical channel. The reader should observe the Transformer row: its spread is at least five times the roughly 0.18 dB that H4's 4.1% high-SNR architecture spread corresponds to, so a single run of that architecture cannot resolve the effect H4 claims. Every other architecture is comfortably inside it. Three seeds on one machine; Section 6.4.2 finds that such spreads change between machines. Values marked $\geq$ are initializations that do not reach 10^{-2} by 20 dB, where DF does; they and the spreads they enter are lower bounds. Penalties follow the definition in Appendix 9.6

Architecture	Params	Per-seed penalty vs DF (dB)	Spread (dB)
Mamba2-3K	3,004	+1.113, +1.094, +1.098	0.019
Mamba-S6-3K	3,027	+1.123, +1.110, +1.082	0.041
MLP-3K	3,004	+0.133, +0.162, +0.189	0.055
VAE-3K	3,037	+0.072, $\geq$+0.228, +0.142	$\geq$0.156
Transformer-3K	3,007	$\geq$+2.039, +1.147, $\geq$+1.133	$\geq$**0.892**

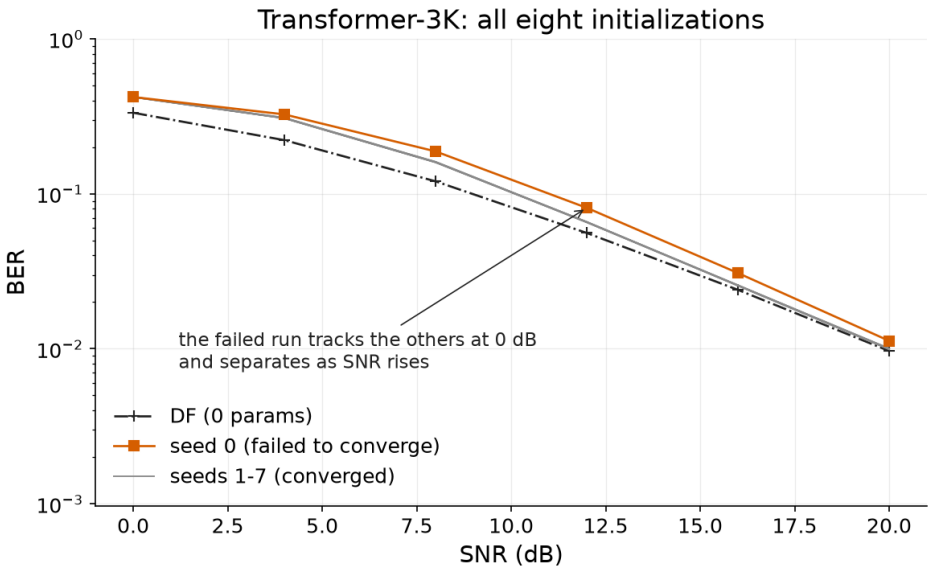


Figure 9.3: Transformer-3K on the canonical channel across eight initializations, against symbol-wise DF. The reader should observe that the failing run is not offset by a constant: it tracks the others at 0 dB and separates as SNR rises, so an evaluation confined to low SNR would not distinguish it from the seven that converge

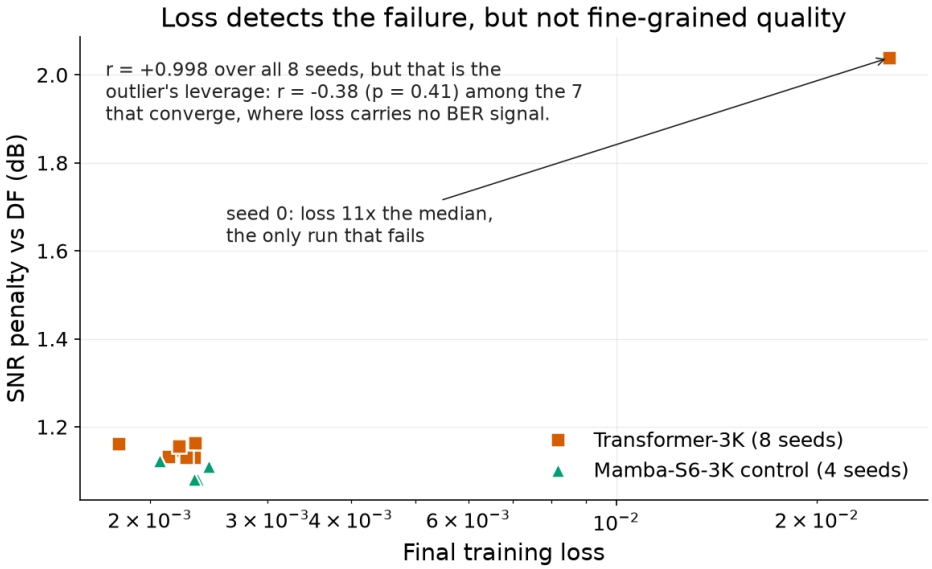


Figure 9.4: Final training loss against SNR penalty for the eight Transformer initializations and the four-seed Mamba-S6 control. The correlation over all eight seeds is $r = +0.998$, but that figure is the outlier's leverage: among the seven runs that converge it is $r = -0.38$ ($p = 0.41$), so loss identifies the failure and carries no information about relative quality once training has succeeded. Three runs, the failing one included, miss 10^{-2} by 20 dB, so their values are lower bounds

9.5 Verification of Headline Claims

Each headline claim of Chapter 6 is restated here with the argument or derivation that supports it and the primary reference against which it was checked. The fixed-ISI channel is the normalized 3-tap FIR $\mathbf{h} = [1, 0.7, 0.5]/\|\cdot\| \approx [0.758, 0.531, 0.379]$.

9.5.1 Matched-receiver bounds and the empirical MLSE comparison

For a fixed observation and supplied model, bit-MAP minimizes expected bit error; MLSE instead minimizes sequence error [32, 33, 31]. Thus a lower measured BER than MLSE would not contradict its optimality. The QPSK BCJR control measures relay-output errors only and must not be compared directly with destination BER. It does not close an end-to-end BCJR benchmark. Learned-receiver studies use different information assumptions and objectives [23, 24, 45, 7, 46]; they should not be summarized as a universal claim that learning wins only under mismatch.

9.5.2 The analytic 0.25 floor and the non-monotonicity of symbol-wise DF

With the normalized taps, the interferer magnitude sum $h_1 + h_2 \approx 0.910$ exceeds the cursor $h_0 \approx 0.758$: the eye is closed. Of the four equiprobable BPSK interferer sign patterns, exactly one (both opposing the cursor) yields a noise-free sample of $0.758 - 0.910 = -0.152 < 0$, a deterministic sign flip, while the remaining three (0.606, 0.910, 1.668) preserve the sign – so a memoryless slicer errs on exactly one pattern in four as noise vanishes, giving the analytic floor of exactly $1/4$. The non-monotonicity follows from the same geometry: at moderate SNR, noise occasionally pushes the flipped sample back across the decision boundary, so the conditional error rate on that pattern rises toward one as SNR grows and total BER climbs toward 0.25 from below – standard behavior for a closed-eye channel under symbol-by-symbol detection [47, 31].

9.5.3 Complexity: the M^L trellis against a fixed forward pass

An unrestricted joint-symbol MLSE implementation maintains M^{L-1} states and evaluates M^L branch metrics for channel length L [32, 31]. For the real-tap QPSK channel, exact I/Q factorization reduces that search to two binary trellises, and for square M-QAM to two $\sqrt{M}$-ary trellises with $2M^{L/2}$ branch metrics (Section 2.6); the joint count is not a necessary cost of matched detection. The separate real $11 \to 13 \to 1$ MLP has 170 parameters and $11 \cdot 13 + 13 = 156$ MACs, plus 14 bias additions and nonlinear activations. Counting a multiply and add as two FLOPs gives approximately 330 FLOPs excluding activation cost. Cost is fixed only for that deployed architecture; longer windows increase it. Reduced-state sequence estimation [44] is another unmeasured lower-cost classical alternative.

9.5.4 The blind regime: CMA against decision-directed MLSE

CMA is a blind equalizer for constant-modulus signals [48, 49]. The measured random composite family also contains nonlinearity and phase uncertainty, so the zeros of one fixed linear tap vector do not establish invertibility or convergence for the whole cascade. The observed CMA and decision-directed MLSE results are implementation-specific. Documented convergence and decision-feedback issues [50, 51] motivate possible explanations but do not identify the cause of these measurements.

9.5.5 The pilot-budget crossover: resolved between twenty and five pilots, with severe degradation at five

A three-tap least-squares fit can be identifiable from five pilots if its design matrix has full column rank. For a correctly specified linear model with white noise, its covariance is $\sigma^2(\mathbf{X}^H\mathbf{X})^{-1}$ [43]; the simplification $\sigma^2 L/N_p$ additionally requires suitably normalized orthogonal pilot columns. The nonlinear composite channel need not satisfy that model. The observed small-pilot degradation is therefore not a measured rank threshold or an unbounded-error theorem. Other estimators, including meta-learned demodulation [52], are not benchmarked here.

9.5.6 The finite-window gap is not causally decomposed

For bit-MAP decisions under the same model and side information, giving the detector a full sequence cannot increase its minimum expected bit error relative to a window. This observation-ordering argument is not a bound on a particular Viterbi-versus-MLP BER gap, because Viterbi optimizes sequence error. No windowed bit-MAP ablation separates truncation, approximation, finite training, and optimization effects here. The BPSK required-SNR penalties range from 0.26 to 2.53 dB at the stated targets. They cannot be attributed solely to the 170-parameter budget.

Supplementary numerical checks. The across-seed spread in SNR penalty against DF in the separate native-architecture replication is 0.004 dB for Mamba-S6, 0.007 for the Hybrid, 0.013 for Mamba-2, 0.043 for the MLP, and at least 0.075 for the Transformer and 0.091 for the VAE (one initialization of each misses 10^{-2}). These finite three-seed ranges are not confidence bounds on failure probabilities.

For the QPSK linear-MMSE detail run, the attained MSE at 16 dB is $0.0666,\ 0.0374,\ 0.0354,\ 0.0333$ for $N = 3, 5, 7, 11$. The required-SNR penalties at targets 10^{-1} and 10^{-2} are respectively $+3.24, +2.98, +2.79, +2.65$ and $+6.34, +4.27, +4.13, +4.02$ dB. These measurements do not imply monotonic BER at every target.

9.6 Channel Models and Metrics of Chapter 6

This appendix specifies the channel models, relay inputs and summary metrics of Chapter 6 as implemented in the scripts that produced the reported data. Unless stated otherwise, every real noise dimension has variance $N_0/2 = 1/(2\gamma)$ with $\gamma = 10^{\mathrm{SNR}/10}$ and unit symbol energy, every tap vector is normalized to unit energy, and the impairments of a hop act in the order listed. Hop-2 fading is coherently compensated: only its magnitude $|h_2|$, with $h_2 \sim \mathcal{CN}(0,1)$ drawn independently per symbol, multiplies the forwarded signal.

Table 9.7: Channel models of the Chapter 6 studies, as implemented. "Per trial" means that a parameter is drawn once per evaluation trial and held for all of its symbols; training data are generated in four blocks per training SNR, each with its own draw

Study (script)	Source and Hop 1	Redraw	Hop 2
Fixed ISI (`e6_sim_ported.py`, `e6_matched_protocol.py`)	BPSK; $y = h * x + n$, $h \propto [1, 0.7, 0.5]$	Fixed	AWGN
Nonlinear bias (`e6_sim_ported.py`)	BPSK; $y = \tanh(1.5x) + 0.5 + n$	Fixed	AWGN
QPSK ISI (`e6_qpsk_unknown_channel.py`)	QPSK; the fixed-ISI taps, then a compensated Rayleigh magnitude $\lvert h_1\rvert$ per symbol, then complex AWGN	Taps fixed; $\lvert h_1\rvert$ per symbol	AWGN (a compensated-Rayleigh run is stored, not reported)
Flat gain (`e6_flat_ported.py`)	BPSK; $y = gx + n$, $g \sim \mathcal{U}[0.3, 2.0]$	Per trial of 10^5 bits	Compensated Rayleigh, see below
Branch asymmetry (`e6_flat_ported.py`)	BPSK; $y = a_+ + n$ if $x = +1$, $y = -a_- + n$ if $x = -1$, with $a_+, a_- \sim \mathcal{U}[0.6, 1.4]$ independent	Per trial of 10^5 bits	As flat gain
Flat phase (`e6_flat_ported.py`)	DBPSK; $y = e^{j\theta}s + n$, $\theta \sim \mathcal{U}[0, 2\pi)$, complex AWGN	Per trial of 10^5 bits	As flat gain
Fixed composite (`e6_composite_ported.py`)	DBPSK; ISI $h \propto [1, 0.6, 0.4]$, then amplifier $A(\cdot)$, then $e^{j\theta}$, $\theta \sim \mathcal{U}[0, 2\pi)$, then complex AWGN	Taps fixed; θ per trial of 10^5 bits	Compensated Rayleigh

Study (script)	Source and Hop 1	Redraw	Hop 2
Random composite (`e6_blind_ported.py`, `e6_partial_ported.py`)	As the fixed composite, with $h \propto [1, u_1, u_2]$, $u_1 \sim \mathcal{U}[0.3, 0.7]$, $u_2 \sim \mathcal{U}[0.2, 0.5]$	Taps and θ per trial of 2×10^4 bits, or per block in the block-length sweep	Compensated Rayleigh, complex AWGN
Coded memory (`joint_latency_memory.py`)	Rate-$1/2$, $K = 3$ convolutional code, QPSK; $h_k \propto 0.7^k$, $k = 0, \ldots, L-1$, $L \in \{3, 5, 7\}$; complex AWGN	Fixed	Complex AWGN

Amplifier. A acts on the amplitude and preserves the phase: $A(r) = r/\sqrt{1 + (r/1.2)^2}$, a Rapp soft limiter with smoothness 1 and saturation 1.2. Because the ISI taps are real and the DBPSK symbols are ± 1, the amplifier input is real, so the rotation $e^{j\theta}$ applied after it is the only source of an imaginary component on Hop 1.

Normalization and noise. Relay outputs are scaled to unit power over the evaluation trial (Eq. (4.8)). In the flat study the Hop-2 noise of real forwarded signals originally had variance $1/\gamma$, twice the convention above, while AF in the flat-phase case forwards a complex signal and received $1/(2\gamma)$ per dimension. The study was rerun with $1/(2\gamma)$ for every relay on the original seeds; Table 6.2 reports the rerun. DF and the MLP had always seen the same Hop-2 noise, so their comparison was unaffected; their absolute BERs roughly halved.

Pilots. Pilot symbols are the first N_p transmitted symbols of a trial or block, which are known to the estimator. The estimator fits three complex taps by least squares to a linear model, so it does not model the amplifier. In the fixed-composite study ($N_p = 200$) the decoded pilot region is scored together with the payload; in the pilot-budget sweep it is excluded, and in its block-length panel every block of L symbols carries 10 pilots and its own channel draw.

Windows and their boundaries. The learned relays of the uncoded Chapter 6 studies read a window of $W = 11$ samples centered on the decided symbol (the coded-memory study sweeps W from 1 to 21, the size sweep from 1 to 7, and the two Mamba relays of Section 6.4.2 use only the causal half of theirs); on complex channels the I and Q windows are concatenated into $2W$ inputs. At both ends of a trial the window is completed with zeros. The canonical MLP of Chapter 5 repeats the edge sample instead.

DBPSK. The source bits $x_i \in \{\pm 1\}$ are differentially encoded as $s_i = s_{i-1} x_i$. The learned relay is trained with the information bit x_i as its target, so it estimates x_i directly rather than the

differential symbol; its real output, scaled to unit power, is forwarded as a BPSK signal, and the destination decides on the sign of its real part without differential decoding. The DF relay performs differential detection, $\hat{x}_i = \operatorname{sign} \operatorname{Re}\{y_i y_{i-1}^*\}$, and forwards $\hat{x}_i$; AF forwards the complex observation and the destination performs the differential detection. The first bit, the differential reference, is excluded from scoring, except for the learned relay in the fixed-composite study, where it is scored (one bit per 10^5).

Training. Each learned relay is trained once, at SNR 5, 10 and 15 dB in equal proportions, on data drawn from the Hop-1 model of its own study, with fixed sample counts and epochs (for example 1.2×10^5 windows and 25 epochs for the fixed ISI, and 1.6×10^5 windows and 30 epochs for the composite studies). The network used for evaluation is the one left after the final epoch; no validation-based checkpoint selection or early stopping is applied. The sequence-model wrappers of the canonical comparison differ: they restore the epoch with the lowest training loss (Section 4.3.3).

Single-number SNR penalties. All required-SNR penalties use one definition, implemented in `ber_metrics.py`. For a BER target t, the SNR at which a curve reaches t is found by linear interpolation of ln BER against SNR in dB between the two measured points that bracket t, taking the last such crossing. A zero-error point cannot be interpolated in log scale: if the preceding point lies above t, the crossing is placed at the zero-error point's SNR, which can only overstate the SNR the curve needs. The penalty at t is the relay's SNR minus the reference's, positive when the relay needs more. The targets are fixed by the reference curve: of 10^{-1}, 10^{-2} and 10^{-3}, those the reference reaches within the tested SNR range form a common set for every curve compared against it. A curve that misses a target in that set needs more than the highest tested SNR there, so its penalty at that target is only bounded below, by that SNR minus the reference's, and is marked as a lower bound rather than dropped. Tables 6.6, 6.7, 9.4, 9.5 and 9.6 report one number per configuration: the largest penalty over the common set, itself a lower bound when a target is missed. Their aggregation over training seeds differs, as follows.

- Table 6.6: the learned-relay column is the best of three seeds of the best size-sweep configuration.
- Table 6.7: the best of three seeds, with the spread in brackets.
- Table 9.4: the candidate's worst-of-seeds BER at each SNR, against the reference relay's mean-of-seeds curve; a candidate with only a lower bound meets no budget.
- Tables 9.5 and 9.6: the largest minus the smallest per-seed penalty, a lower bound when a seed is bounded.

`penalty_common_targets.py` recomputes them from the stored per-seed curves and reproduces the earlier values, which kept only the targets both curves reach. The fixed-ISI comparison with genie MLSE is reported per target (10^{-1} to 10^{-4}), computed from `results/e6_matched_protocol.json` by `e6_isi_penalty.py`.

תקציר

עבודה זו משווה עיבוד קלאסי ועיבוד נלמד במערכת תקשורת דו־דילוגית עם ממסר יחיד וללא נתיב ישיר בין המקור ליעד. היא בוחנת מתי ממסר נלמד קטן משפר את ביצועי קווי הבסיס הקלאסיים שמומשו, וכיצד ההשוואה תלויה במידע על הערוץ ובעלות החישוב.

הניסוי הקנוני משתמש באפנון Gray-QPSK ללא קוד ערוץ, בקישור SISO עם דעיכת ריילי בלתי־תלויה, ובידיעת ערוץ בצד המקלט. שמונה אסטרטגיות ממסר מושוות בסימולציית מונטה קרלו. ממסרים נלמדים בהזנה קדמית משפרים את הביצועים ביחס למימוש של הגבר־והעבר (AF) המנורמל לפי הספק הבלוק ביחס אות־רעש נמוך, אך מול AF קונבנציונלי בהספק סופי הפער של ממסר ה־MLP ב־0 dB הוא 0.003–0.007 בלבד, בהתאם לכלל ההגבר; ממסרים אלה עוקבים אחר פענח־והעבר סימבולי (DF), המשתווה להם או עולה עליהם החל מ־6 dB. סריקות גודל משוכפלות מזהות תצורות קטנות העומדות בקריטריוני הירידה בביצועים שנקבעו; הן אינן מוכיחות מנגנון של התאמת־יתר או ארכיטקטורה מזערית אוניברסלית.

ניסויים נפרדים מוסיפים הפרעה בין־סימבולית, אי־ליניאריות של מגבר ואי־ודאות במידע על הערוץ. בערוץ BPSK הקבוע שנבחר, בעל שלושה מקדמים, ממסר MLP ממוסגר־חלון בן 170 פרמטרים מפחית במידה ניכרת את שיעור שגיאות הסיביות ביחס ל־AF ול־DF בעל סף אפס. גלאי Viterbi MLSE מותאם־מודל עדיין מוביל בתחום שבו ההבדל נמדד בבירור, עם קנס יחס אות־רעש של 0.26–2.53 dB, התלוי ביעד השגיאה. אימות הממסר הנלמד בתקציב קבוע מגיע עד 16 dB; הזנב שמעבר לכך לא אומת. בסריקת פיילוטים נפרדת בערוץ מורכב ב־10 dB, הממסר הנלמד מפגר אחרי הגילוי נעזר־הפיילוטים בעשרים פיילוטים, אינו נבדל ממנו באופן מובהק בעשרה ומוביל בחמישה. ממצאים אלה נוגעים למשפחות הפגמים שנכללו באימון, ולא למשפחות שלא נראו בו.

מחקר זיכרון נפרד בקישור QPSK מקודד משתמש בממסר בן 220 פרמטרים וב־208 פעולות כפל־צבירה לסימבול. גלאי MLSE המופרד במדויק לרכיבי I/Q דורש פחות פעולות כפל מממסר זה עד שישה מקדמים ופי 1.23 יותר בשבעה, ושומר על שיעור שגיאה נמוך יותר; ספירת הטרליס המשותף שמומש, החוצה את עלות הממסר בין שלושה לחמישה מקדמים, היא כלי אבחון של מימוש זה בלבד. במחקר הממסור המקודד, יתרונם של ממסרי ה־MLP הנלמדים ביחס אות־רעש גבוה על פני DF בלוקי נעלם כאשר המפענחים משקללים כל סימבול לפי אמינות הערוץ: עם מדדים מותאמים, ל־DF בלוקי שיעור שגיאות נמוך יותר בכל יחס אות־רעש שנבדק. בקרת BCJR מודדת שגיאות ביציאת הממסר בלבד.

הראיות תומכות ביתרונות מוגבלים ביחס לשרשראות הקלאסיות המסוימות שנבדקו. הן אינן מוכיחות עליונות אוניברסלית של ממסר נלמד, הכללה מחוץ למשפחות האימון או חיסכון בהשהיה ובאנרגיה בחומרה.

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